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A WEB-BASED SERVICE MANAGEMENT SYSTEM WITH DATA ANALYTICS USING MACHINE LEARNING ALGORITHMS FOR SHOELOTSKEY VILLAMOR-PASAY

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CHAPTER 1

INTRODUCTION

Small-scale shoe cleaning, restoration, and other service-oriented establishments play an important role in local economic activity, particularly in urban areas where they provide accessible services to the community. However, many small businesses continue to rely on manual or semi-digital processes for managing day-to-day operations. Such practices may make the organization, retrieval, and analysis of operational records more difficult, particularly as transaction records accumulate over time. In an increasingly data-driven business environment, web-based information systems and data analytics provide opportunities for small enterprises to improve record management, operational monitoring, and decision-making.

In the case of Shoelotskey Villamor-Pasay, owned by Ms. Aileen Rosales, the researchers' pre-assessment of the existing operations, supported by consultation with the shop owner and staff and examination of available business records, identified several limitations associated with the existing manual process. Service information and transaction records were primarily maintained through paper-based records, making the retrieval, monitoring, and consolidation of information more time-consuming. The existing process also presented difficulties in monitoring service status, preparing business reports, reviewing historical transactions, and consistently determining expected service completion dates. These operational conditions established the need for a centralized digital system that could organize service records, provide timely access to operational information, and support more systematic business monitoring.

Despite the availability of web-based management systems and machine learning technologies, there remains an opportunity to develop an integrated solution tailored to the operational requirements of a small-scale service establishment such as Shoelotskey. Existing commercial systems may include features or implementation requirements that do not directly correspond to the business's specific workflow and available resources. Furthermore, integrating centralized service management, inventory monitoring, real-time business analytics, and predictive modeling within a single system may provide a more practical approach to addressing the identified operational requirements of the establishment.

To address these challenges, this study proposes the development of a web-based service management system with data analytics using machine learning algorithms for Shoelotskey Villamor-Pasay. The system centralizes core operational processes, including job order management, sales monitoring, inventory tracking, and report generation. It also provides real-time analytics for monitoring business information and incorporates a machine learning model developed using simulated operational data representing verified service configurations and operational conditions, while historical service records were separately used to evaluate the limitations of the original Order-to-Claim prediction target.

Problem Statement

Shoelotskey Villamor-Pasay currently experiences operational challenges associated with its reliance on manual processes. Based on the pre-assessment interview with the business owner and review of available business records, the following problems were identified:

Service records, customer information, and inventory data are recorded manually or maintained in separate files. This fragmented process makes the retrieval, updating, and monitoring of information time-consuming and increases the possibility of inconsistencies in record management. The business lacks a centralized platform for monitoring daily operations. As a result, the business owner cannot readily access consolidated information regarding ongoing and completed job orders, service demand, sales performance, and other operational records.

Service completion dates are primarily determined through staff judgment and existing business practices rather than through the analysis of historical service data. Consequently, expected completion dates may vary depending on the circumstances of individual job orders and the judgment of the personnel handling them. Report preparation also requires the manual consolidation of transaction records. Daily, weekly, monthly, quarterly, and annual sales and operational reports must be prepared from existing records, requiring additional administrative time and effort.

Based on the pre-assessment interview with the business owner, the retrieval of a job order during claiming was estimated to require approximately 3–10 minutes per inquiry. The business owner also estimated that the manual preparation and consolidation of a report required approximately 1–2 hours per report, depending on the reporting period and volume of records involved. Furthermore, the existing manual process was reported to result in a delay of approximately 2–3 days in reflecting service-status updates, which may affect the timely monitoring of ongoing service orders.

In addition, a review of the establishment's available transaction records indicated an average monthly transaction volume of approximately 280–300

service orders during the period covered by the pre-assessment. The volume of transactions, combined with the continued use of manual and fragmented records, emphasizes the need for a more centralized approach to organizing, retrieving, and monitoring operational information.

Without a centralized and data-driven system, the business may continue to experience limitations in record accessibility, operational monitoring, report preparation, and systematic estimation of service completion dates. These identified problems established the basis for developing a web-based service management system that centralizes operational records, provides real-time analytics, and utilizes historical service data to support the prediction of service turnaround duration and corresponding release dates.

The pre-assessment further identified the following baseline operational conditions:

PRE-ASSESSMENT OF EXISTING SHOLOTSKEY OPERATIONS

Date of Pre-Assessment: March 13-15, 2026
Respondent/s: Business Owner / Staff
Method: Interview, workflow observation, and review of available business records

The pre-assessment interview and observation were guided by the following questions:

1. How are job orders currently recorded and retrieved?
2. Approximately how long does it take to locate a customer's job order during claiming?
3. How are service statuses monitored and updated?
4. Are there delays in reflecting service-status updates? If yes, approximately how long?
5. How are sales and operational reports currently prepared?
6. Approximately how long does preparation of each report take?

7. Approximately how many service orders does the establishment process monthly?
8. How are expected completion/release dates determined?

PRE-ASSESSMENT SUMMARY

Operational Indicator	Pre-Assessment Finding	Source of Evidence
Job order retrieval time during claiming	Approximately **3–10 minutes per inquiry**	Pre-assessment interview and/or workflow observation
Delay associated with manual service-status updating	Approximately **2–3 days**	Owner/staff interview and/or available records
Manual report-preparation time	Approximately **1–2 hours per report**	Owner/staff interview and/or workflow observation
Average monthly transaction volume	Approximately **280–300 service orders**	Available business/service records

Based on the pre-assessment, job orders and related service information were primarily maintained through paper-based records or separate files. The business owner estimated that locating a job order during claiming required approximately 3–10 minutes per inquiry. Service-status information was manually monitored and updated, with an estimated delay of approximately 2–3 days in reflecting status changes. Sales and operational reports required manual consolidation of transaction records, with approximately 1–2 hours required per report, depending on the reporting period and volume of records involved. A review of available business/service records indicated an average monthly transaction volume of approximately 280–300 service orders during the period covered by the pre-assessment. Expected completion dates were primarily determined through staff judgment and existing business practices rather than through analysis of

historical service data. These findings established the baseline operational conditions used in identifying the problems addressed by the developed system.

Research Questions

This study seeks to answer the following questions:

1. How does the existing manual paper-based process compare with the developed centralized digital job order entry and management system in terms of record accessibility, retrieval time, and data accuracy.
2. How does the Random Forest Regressor perform in predicting service turnaround duration and corresponding release dates in terms of:
   2.1 Mean Absolute Error (MAE);
   2.2 Root Mean Square Error (RMSE);
   2.3 Coefficient of Determination (R^2); and
   2.4 Prediction tolerance within ±1, ±2, ±3, and ±5 days;
3. What is the impact of the real-time analytics dashboard on reporting efficiency, accessibility of business information, and decision-making?
4. To what extent does the developed system meet ISO/IEC 25010 software quality standards in terms of:
   4.1 Functional suitability;
   4.2 Performance efficiency;
   4.3 Compatibility;
   4.4 Interaction Capability;
   4.5 Reliability;
   4.6 Security;
   4.7 Maintainability;
   4.8 Flexibility; and

4.9 Safety;

5. What is the level of user acceptance of the developed system based on the Technology Acceptance Model (TAM) in terms of:
   5.1 Perceived usefulness;
   5.2 Perceived ease of use;
   5.3 Behavioral intention to use; and
   5.4 Actual system use;
6. What improvements may be recommended for future enhancement of the system?

Project Context

Micro, small, and medium enterprises (MSMEs) play a significant role in economic development by generating employment opportunities, supporting local commerce, and providing accessible products and services to surrounding communities. In the Philippines, many small service-oriented businesses continue to rely on manual or semi-digital operational processes such as handwritten logbooks, spreadsheet records, and fragmented paper-based documentation. While these traditional practices may be manageable at a small scale, they often become inefficient as transaction volume increases. Common issues include delayed retrieval of records, inaccurate inventory counts, difficulty monitoring service progress, and limited access to timely business information.

The growing use of digital technologies has transformed how businesses manage operations, analyze performance, and improve customer service. Web-based information systems enable centralized data storage, real-time monitoring, automated reporting, and faster transaction processing. Likewise, business analytics tools allow owners and managers to identify sales trends, customer

demand patterns, operational bottlenecks, and performance opportunities. Businesses that fail to adopt these technologies may experience slower decision-making, higher administrative workload, and reduced competitiveness.

Shoelotskey, a shoe cleaning and restoration service business located in Villamor, Pasay City, owned by Ms. Aileen Rosales, currently experiences operational challenges associated with manual record management. Based on initial interviews and observation, service transactions, inventory usage, customer records, and service status updates are recorded manually or through isolated files. This setup makes it difficult to monitor pending and completed services, track frequently availed services, prepare sales summaries, and estimate completion schedules accurately. As a result, managerial decisions may rely more on assumptions than on organized operational data.

Another operational challenge involves predicting service completion dates. Customers commonly expect clear estimates regarding release schedules; however, manual estimation based only on staff experience may produce inconsistent results. Inaccurate estimates may lead to delays, customer dissatisfaction, and inefficient workload distribution. Predictive technologies such as machine learning can support this process by learning patterns from appropriately prepared operational data and generating data-driven completion estimates.

Although many studies discuss business automation, dashboards, and machine learning applications, limited research has focused on integrating these technologies into a single system specifically designed for micro service businesses such as shoe cleaning and restoration shops. Most available systems

target large enterprises or broader retail environments, which may be too costly, complex, or unsuitable for small organizations with limited technical resources.

To address these concerns, this study proposes the development of a Web-Based Service Management System with Data Analytics Using Machine Learning Algorithm for Shoelotskey Villamor-Pasay. The proposed system aims to centralize service transactions, inventory records, reporting processes, and performance monitoring while incorporating predictive analytics for service completion estimation. Through this system, Shoelotskey may improve operational efficiency, data accuracy, customer service responsiveness, and evidence-based decision-making.

Project Purpose and Description

This study aims to develop and evaluate a Web-Based Service Management System with Analytics and Machine Learning for Shoelotskey, a shoe cleaning and restoration service business located in Villamor, Pasay City. The study seeks to address operational issues such as manual record-keeping, fragmented files, delayed reporting, limited service-status monitoring, and inconsistent estimation of service completion dates.

The proposed system is intended to centralize customer records, job orders, inventory data, service transactions, and reporting processes through a web-based platform accessible to authorized users. It includes modules for order entry, inventory monitoring, service status tracking, sales reporting, descriptive analytics, predictive analytics using machine learning, and rule-based release-date computation.

These functions serve distinct purposes within the system. Descriptive analytics summarizes and presents recorded operational and business data through the real-time dashboard to support monitoring and reporting. Predictive analytics is implemented through a Random Forest machine learning model that predicts the Ready-for-Release Duration and derives the corresponding ML Predicted Release Date based on operational input features. Separately, the rule-based component calculates the Standard Release Date using predefined service-duration rules. The Standard Release Date is therefore a rule-based computation, while the ML Predicted Release Date is an output of predictive analytics.

The study will evaluate the developed system by comparing the existing manual paper-based process with the centralized digital system in terms of record retrieval and operational efficiency, evaluating the performance of the predictive analytics component using the machine learning model for Ready-for-Release Duration and corresponding release-date prediction, determining the impact of the descriptive analytics dashboard on reporting efficiency and access to business information, assessing software quality using ISO/IEC 25010, and measuring user acceptance using the Technology Acceptance Model (TAM).

System quality will be assessed using ISO/IEC 25010 standards, while user acceptance will be measured using the Technology Acceptance Model (TAM). The system is intended for internal use by Shoelotskey management and staff only.

This study may provide evidence on the practical application of web-based service management systems, descriptive analytics, predictive analytics using machine learning, and rule-based operational computations in small service-oriented businesses and may serve as a reference for future researchers developing similar systems for MSMEs.

Conceptual Framework

The conceptual framework of the study follows an Input–Process–Output (IPO) model with Evaluation and Feedback components. It illustrates how the required information is processed by the developed system to produce operational outputs, while system quality, user acceptance, machine learning performance, and operational improvements are evaluated separately.

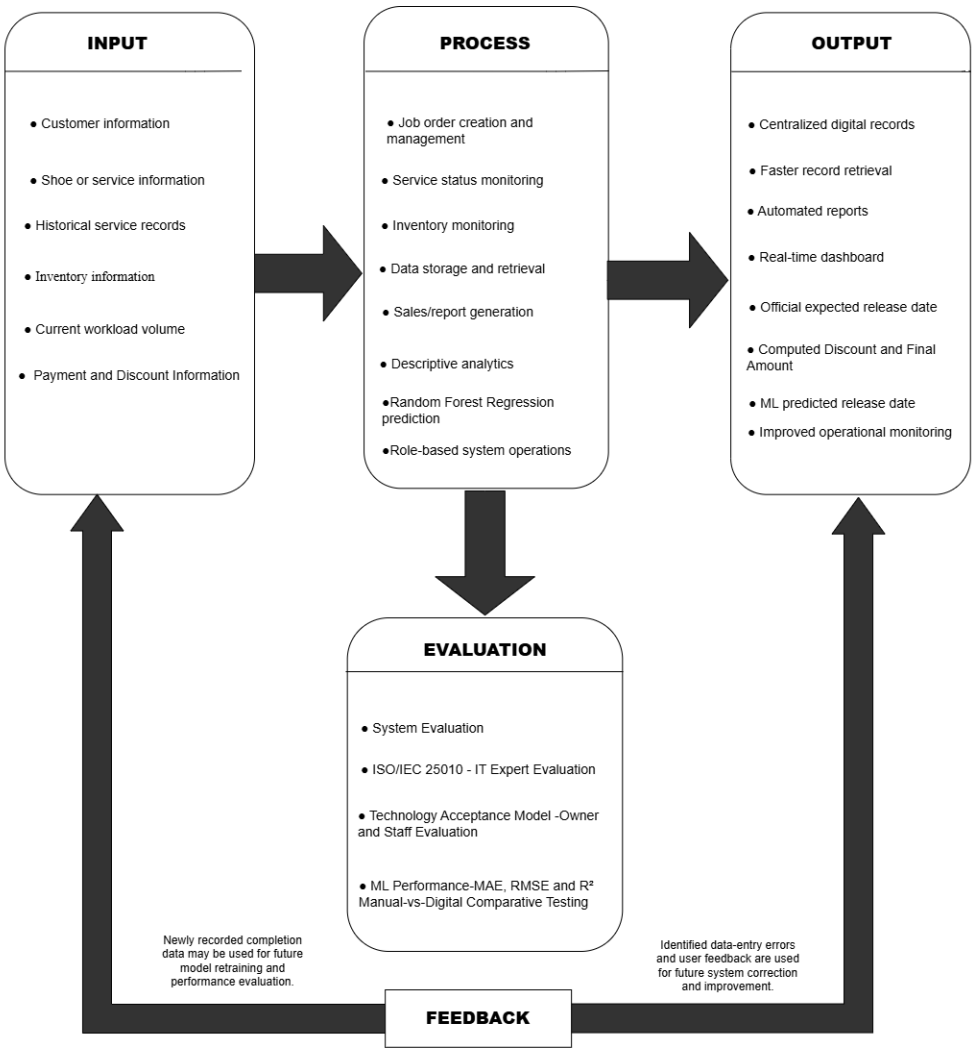


Figure 1: Conceptual Framework

The Input component consists of customer information, shoe or service information, historical service records, inventory information, current workload volume, and payment and discount information. These data serve as the information required by the system for managing service transactions and supporting its operational and analytical functions.

The Process component represents the actual functions performed by the developed system. These include job order creation and management, service status monitoring, inventory monitoring, data storage and retrieval, sales and report generation, descriptive analytics, Random Forest regression for predictive analytics, and rule-based system operations. Descriptive analytics summarizes recorded business and operational information through the dashboard, while the Random Forest model predicts the Ready-for-Release Duration, from which the ML Predicted Release Date is derived. In contrast, rule-based operations use predefined business and service-duration rules to determine outputs such as the Official Expected Release Date. Thus, descriptive analytics, predictive analytics, and rule-based computations are treated as distinct system functions.

The Output component represents the results produced through the system, including centralized digital records, faster record retrieval, automated reports, a real-time dashboard, the Official Expected Release Date, computed discounts and final amounts, the ML Predicted Release Date, and improved operational monitoring.

The Evaluation component is presented separately from the operational processes because the evaluation frameworks are applied to assess the developed system rather than function as system operations. ISO/IEC 25010 is used for evaluation by IT experts, while the Technology Acceptance Model (TAM) is used to assess acceptance among the owner and staff. The machine learning

component is evaluated using MAE, RMSE, and R^2, while manual-versus-digital comparative testing is conducted to assess improvements in selected operational processes.

Finally, the Feedback component represents the continuous improvement of the system. Identified data-entry errors and user feedback may be used for future system corrections and enhancements, while newly recorded completion data may be used for future machine learning model retraining and performance evaluation. The feedback loop therefore allows information obtained from system use and evaluation to contribute to subsequent improvements.

Objectives of the Study

This capstone project aims to design and develop the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms in order to improve operational efficiency, data accuracy, service monitoring, and managerial decision-making.

Specific Objectives

Specifically, the project aims to achieve the following:

1. Conduct a comparative analysis between the existing manual paper-based system and the developed centralized digital job order entry and management system in terms of service record accessibility, retrieval time, and accuracy.
2. Evaluate the predictive performance of the Random Forest Regressor in predicting service turnaround duration and corresponding release dates using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), coefficient of determination (R^2), and prediction tolerance within ±1, ±2, ±3, and ±5 days.

3. Determine the impact of the real-time analytics dashboard on reporting efficiency, accessibility of business information, and decision-making by evaluating report-generation performance and the perceptions of the shop owner and staff regarding the usefulness of the analytics dashboard.
4. Assess the software quality of the developed system based on the IT experts' evaluation using the ISO/IEC 25010 software quality model in terms of:
    - 4.1 Functional suitability;
    - 4.2 Performance efficiency;
    - 4.3 Compatibility;
    - 4.4 Interaction Capability;
    - 4.5 Reliability;
    - 4.6 Security;
    - 4.7 Maintainability;
    - 4.8 Flexibility; and
    - 4.9 Safety;
5. Measure the level of user acceptance of the developed system among the staff and management of Shoelotskey using the Technology Acceptance Model (TAM), measuring perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use.
6. Recommend possible system enhancements by identifying improvements and additional features that may be implemented by future researchers to further enhance system functionality, data analytics capabilities, and user experience.

Scope and Limitations

This study focuses on the design, development, and implementation of a Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey, a shoe cleaning and restoration service business located in Villamor, Pasay City. The system is intended to automate and improve the establishment's internal operations through a centralized and web-based platform accessible to authorized users such as the owner and staff.

The scope of the system includes user authentication and role-based access control, allowing secure login for administrators and staff with designated permissions. It includes a dashboard module that displays real-time operational summaries such as sales, job orders, completed services, pending transactions, and other business performance indicators through charts and reports.

The system also covers a Job Order Management Module that allows digital recording of customer information, shoe details, requested services, and transaction records. A service management module is included for maintaining service categories, pricing, and service descriptions offered by the business.

For inventory operations, the system includes an inventory management module that tracks materials used in services, updates stock quantities, stores inventory history, and provides low-stock notifications. A sales and reporting module is also included to generate daily, weekly, monthly, quarterly, and annual reports related to sales, expenses, and operational performance.

To improve scheduling and customer service accuracy, the system includes a machine learning prediction module that estimates Ready-for-Release Duration and the corresponding ML Predicted Release Date using a Random Forest model

developed from a simulated operational dataset representing verified Shoelotskey service configurations and operational conditions. Historical service records were separately used to assess the limitations of the original Order-to-Claim prediction target.

The system further includes search and record retrieval functions for customers, job orders, transactions, and service history, as well as responsive web access, allowing the system to be used on desktop computers, laptops, tablets, and mobile devices through a browser.

However, the system is limited to the internal operations of Shoelotskey only and is customized according to its current workflow and business requirements. It does not include a customer-facing online booking portal, mobile application, or self-service customer account access. Email functionality is limited to administrative or system-related communications such as account or password reset messages. External integrations such as online payment gateways, accounting software, payroll systems, and third-party delivery services are not included in the present version. The system may record delivery or pickup information associated with an order, including relevant delivery details. However, integration with third-party delivery providers and external courier platforms is outside the scope of the study.

A major limitation of the predictive component concerns the definition of the historical target variable. The paper-based historical records did not contain timestamps identifying when technicians actually completed the service. Consequently, historical turnaround duration was calculated as Claimed Date − Date Received, which represents order-to-claim duration rather than actual service-completion duration. Customer claiming behavior may introduce additional

delay after the shoe has already been completed. In contrast, the developed digital system records the time at which an order is changed to For Release, allowing future records to measure service completion more directly. Therefore, the historical target may contain variability unrelated to actual service processing and may have contributed to the limited predictive performance of the current model.

Additionally, there are privacy and data handling limitations. The system requires careful management of customer personal data and maintains records of customer information and service history within its database. While the system is designed with secure access controls, data privacy measures are limited to user authentication and role-based access restrictions. The system does not include advanced encryption mechanisms for data at rest or comprehensive compliance with external data protection regulations. Therefore, data security relies primarily on institutional safeguards and proper user access management.

The effectiveness and reliability of study findings are also limited by the small sample size of respondents used during system testing and user acceptance testing phases. Due to the localized nature of the business and limited number of staff members and customers involved in the evaluation process, the generalizability of results may be constrained. Feedback and performance metrics obtained from this limited respondent pool may not fully represent scenarios in larger or more complex operational environments.

Multi-branch management, franchise operations, advanced enterprise analytics, and highly customized financial accounting features are beyond the scope of this study but may be considered for future enhancement.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents a thematic review of related literature that supports the development of a Web-Based Service Management System with Data Analytics using Machine Learning Algorithms. All studies cited in this review were published between 2022 and 2026 and were drawn from peer-reviewed journals, conference proceedings, and academic publications. The review of related literature is organized into five alphabetically arranged topics, each focusing on a key component of the study: Business Analytics and Sales Performance Dashboards, Centralized Digital Order Management and Data Accuracy, Machine Learning for Predicting Service Turnaround Duration and Corresponding Release Dates, Software Quality Evaluation Using ISO/IEC 25010 Standards, and the Technology Acceptance Model (TAM). Each section applies the Compare–Contrast–Connect approach by examining the similarities of previous studies, identifying their differences and limitations, and explaining how their findings contribute to and support the present research.

Business Analytics and Sales Performance Dashboards

Analytics dashboards and web-based analytical tools can support operational monitoring and data-informed decision-making. Naprawski [1] demonstrated that web analytics enable managers to monitor service usage patterns and pinpoint improvement areas using real user data. Balolot et al. [2] similarly found that visual charts make performance metrics accessible to non-

technical stakeholders, accelerating responses. These works converge on the value of intuitive presentation.

Yet important divergences emerge when moving beyond display to actionability. Dizon et al. [3] integrated analytics directly with scheduling algorithms, creating proactive workload management rather than retrospective reporting. In contrast, Madlberger and Yesaulov [4] provide a broader review of the business impacts of data analytics in the service sector, while Sharon and Suma [5] discuss the application of predictive analytics in IT service management. Together, these studies demonstrate how analytics can support data-driven service management, although their contexts differ from the resource-constrained environment of a small craft-service business. Dela Vega et al. [6], the most contextually relevant, showed that even a small Philippine print shop could derive meaningful insights from basic regression-based analytics without heavy infrastructure. Canay and Kocabicak [7] focused on high-traffic enterprise frameworks, raising questions about scalability downward to micro-businesses.

A key limitation across several studies is the assumption of abundant data and technical support conditions rarely met by small trade shops like Shoelotskey. This study therefore adopts the pragmatic, low-overhead approach exemplified by Dela Vega et al. [6], incorporating simple daily/weekly/quarterly sales and service trend charts. By grounding the dashboard in the shop's own modest transaction volume, the system adopts a relatively low-overhead approach while applying the data-driven decision-support principles discussed by Madlberger and Yesaulov [4] and Sharon and Suma [5], directly supporting Research Question 3.

Centralized Digital Order Management and Data Accuracy

Web-based information systems provide an alternative to fragmented or manually maintained records by centralizing data entry, storage, retrieval, and reporting. Zeva [8] discussed approaches to website-based information system development, while Szedmák et al. [9] examined digital transformation through a document management application. Agpalo et al. [10] incorporated attendance monitoring and analytics into an event-management system, whereas Banawa and De Jesus [12] combined service-related information with natural language processing and decision support. Kristiani and Marcel [11], in contrast, examined IT service management practices in small application-development firms. Although these studies differ substantially in domain and functionality, they collectively demonstrate how digital information systems can structure operational records and support information management.

However, these findings cannot be transferred directly to Shoelotskey. The reviewed systems were developed for different organizational environments, including public services, event management, and software-related operations, whereas Shoelotskey relies on a relatively small-scale service workflow involving customer records, shoe details, selected services, service status, payment, and release information. Kristiani and Marcel [11] provide a closer small-firm context, although their study focuses on application development firms rather than craft-service operations. This contextual difference indicates that the principles of centralized digital record management require adaptation to the specific workflow and data requirements of a shoe-service establishment.

The Shoelotskey order-entry module directly applies the core principles of Zeva [8], Agpalo et al. [10], and J. A. Banawa and M. J. T. De Jesus [12] while adapting them to low-volume, manual-trade realities. By replacing the physical logbook with a single digital repository for customer, shoe, and service details, the

system mitigates the very errors and delays highlighted in the literature. Alduaij et al. [14] further demonstrate the importance of information quality in evaluating information systems. In the present study, however, software quality is assessed separately using ISO/IEC 25010, as discussed in the software quality evaluation section, addressing Research Question 1.

Machine Learning for Predicting Service Turnaround Duration and Corresponding Release Dates

Accurate estimation of service turnaround duration is important for supporting operational planning and providing customers with more informed release-date estimates. Machine learning offers a data-driven approach by learning patterns from historical records that may influence completion or resolution time. Caturkusuma et al. [15] applied a Random Forest Regressor to IT incident-duration prediction and reported an R^2 of 0.98, demonstrating the potential of ensemble learning for duration-estimation tasks. Tai et al. [16] similarly evaluated multiple machine-learning techniques for predicting support-ticket resolution time and assessed their predictive performance using error-based metrics. Jevsejev and Bereiša [17] further explored the application of machine-learning methods to incident management. More recently, Mulyati et al. [29] compared Linear Regression, Random Forest, and LightGBM regressors for predicting incident resolution time in IT service management and found that Random Forest achieved the strongest performance on their dataset, with an R^2 of 0.8318 and an MAE of 60.67 hours. These studies demonstrate the applicability of supervised machine-learning methods to duration and resolution-time prediction.

However, the performance of a machine-learning algorithm varies according to the characteristics of the dataset and application context. Although Random Forest demonstrated strong performance in the studies of Caturkusuma et al. [15] and Mulyati et al. [29], this does not establish it as universally superior to other regression algorithms. Nakkliang et al. [30], for example, compared Random Forest, XGBoost, CatBoost, and LightGBM for incident-duration prediction and found that LightGBM achieved the best test performance in their dataset. This variation emphasizes the importance of empirically comparing candidate models rather than selecting an algorithm solely based on its reported performance in previous studies. Furthermore, Ahmed et al. [18] identified data quality and organizational factors as important considerations in the practical adoption of machine-learning-driven support systems. Therefore, differences in dataset size, feature availability, data quality, and operational context must be Guided by these studies, the present research considered multiple regression approaches during machine-learning development rather than assuming that Random Forest was inherently superior. The available validated historical service records were first used in Dataset A to assess the feasibility and limitations of predicting Order-to-Claim Turnaround Days. Because the historical records lacked a reliable Ready-for-Release timestamp, this target incorporated both service-processing duration and customer pickup behavior. The identified limitation informed the development of Dataset B, a separate simulated operational dataset using Received-to-Ready-for-Release Duration as the prediction target. Candidate models were subsequently compared under the operational Dataset B framework, and the finalized model was selected based on comparative MAE, RMSE, R^2, and prediction-tolerance performance.

Software Quality Evaluation Using ISO/IEC 25010 Standards

ISO/IEC 25010 provides a standardized, multidimensional framework for evaluating software quality. Rojas et al. [13] documented its application across domains through a bibliometric review of over 1,600 studies, while Azhir et al. [21] demonstrate a more specific data-driven application of ISO/IEC 25010 for predicting software maintainability. In contrast, Alduaij et al. [14] employed the DeLone and McLean Information Systems Success Model, illustrating that information-system evaluation may be approached through frameworks other than ISO/IEC 25010. Other studies [7], [22]–[24] focus more broadly on analytics, artificial intelligence, data processing, and IT service management and therefore provide supporting technical context rather than direct ISO/IEC 25010-based evaluation evidence. Kristiani and Marcel [11], meanwhile, demonstrate the use of a structured IT service management approach within a small-firm setting. Collectively, these studies highlight different perspectives on system quality and evaluation, but they vary considerably in framework, application domain, and evaluation focus.

While ISO/IEC 25010 provides a comprehensive framework for examining software quality, the reviewed literature also reveals contextual and methodological limitations. The reviewed applications span different organizational and technical settings, making their findings not directly transferable to a resource-limited Philippine craft-service business. Few studies combine ISO/IEC 25010-based evaluation with user-centric assessment in such a setting, and even fewer examine it alongside machine-learning predictive performance and everyday system use. Moreover, when ISO/IEC 25010 characteristics are assessed through survey questionnaires, the resulting ratings represent the evaluators' perceptions of software quality and should not be interpreted as objective measurements of technical performance unless supported by corresponding technical tests. This distinction is particularly important in the present study because expert

questionnaire ratings and machine-learning performance metrics represent different forms of evidence and should not be interpreted interchangeably.

In this study, ISO/IEC 25010 is therefore used to assess IT experts' perceptions of the developed system's software quality characteristics, while the machine-learning model's predictive performance and user acceptance are evaluated separately. Predictive performance is assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2), whereas user acceptance is assessed using the Technology Acceptance Model (TAM). These evaluation approaches provide distinct but complementary evidence concerning perceived software quality, predictive performance, and user acceptance, thereby addressing Research Question 4.

Technology Acceptance Model (TAM) for User Acceptance of Web-Based Service Systems

TAM remains a widely used framework for explaining technology adoption through constructs such as Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). The reviewed studies differ in both their evaluation frameworks and application settings. Usita et al. [25] specifically applied TAM to evaluate the acceptance of a web-based inventory system for food-sector MSMEs, making it particularly relevant to the present study's use of TAM in a small-business environment. Estepa et al. [28], meanwhile, examined the technological acceptability and perceived effectiveness of an inventory management application in selected catering services. Adyaputra et al. [27] combined ISO 25010 and TAM in evaluating local government websites, demonstrating how software-quality and user-acceptance perspectives can be examined within the same study while remaining conceptually distinct. These contexts differ from Shoelotskey's craft-

service operations, but they support the relevance of examining user perceptions when introducing digital systems into existing organizational workflows.

Limitations persist because the reviewed evaluations were conducted in government, food, inventory, and other contexts that differ from craft-service trades, while relatively few examine systems that also incorporate machine learning prediction and analytics. Short post-implementation evaluations may also capture users' perceptions and acceptance at a particular period of system use rather than demonstrating sustained long-term adoption. Furthermore, TAM evaluates user acceptance and should not be interpreted as independent evidence of technical performance, system security, reliability, or machine learning predictive accuracy.

Therefore, the Shoelotskey system employs ISO/IEC 25010 and TAM as distinct but complementary evaluation frameworks. ISO/IEC 25010-based survey ratings are used to assess IT experts' perceptions of the system's software quality characteristics, while TAM is used to assess the shop owner's and staff's acceptance of the developed system in terms of Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Behavioral Intention (BI), and Actual System Use (ASU). Technical claims regarding the machine learning component are evaluated separately using model performance metrics rather than inferred from either ISO/IEC 25010 or TAM questionnaire ratings. This distinction allows the study to examine software quality perceptions and user acceptance without treating questionnaire responses as objective technical measurements, thereby addressing Research Question 5.

Synthesis of the Reviewed Literature

Collectively, the five themes establish the relevance and feasibility of developing an integrated web-based service management system for small Philippine service enterprises. Analytics dashboards and data-driven systems [1–7], centralized digital records [8–12, 14], ML-based prediction [15–20, 29, 30], software quality and related evaluation approaches [11, 13, 14, 21–24], and TAM-based user acceptance evaluation [25–28] provide relevant foundations for the present study. However, a critical review reveals persistent contextual gaps. Several reviewed studies were conducted in enterprise, government, educational, IT-service, or other organizational environments whose datasets, technical resources, and workflows differ from those of a small craft-service establishment. Philippine shoe-service businesses remain underrepresented in the reviewed literature, while studies integrating service management, machine learning prediction, analytics, software-quality assessment, and user-acceptance evaluation within a single small-business setting remain limited.

Prior work also reveals important limitations. Enterprise-oriented approaches such as [7] may introduce complexity that is difficult to apply in resource-constrained micro-businesses, while predictive approaches developed in other operational contexts [16, 19] may not directly transfer to the data and workflow characteristics of a small craft-service business. Studies on technology acceptance report findings concerning perceived usefulness, ease of use, behavioral intention, and related measures of user acceptance, but short evaluation periods may not adequately reflect sustained adoption under actual operational conditions. Moreover, ISO/IEC 25010 questionnaire ratings represent evaluators' perceptions of software quality unless supported by objective technical tests, while TAM measures user acceptance rather than technical performance, security, reliability, or predictive accuracy.

By designing a unified platform in which digitized service records support machine learning prediction and analytics, the present study addresses the identified gap through its application in a small shoe-service business. The study employs distinct but complementary forms of evaluation: ISO/IEC 25010-based questionnaires assess IT experts' perceptions of software quality characteristics, TAM assesses the shop owner's and staff's acceptance of the developed system, and model evaluation metrics independently assess the predictive performance of the machine learning component.

The machine-learning component further recognizes the distinction between the available historical records and the operational prediction objective of the developed system. Dataset A, consisting of validated historical service records, was used for the empirical historical assessment in which Order-to-Claim Turnaround Days served as the target because the available paper records contained Date Received and Claimed Date but lacked a reliable actual Ready-for-Release timestamp. This historical target therefore incorporated both service-processing duration and possible customer pickup delay. The limitation identified through Dataset A informed the subsequent operational machine-learning development using a separate Dataset B, in which Received-to-Ready-for-Release duration was used as the prediction target. This distinction aligns the finalized operational model with the service-completion event that the developed system is intended to estimate while retaining Dataset A as evidence of the limitations of the available historical data.

The findings remain context-specific and should not be interpreted as establishing a universally generalizable model for Philippine MSMEs. The study is limited to one establishment with six intended users, and the historical machine-learning assessment was constrained by the availability and quality of the original

paper-based records and by the absence of a reliable historical Ready-for-Release timestamp. Furthermore, the finalized operational machine-learning model was developed and evaluated using a simulated operational dataset based on the verified service configurations and operational conditions of Shoelotskey; therefore, its predictive performance should not be interpreted as equivalent to validation on a large independent dataset of actual Ready-for-Release transactions. Consequently, the study provides context-specific evidence regarding the opportunities and limitations of integrating service management, analytics, and machine-learning prediction within a resource-constrained small-business environment. Future studies may validate and extend the approach using larger collections of actual operational records containing reliable Ready-for-Release timestamps, longer implementation periods, more users, additional establishments, and continued evaluation and retraining of the predictive model.

CHAPTER 3

TECHNICAL BACKGROUND

Web-based information systems are widely used by service-oriented businesses to manage daily operations such as job orders, customer transactions, service records, scheduling, and report generation. Compared with manual logbooks or spreadsheet-based methods, web-based systems provide faster data access, centralized storage, real-time updates, and improved accuracy. In this study, a Web-Based Service Management System with Analytics and Machine Learning was developed for Shoelotskey to improve service transaction management, monitor service progress, generate operational reports, provide descriptive analytics, and support machine learning-based prediction of service completion.

The proposed system follows a three-tier architecture, consisting of the presentation layer, application layer, and data layer. This architecture was selected because it separates the user interface, business logic, and database operations into organized components, making the system easier to maintain, troubleshoot, and expand. It also supports scalability, allowing additional modules or users to be accommodated in the future without major system redesign.

The presentation layer is the front-end component accessed by users through standard web browsers. It was developed using React with TypeScript to support structured, reusable, and maintainable code. Vite was utilized as the build tool to provide efficient development and optimized application loading. Tailwind CSS was implemented to create a responsive and user-friendly interface across desktop and mobile devices, while Lucide React was integrated to provide

consistent icons throughout the system. Through this layer, authorized users can record transactions, create and manage job orders, update service statuses, search records, monitor inventory, view analytics, and generate reports.

The application layer contains the core system logic and server-side processing. It was developed using Python with the FastAPI framework, which supports RESTful API development, asynchronous processing, and integration with database and analytical components. This layer handles user authentication, input validation, business rules, report generation, system permissions, and communication between the front-end interface and the database. It also integrates the rule-based computation of the Official Expected Release Date separately from the machine learning component responsible for generating the ML Predicted Release Date.

The data layer uses PostgreSQL, an open-source relational database management system for storing and managing the operational data of the system. PostgreSQL was selected because of its reliability, scalability, querying capabilities, and security controls. SQLAlchemy ORM was used to facilitate interaction between the application and database and improve the maintainability of database operations. The database contains structured records related to users, customers, job orders, services, transactions, inventory, predictions, and audit logs. Relational constraints and organized data structures are used to maintain data consistency and support efficient storage and retrieval.

Different user roles and permissions were implemented to control access to system functions. The Owner has access to administrative and management functions, including dashboards, user management, reports, service monitoring, and other authorized business information. Staff access is limited according to assigned permissions for operational functions such as job order management,

payment processing, service status updates, discounts, and inventory activities. This role- and permission-based access control helps prevent unauthorized access or modification of system information.

The system incorporates descriptive analytics, predictive analytics, and rule-based release-date computation as distinct functions. Descriptive analytics summarizes recorded operational and business information through the dashboard to support monitoring and reporting. Predictive analytics is implemented through a Random Forest Regressor to estimate the Ready-for-Release Duration and derive the corresponding ML Predicted Release Date. In contrast, the rule-based component determines the Official Expected Release Date using Shoelotskey's predefined service-duration rules. Therefore, the rule-based release date and machine learning prediction are presented as separate outputs and are not treated as interchangeable forms of analytics.

The machine learning component was developed using the Scikit-Learn library in Python, with a Random Forest Regressor used as the finalized prediction algorithm. The finalized prediction target is Received-to-Ready-for-Release Duration, representing the period from the date an order is received until the service is ready for release. The predicted duration is used to derive the corresponding ML Predicted Release Date. The Random Forest algorithm was selected for development because it can model nonlinear relationships and interactions among structured operational features, while its suitability for the prediction task was determined through model evaluation.

The development of the machine learning component involved two distinct datasets with different research purposes. Dataset A consisted of 119 validated historical orders obtained from 714 digitized paper receipts covering August 2025 to January 2026. These historical records contained verified Date Received and

Claimed Date values and were assessed using Order-to-Claim Turnaround as the historical target. However, the available paper receipts did not contain a reliable actual Ready-for-Release timestamp. Consequently, the historical target included both the shop's service-processing period and the time associated with customer pickup behavior, meaning that it did not directly represent the operational service-completion duration that the system was intended to predict.

The historical assessment was retained as part of the research process and provided evidence of the limitation of the available historical target. Based on this limitation, the prediction target for the finalized operational model was refined from Order-to-Claim Turnaround to Received-to-Ready-for-Release Duration, which more directly represents the service-completion process. The 119 historical records were therefore retained for the historical assessment rather than being treated as the training dataset of the finalized active model.

Following the identification of this limitation, Dataset B, a separate 10,000-record Simulated Operational Model Development and Evaluation Dataset, was developed to represent the operational conditions required for the finalized prediction task. Unlike Dataset A, Dataset B uses Received-to-Ready-for-Release Duration as the prediction target, thereby separating operational service completion from customer pickup behavior. The simulated dataset contains operational variables relevant to service-completion prediction and was used specifically for the development and evaluation of the finalized Random Forest model.

The 10,000-record simulated operational dataset was divided into 7,000 training records, 1,500 validation records, and 1,500 final test records. The training set was used to fit the Random Forest model, the validation set supported model development and selection, and the final test set was reserved for evaluating the

finalized model on unseen simulated operational records. Predictive performance was assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). Detailed procedures and results of the model development and evaluation are presented in the succeeding sections of the study.

Dataset A and Dataset B therefore serve separate but related purposes in the research. Dataset A documents the assessment and identified limitation of using the available historical Order-to-Claim target, while Dataset B supports the development and evaluation of the finalized Received-to-Ready-for-Release prediction model. This distinction preserves the historical assessment while ensuring that the active machine learning component corresponds to the operational outcome that the system is intended to predict.

During system operation, the finalized Random Forest model generates a Predicted Duration, which is used to derive the ML Predicted Release Date. This predictive output serves as an advisory machine learning estimate and remains separate from the Official Expected Release Date, which is generated through Shoelotskey's predefined service-duration rules. Newly recorded actual completion data may be retained for future model retraining and performance evaluation as additional Ready-for-Release observations become available.

To ensure system security, user passwords are protected using the Bcrypt hashing algorithm, while authentication is implemented using JSON Web Tokens (JWT). Additional safeguards include input validation, role- and permission-based access controls, account lockout mechanisms, inactivity controls, audit logging, and database backup procedures. These measures help protect system records and reduce risks related to unauthorized access, modification, data loss, and misuse.

By integrating the React and TypeScript front-end, FastAPI and Python backend, PostgreSQL database, descriptive analytics dashboard, rule-based service-duration computation, and Random Forest predictive model, the proposed system provides a centralized platform for Shoelotskey's service operations. Descriptive analytics supports the monitoring and summarization of recorded business information, the rule-based component determines the Official Expected Release Date based on predefined service-duration rules, and predictive analytics estimates the Ready-for-Release Duration and corresponding ML Predicted Release Date. The selected technologies and system architecture support the maintainability, scalability, performance, and future expansion of the developed system.

Research and Development Methodology

This study adopted the Developmental Research Method combined with the Waterfall Software Development Life Cycle (SDLC) model. Developmental research is appropriate because the study focuses on designing, developing, and evaluating a web-based service management system integrated with data analytics and machine learning algorithms.

The Waterfall model was selected to ensure a structured, systematic, and well-documented development process suitable for this capstone project. It follows a linear and sequential approach where each phase, Requirements Analysis, System Design, Implementation, Testing, Deployment, and Maintenance, is completed and formally documented before proceeding to the next, ensuring that requirements are clearly defined and approved early in the project.

Software Development Life Cycle Model

The development of the Web-Based Service Management System with Data Analytics using Machine Learning Algorithms adopted the Waterfall Software Development Life Cycle (SDLC) model. This model was selected because it follows a systematic and phase-by-phase process that is appropriate for academic and capstone projects requiring clear documentation, defined outputs, and controlled implementation. Each development stage must be completed before proceeding to the next, allowing better requirement tracking, organized project management, and formal validation of deliverables.

The business operations of Shoelotskey, including job order processing, scheduling, service monitoring, and performance tracking, follow predictable and well-defined workflows. Since the system requirements were identified during the initial planning stage and were not expected to change significantly, the Waterfall model was considered more suitable than iterative approaches such as Agile, which are more effective for projects with rapidly changing requirements. For this study, a structured development process was more appropriate than frequent redesign cycles.

The model was also suitable because the system integrates data analytics and machine learning components that require organized preparation of datasets, database structures, and processing logic before implementation. Through the Waterfall approach, these technical foundations were properly designed during the earlier phases, reducing development errors and improving system consistency.

In terms of software quality assurance, the Waterfall model supports systematic testing and evaluation in every major phase of development. This strengthens compliance with ISO/IEC 25010 standards by enabling structured

assessment of functionality, reliability, usability, maintainability, and performance efficiency.

Although the Waterfall model has limited flexibility when changes arise during later stages, this limitation was minimized through early requirement gathering, stakeholder interviews, and detailed planning conducted before development began. These measures helped ensure that the identified system requirements remained stable throughout the project lifecycle.

Figure 2 illustrates the Waterfall development workflow adopted in this study.

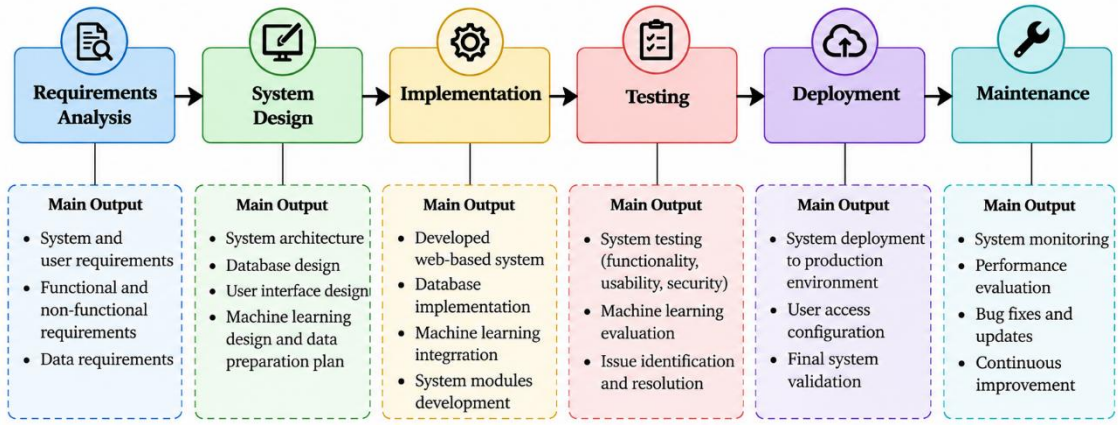


Figure 2: Waterfall Development Process

The figure presents the sequential progression of the study from Requirements Analysis, System Design, Implementation, Testing, Deployment, to Maintenance. Each phase generated specific outputs that were reviewed or validated before they were used in the succeeding phase.

Waterfall Development Phases

Requirements Analysis

The requirements analysis and planning phase focused on understanding the existing operational procedures of Shoelotskey and determining the functional and non-functional requirements of the proposed system. The researchers conducted interviews, discussions, and consultations with the shop owner and staff to document the existing procedures for customer intake, job order processing, service tracking, inventory management, payment and claim processing, and reporting.

The identified requirements included centralized Job Order Management, Service Management, Release Calendar, Inventory Management, Sales Reporting, User Management, Analytics Dashboard, role-based access control, audit trails, and machine learning-based service completion prediction. Non-functional requirements concerning security, usability, reliability, performance, maintainability, and other applicable software quality characteristics were also identified.

For the machine learning component, historical transaction records were collected and examined to determine whether the available data could support predictive modeling. A total of 714 paper receipts covering August 2025 to January 2026 were digitized. After screening and validation, 119 historical orders containing verified Date Received and Claimed Date information were retained for the historical machine learning assessment.

The primary deliverable of this phase was the System Requirements Specification (SRS) containing the functional and non-functional requirements of the proposed system. These requirements were reviewed to ensure that the

proposed functions reflected the operational needs of Shoelotskey before proceeding to system design.

System Design

During the system and architectural design phase, the documented requirements were translated into the technical, process, database, security, machine learning, and user interface designs of the proposed system.

The researchers developed the Context Diagram, Data Flow Diagram (DFD) Level 0, Data Flow Diagram (DFD) Level 1, procedural flowcharts, use case models, Entity Relationship Diagram (ERD), and Logical Database Design. These models established the interaction between users and the system, the flow of information among processes, and the relationships among the system's data entities.

The overall architecture was organized into the presentation, application, and data layers. User interfaces and prototypes were also prepared for the Dashboard, Job Order Management, Service Management, Release Calendar, Inventory Management, Sales Reporting, User Management, and Analytics Dashboard.

The machine learning architecture was designed during this phase by identifying the required job-order features, prediction workflow, model integration, and release-date logic. The system was designed to distinguish the operational release-date mechanism from the machine learning-generated prediction, allowing the ML output to function as prediction date estimate.

The primary deliverables of this phase were the system architecture, process models, database design, user interface designs, security design, and machine learning architecture, which served as the technical basis for implementation.

Implementation (Coding)

During the coding and implementation phase, the approved designs were transformed into a functional web-based application.

The presentation layer was developed using React, TypeScript, Vite, and Tailwind CSS. The application layer utilized FastAPI, Uvicorn, Python, and SQLAlchemy ORM, while PostgreSQL was used for persistent data storage.

The major system modules implemented included the Dashboard, Job Order Management, Service Management, Release Calendar, Inventory Management, Sales Reporting, User Management, and Analytics Dashboard. Security mechanisms such as JWT Bearer Token authentication, Bcrypt password hashing, role-based access control, account lockout, inactivity/session controls, permissions, and audit logging were also incorporated.

The machine learning component was developed using Scikit-learn, Pandas, NumPy, and the Random Forest Regressor. The historical assessment initially used the 119 validated historical records and Order-to-Claim Turnaround as the available target. The assessment revealed a limitation in the historical data because the paper receipts recorded Date Received and Claimed Date but did not provide a reliable actual Ready-for-Release timestamp.

Consequently, a separate 10,000-record Simulated Operational Dataset was developed for the finalized operational machine learning model using Received-to-Ready-for-Release as the prediction target. The dataset was divided into 7,000 training records, 1,500 validation records, and 1,500 final test records. The finalized Random Forest model was then integrated into the system to generate advisory service completion predictions.

The primary deliverable of this phase was a functional integrated prototype containing the required system modules, database functions, security controls, analytics features, and machine learning component.

Testing

The system integration and verification testing phase determined whether the implemented system satisfied its documented functional and non-functional requirements.

Use case and functional testing were conducted on the major system processes, including authentication, creation and updating of job orders, service processing, record retrieval, inventory transactions, payment processing, report generation, dashboard access, user permissions, and access restrictions. Integration testing was performed to verify that interconnected modules correctly exchanged and processed information.

Security controls were also verified through dedicated security test cases covering authentication, authorization, role-based restrictions, account lockout, session handling, input validation, protected routes, and unauthorized access.

Operational performance was evaluated by comparing selected manual and digital procedures, including record retrieval and information processing. The

results were documented as part of the quantitative evaluation of the developed system.

The machine learning component was evaluated separately using appropriate regression performance measures, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2). The 119-record historical dataset was used to assess the limitations of the available historical Order-to-Claim target, while the separate simulated operational dataset was used for development, validation, and final testing of the active Random Forest model.

The primary deliverables of this phase were the accomplished functional, integration, security, and machine learning test results; documented system issues; implemented corrections; and verified system version prepared for deployment.

Deployment

Following system verification and the implementation of required corrections, the application was deployed and configured for operational use at Shoelotskey Villamor–Pasay. The deployment environment, PostgreSQL database connection, authorized user accounts, permissions, and required system configurations were established.

The owner and staff were oriented on the system's major functions, including job order processing, service status monitoring, release scheduling, inventory operations, report access, and other functions relevant to their assigned roles. The deployed system was used during the pilot-testing period from September 2 to September 16, 2026. During this period, the intended users interacted with the system through actual and simulated operational activities,

which enabled the researchers to identify system issues and gather feedback based on the users' hands-on experience with the developed system. The pilot-testing period was intended to provide short-term operational exposure and system evaluation and was not treated as evidence of sustained long-term organizational adoption.

Following pilot testing, the formal system evaluation was conducted. Five (5) IT experts evaluated the system using an instrument based on the applicable ISO/IEC 25010 software quality characteristics. The user evaluation involved six (6) respondents consisting of the shop owner, co-owner, and staff, who evaluated the system using the Technology Acceptance Model (TAM)-based instrument. The questionnaire administration and evaluation activities were conducted from September 17 to September 19, 2026.

The quantitative responses were summarized using the statistical procedures specified in the study, including the mean and standard deviation. Evaluator comments and recommendations were also documented and used as the basis for applicable system revisions.

The primary deliverables of this phase were the deployed system, pilot-testing records, accomplished evaluation instruments, statistical evaluation results, and documented evaluator recommendations.

Maintenance

The operations and maintenance phase began after deployment and evaluation. This phase focused on addressing issues, corrections, and recommendations identified during pilot testing and formal system evaluation.

The researchers reviewed reported errors and evaluator feedback and implemented applicable bug fixes, interface refinements, security improvements, permission controls, inventory controls, audit-trail improvements, and other functional revisions. Recommendations that could not be implemented within the scope and duration of the study were documented as proposed enhancements for future versions of the system.

Maintenance provisions were also established for system administration, database backup, audit-log monitoring, security maintenance, and future machine learning model evaluation and retraining. As additional actual operational records containing reliable Ready-for-Release information become available, these records may be used for future retraining and evaluation of the prediction model.

The primary deliverables of this phase were the documented system revisions, maintenance procedures, corrected system functions, and final version of the developed system incorporating the applicable recommendations obtained during testing and evaluation.

Machine-Learning Development

Machine learning was integrated into the developed Web-Based Service Management System with Data Analytics to provide an advisory prediction of the expected service completion duration and corresponding Ready-for-Release Date. The machine-learning development process involved historical data preparation and validation, feature engineering, model development, baseline comparison, model evaluation, and subsequent refinement of the prediction target.

Initially, 714 historical paper transaction records were digitized and examined. The historical records were subjected to preprocessing and manual validation to determine whether the information required for machine-learning

development could be reliably obtained. Particular attention was given to the Date Received and Claimed Date because these fields were required to calculate the historical Order-to-Claim turnaround duration.

Table 1

PREPROCESSING AND VALIDATION OF HISTORICAL RECORDS

Preprocessing Stage / Reason	No. of Records	Treatment	Records Remaining
Initial digitized historical records	714	Initial dataset subjected to preprocessing	714
Unreadable or unverifiable dates	576	Excluded because the turnaround target could not be reliably calculated	138
Incomplete records	18	Excluded because required information was incomplete	120
Candidate records for manual validation	120	Individually reviewed for consistency	120
Validated without correction	35	Retained	120
Inconsistent entries that could be verified	84	Corrected through manual validation and retained	120
Record that could not be reliably validated	1	Excluded during final validation	119
Final validated historical dataset	119	Retained as Dataset A	119

As shown in Table 1, 595 of the original 714 digitized records were excluded from the final historical dataset. Of these, 576 records contained unreadable or unverifiable dates, 18 contained incomplete information required for the analysis, and one additional record was rejected during final validation. The 84 inconsistent entries were not counted as excluded records because their inconsistencies could be resolved through manual verification and the corrected records were retained. This resulted in a final Dataset A consisting of 119 validated historical orders.

No statistical imputation was performed for an unreadable or unavailable target date. Records without sufficient information to reliably calculate the Order-to-Claim turnaround were excluded instead of estimating or artificially generating the missing dependent variable. This procedure was applied to preserve the integrity of the historical target.

Table 2

DATA PREPROCESSING AND FEATURE PREPARATION PROCEDURES

Preprocessing Component	Procedure Applied
Missing or unreadable target dates	Records were excluded when the dates required to calculate turnaround could not be reliably verified.
Incomplete records	Records lacking information required for model development were excluded.
Inconsistent entries	Entries that could be resolved from the available source information were manually corrected and retained; one unverifiable record was rejected.
Imputation	No statistical imputation was performed for the target variable.
Numerical variables	Quantity and count variables were represented as numerical values.
Service information	Individual service quantities/counts were represented as model input features rather than raw service-name text.
Rush priority	Represented as a binary indicator distinguishing regular and rush orders.
Intake date	The raw intake date was transformed into calendar-related predictors such as day of the week and month.

Preprocessing Component	Procedure Applied
Historical target	Calculated as the number of days between Date Received and Claimed Date.
Data partitioning	Historical records were chronologically separated to prevent future records from being used to predict earlier observations.

The 119 validated historical records were then chronologically divided into 93 development records and 26 final temporal holdout records. The development records were used for model development and expanding-window cross-validation, while the 26 most recent historical records were reserved as an unseen final test set. Chronological separation was used instead of random splitting to preserve the temporal order of the transaction records and reduce the possibility of future information influencing earlier predictions.

The historical model used 13 input features representing order size, service composition, priority, and intake-calendar characteristics. These included total pairs, distinct service types, total service applications, rush priority, intake day of the week, intake month, and service-specific quantities for the services represented in the historical dataset. The target variable for Dataset A was Order-to-Claim Turnaround Days, computed from the difference between Claimed Date and Date Received. To determine whether machine learning provided an improvement over simpler prediction approaches, the Random Forest model was compared with the mean-duration baseline, median-duration baseline, Shoelotskey's existing business-rule estimates, Linear Regression, and Decision Tree. All methods were evaluated using the same 26-record historical temporal holdout set.

Table 3

BASELINE AND MACHINE-LEARNING MODEL COMPARISON ON HISTORICAL DATASET A

Model / Benchmark	MAE (Days)	RMSE (Days)	R^2	±1 Day	±2 Days	±3 Days	±5 Days
Mean-Duration Baseline	9.34	10.13	−0.0006	3.9%	3.9%	7.7%	11.5%
Median-Duration Baseline	8.96	10.36	−0.0469	7.7%	7.7%	15.4%	34.6%
Shoelotskey Business Rules	**6.62**	**8.45**	**0.3033**	**23.1%**	**30.8%**	**38.5%**	**42.3%**
Linear Regression	18.50	20.51	−3.1065	0.0%	0.0%	0.0%	7.7%
Decision Tree Regressor	10.23	13.29	−0.7237	19.2%	23.1%	26.9%	42.3%
Random Forest Regressor	8.35	10.19	−0.0125	11.5%	23.1%	30.8%	42.3%

The historical evaluation showed that the Random Forest Regressor achieved an MAE of 8.35 days and RMSE of 10.19 days. It performed better in MAE than the mean-duration baseline of 9.34 days and median-duration baseline of 8.96 days. However, it did not outperform Shoelotskey's existing business-rule estimates, which achieved an MAE of 6.62 days and RMSE of 8.45 days. The Random Forest obtained an R^2 of −0.0125 on the historical holdout set. This value was not interpreted as “negative accuracy”; rather, it indicated that the model did not explain the variation in the historical Order-to-Claim target better than a simple reference prediction on the unseen records. These historical results

Further examination of Dataset A identified an important limitation in the historical target. The available paper records contained Date Received and Claimed Date but did not provide a reliable actual Ready-for-Release timestamp indicating when the shop had completed the service. Consequently, the historical target represented Order-to-Claim Turnaround, which included not only service-processing time but also the period between service completion and the customer's eventual collection of the item. Therefore, the model was partially attempting to predict customer pickup behavior, which was outside the operational control of Shoelotskey.

For this reason, Dataset A was retained as an empirical historical assessment rather than discarded. Its results demonstrated the limitation of using Claimed Date as a proxy for actual service completion. The subsequent operational machine-learning development therefore changed the prediction target from Order-to-Claim to Order-to-Ready-for-Release, which corresponds directly to the operational event that the system is intended to estimate and baseline values are documented in the machine-learning evidence report.

Operational Machine-Learning Dataset

To develop the finalized operational model, a separate Dataset B containing 10,000 simulated service orders was generated based on the verified operational rules and service configurations of Shoelotskey. Dataset B was designed specifically around the Received-to-Ready-for-Release target rather than customer claiming behavior.

The 10,000 records were separated into three independent partitions: 7,000 records (70%) for model training, 1,500 records (15%) for model validation and selection, and 1,500 records (15%) as a sealed final test set. The final test partition remained separate from model training and model selection and was used only

after the selected model had been finalized. This partitioning is documented in the machine-learning methodology evidence.

Table 4

DATASET B PARTITIONING

Dataset Partition	No. of Records	Percentage	Purpose
Training Set	7,000	70%	Model training and parameter development
Validation Set	1,500	15%	Candidate-model evaluation and model selection
Final Test Set	1,500	15%	Final evaluation of the selected and frozen model
Total	**10,000**	**100%**	**Operational machine-learning dataset**

For Dataset B, the finalized Feature Set C consisted of 12 operational predictors available at the time a job order is received. These were total pairs, Basic Cleaning quantity, Full Reglue quantity, Minor Reglue quantity, Unyellowing quantity, Minor Restoration quantity, Color Renewal quantity, rush indicator, number of distinct service types, total service applications, intake day of the week, and intake month. The feature set intentionally excluded variables that directly revealed the target or internal target-generating rules to reduce direct target leakage.

Table 5

FINAL OPERATIONAL INPUT FEATURES – DATASET B

No.	Input Feature	Description
1	Total Pairs	Number of shoe pairs included in the job order
2	Basic Cleaning Quantity	Number of Basic Cleaning service applications
3	Full Reglue Quantity	Number of Full Reglue service applications
4	Minor Reglue Quantity	Number of Minor Reglue service applications

No.	Input Feature	Description
5	Unyellowing Quantity	Number of Unyellowing service applications
6	Minor Restoration Quantity	Number of Minor Restoration service applications
7	Color Renewal Quantity	Number of Color Renewal service applications
8	Rush Indicator	Indicates whether the job order is regular or rush
9	Distinct Service Types	Number of different service types included in the order
10	Total Service Applications	Total number of service applications across the order
11	Intake Day of Week	Day of the week when the order was received
12	Intake Month	Month when the order was received

Three regression algorithms were evaluated for the operational dataset: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. Model selection was performed using the separate 1,500-record validation set after training on the 7,000-record training partition.

Table 6

CANDIDATE MODEL VALIDATION – DATASET B

Candidate Model	Validation MAE	Validation RMSE	Validation R^2	±1-Day Accuracy	Selection
Linear Regression	2.4764 days	3.1062 days	0.8619	20.33%	Not selected
Decision Tree Regressor (depth = 8)	**0.0616 days**	0.3093 days	0.9986	97.53%	Not selected
Random Forest Regressor (300 trees, depth = 12)	0.0641 days	**0.2994 days**	**0.9987**	**97.73%**	**Selected**

The Decision Tree produced the lowest validation MAE at 0.0616 days, compared with 0.0641 days for Random Forest. Therefore, Random Forest was

not selected on the basis of MAE alone. Random Forest produced the lower RMSE of 0.2994 days compared with 0.3093 days for the Decision Tree, the higher R^2 of 0.9987, and the higher ±1-day tolerance accuracy of 97.73%. The ensemble structure of Random Forest was also preferred to reduce dependence on a single decision tree. The finalized configuration contained 300 trees with a maximum depth of 12. These validation results and the model-selection rationale are documented in the model evidence report.

After model selection, the Random Forest model was frozen and evaluated once against the independent 1,500-record final test partition. The final evaluation produced an MAE of 0.0815 days, RMSE of 0.4042 days, and R^2 of 0.9976. Of the 1,500 final test cases, 1,458 predictions, equivalent to 97.20%, were within ±1 day of the target value.

The complete machine-learning development process therefore followed two distinct but related stages. Dataset A provided the empirical historical assessment based on the available Shoelotskey paper records and demonstrated the limitation of using customer Claim Date as the prediction target. Dataset B subsequently addressed the operational prediction objective by using Ready-for-Release as the target and was used to develop and evaluate the finalized Random Forest model. The two datasets were maintained separately because they represented different target definitions and served different methodological purposes.

Random Forest Hyperparameter Tuning

For the finalized operational machine-learning model, Random Forest hyperparameter tuning was conducted using Dataset B. The tuning procedure

used an exhaustive pre-declared parameter grid on the 7,000-record training set, while the 1,500-record validation set was used for model selection. The remaining 1,500 records were maintained as a sealed final test set and were not used during hyperparameter selection. A total of 324 Random Forest parameter combinations were evaluated.

Table 7

RANDOM FOREST HYPERPARAMETER SEARCH AND SELECTED CONFIGURATION

Hyperparameter	Values Evaluated	Selected Value
Number of Trees (n_estimators)	100, 200, 300	300
Maximum Tree Depth (max_depth)	None, 8, 12, 16	12
Minimum Samples Split (min_samples_split)	2, 5, 10	10
Minimum Samples Leaf (min_samples_leaf)	1, 2, 4	4
Maximum Features (max_features)	sqrt, 0.7, 1.0	1.0

The hyperparameter search evaluated three candidate values for the number of trees, four values for maximum tree depth, three values for minimum samples split, three values for minimum samples leaf, and three values for maximum features. This resulted in 324 possible parameter combinations (3 × 4 × 3 × 3 × 3). The selected Random Forest configuration consisted of 300 trees, a maximum depth of 12, minimum samples split of 10, a minimum samples leaf of 4, and a maximum-features value of 1.0. The finalized model used random_state=20261002 to support reproducibility.

The selected configuration was additionally verified through direct inspection of the serialized operational Random Forest model. The model

contained n_estimators=300, max_depth=12, min_samples_split=10, min_samples_leaf=4,max_features=1.0,bootstrap=True,criterion='squared_error', and random_state=20261002.

It should be noted that this exhaustive hyperparameter search applies to Dataset B, which was used for the finalized operational Ready-for-Release prediction model. The earlier historical Dataset A experiment did not undergo the same grid-search procedure. For Dataset A, a pre-fixed Random Forest configuration consisting of 200 trees, unrestricted maximum tree depth (max_depth=None), max_features='sqrt', and random_state=42 was used for the historical empirical assessment. Maintaining this distinction prevents the hyperparameter tuning performed on the simulated operational dataset from being incorrectly attributed to the historical dataset.

Expanding-Window Cross-Validation

For Dataset A, model development and validation were performed using the 93 historical development records. A five-fold expanding-window cross-validation strategy was applied to preserve the chronological sequence of the transaction data. In each succeeding fold, the training set increased by incorporating earlier observations, while the corresponding validation set consisted only of records occurring after the training period. The 26 historical records from January 2026 were excluded from cross-validation and maintained as the final temporal holdout set.

Table 8

EXPANDING-WINDOW CROSS-VALIDATION RESULTS FOR HISTORICAL DATASET A

Fold	Training Records	Validation Records	Original MAE (Days)
Fold 1	18	15	7.97
Fold 2	33	15	5.52
Fold 3	48	16	8.34
Fold 4	64	14	11.41
Fold 5	78	15	7.54
Mean	—	—	**8.1541**

The original persisted cross-validation evidence produced MAE values of 7.97, 5.52, 8.34, 11.41, and 7.54 days across the five folds, respectively. The resulting mean cross-validation MAE was 8.1541 days. The original aggregate cross-validation results further recorded an RMSE of 10.3225 days and an R^2 of 0.2679. These values represent the original historical model-development results and are therefore used as the primary cross-validation results reported in the study.

Table 9

SUMMARY OF ORIGINAL RANDOM FOREST CROSS-VALIDATION PERFORMANCE

Evaluation Measure	Result
Historical Dataset	Dataset A
Development Records	93
Validation Method	5-Fold Expanding-Window Cross-Validation
Random Forest Trees	200
Maximum Depth	None
Maximum Features	sqrt
Random State	42
Mean MAE	**8.1541 days**
Mean RMSE	**10.3225 days**
Mean R^2	**0.2679**
Final Temporal Holdout	26 records

The cross-validation results were generated during model development before the 26-record temporal holdout was used for final testing. This separation ensured that the final historical test records were not used during model

development or cross-validation. The original development results reported MAE = 8.1541 days, RMSE = 10.3225 days, and R^2 = 0.2679.

Cross-Validation Reproducibility Verification

As part of the subsequent evidence audit, the Dataset A expanding-window cross-validation procedure was independently reproduced using the preserved historical dataset and the original Random Forest configuration of 200 trees, max_features='sqrt', and random_state=42. This reproduction was performed as a verification procedure and was not used to replace the original persisted results.

Table 10

REPRODUCED EXPANDING-WINDOW CROSS-VALIDATION VERIFICATION

Fold	Training Period (N)	Validation Period (N)	MAE (Days)	RMSE (Days)	R^2
Fold 1	Aug. 15–Oct. 14, 2025 (18)	Oct. 16–Nov. 13, 2025 (15)	7.7417	9.7380	0.0699
Fold 2	Aug. 15–Nov. 13, 2025 (33)	Nov. 14–Nov. 20, 2025 (15)	5.8007	6.3961	0.6059
Fold 3	Aug. 15–Nov. 20, 2025 (48)	Nov. 21–Dec. 2, 2025 (16)	8.3122	12.3663	0.1914
Fold 4	Aug. 15–Dec. 2, 2025 (64)	Dec. 4–Dec. 12, 2025 (14)	11.6223	14.1517	−0.1329
Fold 5	Aug. 15–Dec. 12, 2025 (78)	Dec. 13–Dec. 29, 2025 (15)	7.8474	9.6003	0.5435
Mean	—	—	**8.2649**	**10.4505**	**0.2556**

The reproduced execution yielded a mean MAE of 8.2649 days, mean RMSE of 10.4505 days, and mean R^2 of 0.2556. The audit confirmed that the expanding-window sequence maintained chronological ordering and that no observations from a validation period were included in its preceding training set.

The slight numerical differences between the original persisted results (MAE = 8.1541 days, RMSE = 10.3225 days, R^2 = 0.2679) and the reproduced audit results (MAE = 8.2649 days, RMSE = 10.4505 days, R^2 = 0.2556) were documented separately rather than combining values from the two executions. The original persisted results remain the primary values reported for the historical experiment, while the reproduced values serve as independent verification of the cross-validation procedure. The evidence report attributes the slight variation to differences in the computational environment and scikit-learn tree-construction numerical behavior.

Final Methodological Distinction

The machine-learning development therefore involved two separate validation strategies corresponding to the two research datasets. Dataset A, consisting of 119 validated historical orders, used 93 development records with five-fold expanding-window cross-validation and a separate 26-record temporal holdout. Its Random Forest configuration was pre-fixed at 200 trees with max_features='sqrt' and random_state=42; no hyperparameter grid search was conducted for Dataset A. Dataset B, consisting of 10,000 simulated operational orders, used 7,000 training records, 1,500 validation records, and 1,500 sealed final test records. Dataset B underwent the 324-combination Random Forest hyperparameter search, from which the finalized configuration of 300 trees, maximum depth 12, minimum samples split 10, minimum samples leaf 4, and max_features=1.0 was selected.

This separation ensures that the historical cross-validation evidence is not conflated with the hyperparameter tuning and performance evaluation of the finalized operational model. Dataset A serves as the empirical historical assessment of the available Order-to-Claim target, whereas Dataset B serves as

the development and evaluation dataset for the active Ready-for-Release prediction model.

Methodology Justification using Waterfall Model

The Waterfall development model was selected as the methodology for this study because the system requirements were clearly defined and stable at the start of the project. The proposed Web-Based Service Management System for Shoelotskey was developed based on identified business processes, documented service records, and user requirements gathered through interviews and focus group discussions. Since the scope and functionality of the system were well understood early in the study, the sequential nature of the Waterfall model was appropriate.

The Waterfall model follows a structured and systematic approach that includes requirements analysis, system design, development, testing, implementation, and maintenance. This step-by-step process allowed the researchers to focus on completing each phase thoroughly before moving to the next, reducing errors and ensuring that the system design aligned with the actual operational needs of the business. It also made documentation easier to manage, which is essential for academic research and capstone projects.

Furthermore, the Waterfall model was suitable for integrating the machine learning component of the system. The availability of historical service data made it possible to define the machine learning requirements during the early stages of development. This allowed the researchers to design and implement the service completion date prediction model during the development phase and evaluate it during testing. Overall, the Waterfall methodology provided a clear development

structure, supported proper documentation, and ensured that the proposed system met the objectives of the study within the given timeframe.

System Development Framework

The system development framework follows a Three-Tier Architecture, integrating the presentation layer, application layer, and data management layer to support efficient system operation and organized data processing. The development workflow includes requirements gathering, system design, module development, integration testing, and system validation to ensure that the proposed platform meets the operational needs of Shoelotskey.

The system was developed using modern web technologies, including server-side scripting, relational database management systems, and client-side interface frameworks to support web-based access and centralized service management. The layered architecture allows the system to separate user interface functions, system processing logic, and database management, improving maintainability, scalability, and system performance.

Machine learning components are integrated within the application layer to support predictive analytics and service completion estimation. These components analyze historical service records and job order data to generate predicted completion times and identify service trends, assisting administrators in improving operational planning and decision-making.

Analysis of the Existing System

The analysis of the existing system examines Shoelotskey's current operational procedures for managing service transactions, job orders, and business records. Understanding the current workflow is essential in identifying

limitations that affect service monitoring, data management, and overall business efficiency. This analysis focused on how service requests are recorded, how service completion is monitored, and how transactions are documented within the organization.

Currently, Shoelotskey relies on manual recording procedures and basic spreadsheet tools to manage daily operations. Service transactions and job orders are encoded manually, while service progress updates are monitored through handwritten notes or verbal coordination among staff. Although these methods allow basic record keeping, they do not provide an integrated platform for storing, updating, and retrieving operational data efficiently.

Based on interviews and observations conducted during the requirements-gathering process, the owner and staff identified record retrieval, manual encoding, and service-status monitoring as recurring operational difficulties. Previous customer and service records may require searching through separate physical records or spreadsheet files, which can delay transaction inquiries. Manual recording may also result in duplicate entries, incomplete information, or details that require subsequent correction. During busy periods, the absence of centralized service-status monitoring may also contribute to delayed updates and miscommunication among staff and customers.

The current system also has limited capability for generating reports and analyzing business performance. Daily transactions, service volume, turnaround time, and customer trends must be summarized manually, consuming additional staff time and reducing management efficiency. Since historical records are not centralized, management has difficulty identifying service demand patterns and making timely operational decisions.

The identified operational problems were used as the basis for defining the requirements of the proposed system. In response, the proposed system centralizes job order and customer records, provides service-status monitoring, supports inventory and sales management, and provides reporting and data analytics functions. These requirements establish a direct connection between the limitations of the existing process and the functions incorporated into the proposed system.

These findings indicate the need for a more efficient and automated system that can centralize records, improve service monitoring, designed to reduce manual errors, and support data-driven decision-making for Shoelotskey.

Current Service Management Procedures

The current service management process at Shoelotskey primarily relies on manual recording of service transactions and job orders. When customers request services, staff members manually record service details using paper-based forms or spreadsheet files. Staff members manually enter information such as customer name, service type, service cost, and service date manually and store it in separate records.

After a service request is recorded, the job order is processed and performed in accordance with the requested service. Once the service is completed, staff members manually update the service status and record the transaction details in the existing record-keeping system. However, this process often requires staff members to review multiple documents or files to verify whether a service request has already been completed or is still pending.

Monitoring service schedules and completed services is also performed manually. Staff members must manually track which services have been released

or completed, which can delay record updates. This manual monitoring process becomes more challenging as the volume of service transactions increases.

Sales monitoring and reporting are conducted by manually reviewing recorded transactions. Staff members must compile service records and calculate totals manually to generate reports for business monitoring. This process requires considerable time and effort, particularly when reviewing service performance or preparing transaction summaries. As a result, the existing system limits Shoelotskey's ability to quickly access operational information and evaluate business performance.

Limitations and Problem Analysis

Analysis of current service management procedures revealed several operational and administrative limitations affecting the efficiency and reliability of business operations. One of the primary issues is the heavy reliance on traditional record-keeping, which increases staff workload and the risk of data entry errors. This approach also reduces efficiency in tracking service transactions and updating service records.

Another significant limitation is the difficulty in monitoring job orders and service completion. Since service status updates are recorded manually, staff members must check multiple records to determine whether services are completed, pending, or scheduled for release. This process makes it difficult to track service progress in real time and may delay service record updates.

The absence of a centralized database also presents a major challenge. Service records, job orders, and sales information are stored in separate documents or spreadsheet files, making information retrieval inefficient. When the

business owner needs to review service history or analyze operational performance, staff members must search through multiple records to gather the required data.

In addition, the current system lacks analytical capabilities to identify trends in service demand, customer preferences, and service completion performance. Without automated data analysis tools, the organization cannot effectively use operational data for informed decision-making. These limitations highlight the need for an integrated and automated system that improves data management, monitoring, and overall operational efficiency.

Requirements Gap Analysis

Requirements gap analysis was conducted to determine the differences between Shoelotskey's existing manual service management procedures and the functional capabilities required for an efficient automated system. The analysis identifies missing features, process limitations, and operational needs necessary to improve the current workflow.

The findings show that the current system lacks a centralized platform for recording and managing service transactions. Service records are stored in separate logbooks and spreadsheet files, making it difficult for staff to access updated information and monitor job orders in real time. This results in fragmented records, slower transaction processing, and limited data accuracy.

Another major gap is the absence of automated tools for service monitoring and scheduling. Staff members manually check service progress and update completion status, which increases the possibility of delayed updates, missed schedules, and inefficient workflow coordination.

The existing process also lacks automated report generation and data analytics capabilities. Reports on daily transactions, service volume, and turnaround time must be prepared manually, consuming time and limiting management's ability to evaluate performance trends and make timely decisions.

These identified gaps indicate the need for a web-based service management system that can centralize records, automate service monitoring, generate real-time reports, and provide analytical insights for improved decision-making.

Current State	Required State
Service records stored in separate logbooks and spreadsheets	Centralized digital database for all service records
Manual monitoring of job orders and completion status	Real-time tracking of job orders and service progress
Manual scheduling and status updates	Automated scheduling and notification system
Reports prepared manually	Automated report generation with dashboards
Limited historical data analysis	Data analytics for trends, demand forecasting, and performance evaluation
High risk of duplicate or incomplete entries	Accurate, validated, and consistent data management

Alternative Solutions and Feasibility Analysis

To address the identified limitations of the current system, several alternative solutions were evaluated to improve Shoelotskey's service management operations. Each option was assessed using key criteria such as functionality, scalability, cost efficiency, automation capability, reporting features, and compatibility with business operations. The purpose of this analysis was to determine the most suitable and sustainable solution for the organization.

The first alternative was to continue using the existing manual record-keeping process with minor improvements in documentation practices. This option requires minimal cost and no major system changes; however, it offers limited improvement in operational efficiency, data accessibility, and service monitoring. Manual processes also remain prone to delays, missing records, and encoding errors.

The second alternative was the use of spreadsheet-based management tools. Spreadsheet software can improve data organization, record storage, and basic reporting compared to purely manual methods. However, it still lacks real-time automation, centralized multi-user access, advanced analytics, and efficient transaction monitoring. As service transactions increase, spreadsheet files may become difficult to maintain and more susceptible to version control issues.

The third alternative was the development of a custom web-based service management system specifically designed for Shoelotskey. This solution offers the highest level of flexibility because it can be tailored to the organization's workflow and operational requirements. It also supports centralized database management, real-time service monitoring, automated reporting, user access control, and future integration of data analytics and machine learning features.

Comparative Evaluation of Alternatives

This section was used to determine the proposed solution during the feasibility and planning stage. The actual effectiveness and performance of the developed system were evaluated separately in Chapter 4.

Criteria	Weight	Manual Process	Spreadsheet System	Custom Web-Based System
Functionality	30%	2	3	5
Scalability	20%	1	2	5

Criteria	Weight	Manual Process	Spreadsheet System	Custom Web-Based System
Cost Efficiency	15%	5	4	3
Automation Capability	20%	1	2	5
Reporting & Analytics	10%	1	3	5
Compatibility with Operations	5%	3	4	5

Rating Scale: 1 = Poor, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent

Weighted Decision Matrix

Alternative	Weighted Score
Manual Process	2.00
Spreadsheet System	2.75
Custom Web-Based System	4.70

Based on the weighted decision matrix, the custom web-based service management system obtained the highest weighted score among the evaluated alternatives and was selected as the proposed solution for the study.

Proposed Technology Alternatives

Several technology solutions were evaluated to address the operational limitations identified in the existing service management procedures. Each technology alternative was examined based on its ability to support automation, centralized data management, scalability, and operational efficiency.

One alternative considered was enhancing the existing spreadsheet-based record management system. While spreadsheets can improve data organization, they still rely heavily on manual data entry and lack automated monitoring capabilities. Additionally, spreadsheet systems do not provide integrated dashboards, centralized database management, or advanced analytics features.

Another alternative was to adopt commercial service management platforms available on the market. These platforms offer automated features, including service tracking, reporting tools, and data storage. However, many commercial solutions require subscription fees and provide limited customization options. These limitations may prevent the system from fully addressing Shoelotskey's unique operational requirements.

The most suitable solution identified was the development of a custom web-based service management system. This approach enables the system to be specifically designed to meet the organization's operational needs. The proposed system includes modules for job order summary management, service completion tracking, sales reporting, dashboard monitoring, and user management. Additionally, the system integrates data analytics and machine learning capabilities to analyze service completion data and identify operational trends.

By implementing a custom web-based solution, Shoelotskey can improve service monitoring, centralize business records, and enhance decision-making through data-driven insights.

Feasibility Study

The feasibility study evaluates the practicality and sustainability of developing and implementing the proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithm for Shoelotskey. The objective of this assessment is to determine whether the proposed system can be successfully developed using available resources while meeting the organization’s operational requirements. The feasibility study covers four major dimensions: technical, operational, economic, and schedule feasibility.

Technical Feasibility

Technical feasibility examines whether the required hardware, software, and technical resources are available to support system development and deployment. Shoelotskey currently has access to one computer unit, standard printer, and stable internet connectivity as identified during the assessment of the shop's available resources. These existing resources were considered adequate for accessing the proposed web-based system.

The proposed system will be developed using widely available technologies such as Python, PostgreSQL, Tailwind CSS, JavaScript, and open-source development frameworks. These technologies are commonly used for small- to medium-scale business systems because they are cost-effective, reliable, and easy to maintain.

The system will use a three-tier architecture consisting of the presentation layer, application layer, and database layer. This structure improves maintainability, security, and system performance. The centralized database will store service records, job orders, transactions, and user accounts in a single platform.

Potential risks include increased data volume, server downtime, and database management complexity. However, these risks can be mitigated through regular backups, optimized database design, and preventive maintenance. Based on available infrastructure and accessible technologies, the proposed system is considered technically feasible.

Operational Feasibility

Operational feasibility assesses whether the proposed system fits Shoelotskey’s daily workflow and whether users can effectively adopt the system.

At present, service transactions, job orders, and reports are managed manually using notebooks and spreadsheets, resulting in delays, duplicate entries, and difficulty tracking service status.

Based on interviews and observations, six (6) shop personnel, consisting of the owner, co-owner, and staff, participated in the requirements-gathering activities. They identified faster record retrieval, easier service monitoring, and automated report generation as expected benefits of adopting the proposed digital system. The respondents also indicated willingness to use the system with appropriate orientation and basic training.

The proposed system will allow users to record transactions, update job order status, track service completion, and generate reports through a user-friendly web interface accessible via standard browsers. The proposed system was designed to support the shop's existing operational workflow while providing digital tools for record management, monitoring, and reporting.

Economic Feasibility

Economic feasibility evaluates whether the expected benefits justify the cost of development and implementation. Since the system will use existing computer units and open-source software, implementation cost is limited mainly to development, internet usage, and maintenance expenses.

The current manual process requires significant administrative time for recording transactions and preparing reports. Based on the requirements-gathering activities, manual encoding, record retrieval, and report preparation were identified as activities that require staff time under the existing process. The proposed system is expected to reduce the manual effort required for these activities through centralized record storage, automated retrieval, and report

generation. Any estimated time or monetary savings presented in the Cost–Benefit Analysis are treated as projected values based on documented assumptions rather than verified financial results.

Tangible benefits include reduced paper usage, faster report generation, improved record accuracy, and lower administrative workload. Intangible benefits include better customer service, improved monitoring of operations, and enhanced decision-making through data analytics. Given the manageable cost and long-term efficiency gains, the proposed system is considered economically feasible.

Schedule Feasibility

Schedule feasibility evaluates whether the proposed system can be developed and implemented within the planned project timeline. The development of the Web-Based Service Management System with Data Analytics using Machine Learning Algorithm will follow the Waterfall Software Development Life Cycle (SDLC), which provides a structured and sequential approach to project execution. This methodology ensures that each phase is completed before proceeding to the next stage, allowing better control of project activities, deliverables, and deadlines.

The estimated duration for completing the project is approximately five months. The first phase, requirements analysis, is expected to take three weeks and will focus on gathering system requirements, identifying user needs, and analyzing existing business processes. This will be followed by the system design phase, which is estimated to take four weeks and will cover database design, interface layouts, and overall system architecture.

The system development phase is projected to require eight weeks, during which the core modules, database functions, and user interfaces will be coded and

integrated. After development, testing and debugging activities are expected to take three weeks to verify system functionality, identify errors, and ensure that all requirements are met. The final phase, deployment and user training, is estimated to take two weeks and will involve installing the system, orienting users, and preparing the organization for operational use.

Possible scheduling risks include delays in coding, technical issues during testing, and revisions in user requirements. However, these risks can be minimized through effective planning, clear task allocation among team members, regular progress monitoring, and adherence to the proposed schedule. Based on the defined scope, available resources, and structured development methodology, the proposed system is considered schedule feasible and can be completed within the planned timeframe.

Cost–Benefit Analysis

Overview

The Cost–Benefit Analysis (CBA) evaluates the economic feasibility of implementing the proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithm for Shoelotskey Villamor-Pasay. The analysis compares the estimated costs of system development, deployment, and operation with the measurable and non-monetary benefits expected from the proposed system.

To provide a transparent basis for the analysis, development labor costs were estimated using published Philippine salary benchmarks for comparable information technology positions. Hosting costs were based on the published pricing of the cloud services used by the system. The monetary value of

operational time savings was estimated using the applicable National Capital Region (NCR) minimum wage as a conservative labor-value basis. Other benefits that could not be reliably converted into monetary amounts were classified as intangible benefits rather than assigned arbitrary financial values.

The monetary values presented in this analysis are estimates for academic feasibility assessment and should not be interpreted as actual salaries paid by Shoelotskey or guaranteed financial returns.

A. Cost Analysis

The cost analysis includes the estimated expenses for documentation materials, system development, operational needs, implementation, and web hosting.

1. Documentation and Office Supply Costs

Item	Quantity/Assumption	Unit Cost	Estimated Cost
Bond Paper A4	2 reams	350	700
Long Folder	10 pieces	10	100
Ballpen (Tri Quick Dry Gel 0.5)	5 pieces	20	100
Pentel Pen (Mark-4-all Permanent)	2 pieces	50	100
Logbook	2 pieces	150	300
Correction Tape (Getwin 6m)	2 pieces	27	54
Clip	10 pieces	8	80

Subtotal – Documentation and Office Supplies: ₱1,434

The estimated documentation and office supply cost is **₱1,434**. The amounts are based on the quantities and unit prices used in the project cost

estimate. Receipts or price documentation should be retained as supporting evidence where available.

2. Development Labor Cost

Development labor was estimated according to the activities performed throughout the Waterfall Software Development Life Cycle. Instead of assigning arbitrary hourly rates, published Philippine salary benchmarks for comparable IT positions were used.

Development Role	Comparable Salary Benchmark	Monthly Salary Benchmark (₱)
Project Manager	IT Project Manager	65,603
System Analyst	Systems Analyst	34,888
Developer/Programmer	Software Engineer	42,712
QA Tester	Quality Assurance Engineer	25,759

For consistency, the monthly salary benchmarks were converted into hourly labor-value equivalents using an assumed 22 working days per month and eight working hours per day:

Monthly Working Hours = 22 days × 8 hours = 176 hours

Hourly Labor Rate = Monthly Salary Benchmark ÷ 176 hours

Therefore:

Project Manager

₱65,603 ÷ 176 = **₱372.74/hour**

System Analyst

₱34,888 ÷ 176 = **₱198.23/hour**

Developer/Programmer

₱42,712 ÷ 176 = **₱242.68/hour**

QA Tester

₱25,759 ÷ 176 = **₱146.36/hour**

Table 11. Development Labor Rate Basis

Role	Monthly Benchmark (₱)	Monthly Hours	Estimated Hourly Rate (₱)
Project Manager	65,603	176	372.74
System Analyst	34,888	176	198.23
Developer/Programmer	42,712	176	242.68
QA Tester	25,759	176	146.36

The estimated hourly rates represent the economic value of the development effort and do not represent actual compensation paid to the student researchers.

3. Waterfall Development Cost

The following formula was applied to each development activity:

Development Cost = Number of Persons × Estimated Hours × Hourly Labor Rate

Phase 1 – Requirements Analysis

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
Project Planning Meeting	Project Manager	6	372.74	2,236.44
Business Process Analysis	System Analyst	10	198.23	1,982.30
Technical Consultation	Developer	5	242.68	1,213.40
Initial Validation	QA Tester	4	146.36	585.44

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
Phase 1 Total				**₱6,017.58**

Phase 2 – System Design

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
System Architecture Planning	Project Manager	8	372.74	2,981.92
Database Design	System Analyst	10	198.23	1,982.30
UI/UX Design Review	Developer	8	242.68	1,941.44
Design Validation	QA Tester	4	146.36	585.44
Phase 2 Total				**₱7,491.10**

Phase 3 – System Development

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
Development Supervision	Project Manager	10	372.74	3,727.40
Database and Logic Review	System Analyst	8	198.23	1,585.84
Front-End Development	Developer	35	242.68	8,493.80
Back-End Development	Developer	40	242.68	9,707.20
ML Module Integration	Developer	15	242.68	3,640.20
Code Inspection	QA Tester	10	146.36	1,463.60
Phase 3 Total				**₱28,618.04**

Phase 4 – Testing and Deployment

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
Deployment Oversight	Project Manager	6	372.74	2,236.44
Functional Review	System Analyst	8	198.23	1,585.84
Bug Fixing	Developer	15	242.68	3,640.20

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
Full Testing and UAT	QA Tester	18	146.36	2,634.48
Phase 4 Total				**₱10,096.96**

Phase 5 – System Evaluation

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
Project Review	Project Manager	5	372.74	1,863.70
Performance Evaluation	System Analyst	8	198.23	1,585.84
Minor Adjustments	Developer	10	242.68	2,426.80
Feedback Testing	QA Tester	6	146.36	878.16
Phase 5 Total				**₱6,754.50**

Phase 6 – Maintenance and Support

Activity	Role	Hours	Rate (₱/hr)	Estimated Cost (₱)
Maintenance Supervision	Project Manager	4	372.74	1,490.96
System Monitoring	System Analyst	6	198.23	1,189.38
Bug Fixes/Updates	Developer	15	242.68	3,640.20
Regression Testing	QA Tester	8	146.36	1,170.88
Phase 6 Total				**₱7,491.42**

Overall Waterfall Development Cost

Development Phase	Estimated Cost (₱)
Requirements Analysis	6,017.58
System Design	7,491.10
System Development	28,618.04
Testing and Deployment	10,096.96
System Evaluation	6,754.50
Maintenance and Support	7,491.42

Development Phase	Estimated Cost (₱)
Total Estimated Development Labor Cost	**₱66,469.60**

The estimated development labor cost is therefore approximately **₱66,470**. This represents the estimated economic value of the development effort based on published Philippine salary benchmarks and the estimated hours allocated to each Waterfall activity.

4. Hosting and Deployment Cost

The system is hosted using Heroku. For a small deployed application, the cost estimate uses a Basic Dyno at US$7 per month and Heroku Postgres Essential-0 at US$5 per month, resulting in:

Monthly Cloud Cost

= US$7 + US$5
= **US$12/month**

Annual Cloud Cost

= US$12 × 12 months
= **US$144/year**

Using an exchange rate of approximately **₱62.5835 per US$1** for October 2, 2026:

US$144 × ₱62.5835
= **₱9,012.02 per year**

The domain registration/renewal cost is budgeted at ₱1,000 per year based on the project's domain-cost estimate. The actual registrar invoice or renewal price should be used as supporting documentation and should replace this amount if the actual cost differs.

Table 12. Annual Hosting and Deployment Cost

Category	Basis	Annual Estimated Cost (₱)
Heroku Basic Dyno	US$7/month × 12	5,257.01
Heroku Postgres Essential-0	US$5/month × 12	3,755.01
Domain Registration/Renewal	Project domain-cost estimate	1,000.00
Total Annual Hosting and Deployment Cost		**₱10,012.02**

5. Total Initial Project Cost

To avoid double counting, separate maintenance, training, software development tools, database setup, machine learning integration, and server-configuration labor were not added again because the corresponding development activities are already represented in the Waterfall labor calculation.

Existing computer equipment and internet connectivity are also not treated as new project acquisition costs because these resources are already available to the business.

Table 13. Estimated Initial Project Cost

Cost Category	Estimated Cost (₱)
Documentation and Office Supplies	1,434.00
Waterfall Development Labor	66,469.60
First-Year Hosting and Database	9,012.02
First-Year Domain Cost	1,000.00
Total Estimated Initial Project Cost	**₱77,915.62**

Therefore:

Total Initial Project Cost

= ₱1,434 + ₱66,469.60 + ₱9,012.02 + ₱1,000

= **₱77,915.62**

6. Estimated Three-Year Cost

The first year of hosting and domain expenses is already included in the initial project cost. Therefore, only the recurring hosting and domain costs for Years 2 and 3 are added.

Annual Recurring Cost

= ₱9,012.02 + ₱1,000

= **₱10,012.02**

Three-Year Cost

= Initial Project Cost + (Annual Recurring Cost × 2)

= ₱77,915.62 + (₱10,012.02 × 2)

= ₱77,915.62 + ₱20,024.04

= **₱97,939.66**

Thus, the estimated total project cost over three years is approximately **₱97,940**.

B. Benefit Analysis

To avoid unsupported financial estimates, only operational benefits with a measurable basis were assigned monetary values. Benefits such as better monitoring, customer satisfaction, record organization, and data-driven decision-making were retained as intangible benefits.

1. Estimated Administrative Time-Saving Benefit

Based on the researchers' observation of the existing process, approximately 2 hours per operating day are spent on manual encoding, record searching, and preparation of summaries. With the proposed centralized system, the same administrative activities are estimated to require approximately 30 minutes per day**.**

Therefore:

Estimated Daily Time Saved

= 2.0 hours − 0.5 hour

= **1.5 hours per operating day**

Shoelotskey is closed on Sundays; therefore, approximately 26 operating days per month were used for the estimate. The applicable NCR minimum wage for service/retail establishments employing 15 workers or fewer was used as a conservative basis for valuing staff time:

Daily Wage = ₱718

Assuming an eight-hour workday:

Hourly Labor Value

= ₱718 ÷ 8
= **₱89.75/hour**

Estimated monthly labor-time value:

= 1.5 hours × 26 operating days × ₱89.75

= **₱3,500.25/month**

Estimated annual labor-time value:

= ₱3,500.25 × 12 months

= **₱42,003.00/year**

Table 14. Estimated Annual Measurable Benefit

Measurable Benefit	Computation Basis	Annual Estimated Value (₱)
Administrative time savings	1.5 hrs/day × 26 days/month × 12 months × ₱89.75/hr	42,003.00
Total Estimated Annual Measurable Benefit		**₱42,003.00**

The ₱42,003 annual amount represents the estimated economic value of staff time made available for other productive activities. It should not be interpreted as a direct cash saving or reduction in employee salary.

2. Intangible Benefits

The following anticipated benefits were not assigned monetary values because sufficient evidence was not available to reliably convert them into direct financial savings:

Intangible Benefit	Expected Operational Contribution
Improved record organization	Centralized storage of customer, job order, and transaction records
Faster record retrieval	Allows authorized users to locate service information more efficiently
Better service monitoring	Provides centralized visibility of job order status
Improved reporting	Provides easier access to operational and management reports
Data-driven decision support	Provides analytics that can support management decisions

Intangible Benefit	Expected Operational Contribution
Improved customer service	Supports more timely access to service and order information
Reduced recording errors	Digital validation can reduce incomplete or inconsistent entries

These benefits are recognized as operational improvements but are excluded from the monetary benefit calculation to avoid overstating the financial return of the proposed system.

C. Three-Year Benefit

Assuming that the estimated administrative time-saving benefit remains consistent throughout the three-year analysis period:

Three-Year Measurable Benefits

= ₱42,003 × 3 years

= **₱126,009**

Therefore, the total estimated measurable benefit over three years is **₱126,009**.

D. Financial Evaluation

1. Projected Net Benefit

Projected Net Benefit = Total Measurable Benefits − Total Estimated Costs

= ₱126,009 − ₱97,939.66

= **₱28,069.34**

2. Projected Return on Investment (ROI)

The projected ROI was calculated as:

ROI = (Projected Net Benefit ÷ Total Estimated Cost) × 100

= (₱28,069.34 ÷ ₱97,939.66) × 100

= **28.66%**

Therefore, based on the documented assumptions used in the analysis, the proposed system has a **projected three-year ROI of approximately 28.66%**.

3. Projected Benefit–Cost Ratio (BCR)

The projected Benefit–Cost Ratio was calculated as:

BCR = Total Measurable Benefits ÷ Total Estimated Costs

= ₱126,009 ÷ ₱97,939.66

= **1.29 : 1**

Therefore, the analysis produces a **projected BCR of approximately 1.29:1**. Under the stated assumptions, this means that approximately **₱1.29 in measurable economic value is projected for every ₱1.00 of estimated project cost** over the three-year analysis period.

The BCR represents a projection based on the assumptions used in the feasibility analysis and should not be interpreted as a guaranteed financial return.

E. Cost–Benefit Summary

Financial Measure	Estimated Value
Initial Project Cost	**₱77,915.62**
Annual Recurring Cost	**₱10,012.02**
Three-Year Total Cost	**₱97,939.66**
Annual Measurable Benefit	**₱42,003.00**
Three-Year Measurable Benefits	**₱126,009.00**
Projected Net Benefit	**₱28,069.34**
Projected ROI	**28.66%**

Financial Measure	Estimated Value
Projected BCR	**1.29 : 1**

Conclusion

Based on the documented assumptions and computation bases used in the Cost–Benefit Analysis, the proposed system has an estimated initial project cost of **₱77,915.62** and an estimated three-year cost of **₱97,939.66**. The measurable economic benefit associated with reduced administrative processing time is estimated at **₱42,003 per year**, or **₱126,009 over three years**.

Using these estimates, the system produces a projected net benefit of **₱28,069.34**, a projected three-year ROI of **28.66%**, and a projected Benefit–Cost Ratio of approximately **1.29:1**.

These results indicate potential economic feasibility under the assumptions used in the analysis. However, the calculated benefits represent the estimated economic value of time savings rather than guaranteed cash savings. Other expected improvements, including centralized records, better service monitoring, improved reporting, reduced recording errors, data-driven decision support, and improved customer service, were treated as intangible benefits and were not assigned monetary values to avoid overstating the financial benefits of the proposed system.

System Requirements Specification

The System Requirements Specification details the operational capabilities and technical integrations of the proposed Web-Based Service Management System. Developed specifically for Shoelotskey in Barangay 183 Villamor Pasay, the system integrates modern data analytics and machine learning algorithm to

transition the business from a manual ledger process into a fully digitized platform. The primary objective is to automate the recording of customer orders, calculate exact financial reports, and formulate accurate predictions regarding service completion timelines. By utilizing the Python Scikit-Learn library, the system actively analyzes historical processing times to provide the business owner with reliable service completion forecasts. The requirements specification serves as a guide for system development, ensuring that the implemented system meets Shoelotskey's operational needs while maintaining system reliability, security, and efficiency.

Stakeholder Identification and User Roles

The proposed system involves two primary stakeholders who interact with it according to their operational responsibilities. Identifying user roles ensures that system functionalities align with the business workflow while maintaining proper access control and system security.

The Owner serves as the primary administrator of the system and is responsible for monitoring overall service operations and business performance. The owner has full access to the system features, including viewing operational dashboards, generating sales reports, managing service information, analyzing service performance trends and managing the users. Through the system dashboard, the owner can review service completion statistics, transaction summaries, and analytics generated from historical service records.

The Staff users are responsible for managing the business's daily service operations. Staff members use the system to record customer service requests, create job orders, update service status, and monitor service completion. Staff

users ensure that service transactions are accurately recorded in the system and that job order information is properly maintained.

The implementation of role-based access control ensures that system functions are accessible only to authorized users according to their responsibilities, thereby improving system security and operational efficiency.

Functional Requirements

The functional requirements describe the core system operations that support Shoelotskey's service management activities. These requirements define how users interact with the system and how the system processes service-related data.

The system shall provide secure user authentication, allowing authorized users to access the system using login credentials. This feature ensures that only the owner and staff members can access system functionalities based on their assigned roles.

The system shall provide a job order summary management function that allows staff members to record service requests, store customer information, and maintain job order details. Staff users shall be able to update service status and record service completion once the requested service has been performed.

The system shall include a service management module that allows the owner to manage available services and update service pricing information. This functionality ensures that service details are maintained accurately within the system database.

The system shall provide a dashboard module that displays summarized operational information, including the number of completed services, pending job orders, and sales performance indicators. This dashboard enables the owner to monitor business activities in real time.

The system shall include a released calendar module that allows users to track scheduled service releases and monitor service completion timelines.

The system shall generate sales reports that summarize service transactions and revenue performance. These reports allow the owner to evaluate business performance and identify trends in service demand.

The system shall integrate machine learning-based data analytics to analyze historical service records and identify patterns related to service completion, transaction frequency, and operational performance. The generated insights support better monitoring and decision-making for the business.

Non-Functional Requirements

The non-functional requirements define the quality attributes necessary to ensure that the proposed system operates efficiently, securely, and reliably. These requirements are based on recognized software quality characteristics such as security, usability, reliability, performance efficiency, and maintainability.

The system shall ensure security by implementing authentication mechanisms and role-based access control to prevent unauthorized access to system data. User login credentials will be required before accessing system features.

The system shall maintain usability by providing a user-friendly interface that allows both the owner and staff members to easily navigate system modules and perform operational tasks with minimal training.

The system shall ensure reliability by maintaining stable system performance during service transactions and database operations. Data backup mechanisms will be implemented to prevent data loss.

The system shall support performance efficiency by enabling fast retrieval of service records, efficient data processing, and quick generation of operational reports and dashboards.

The system shall also support maintainability by implementing a modular system architecture and a structured database design that allows developers to update, modify, and maintain the system efficiently.

Proposed System Architecture and Design

The proposed system architecture defines the overall structure of the Web-Based Service Management System for Shoelotskey. The architecture is designed to support centralized service data management, secure system access, and efficient processing of service transactions.

The system adopts a Three-Tier Architecture, which separates system components into three layers: the presentation layer, application layer, and data management layer. This layered structure improves system scalability, maintainability, and performance by separating user interface functions, application processing logic, and database operations.

Through this architecture, users interact with the system through web-based interfaces, while system processing and data management operations are handled within the application and database layers.

System Architecture Overview

The system architecture provides a structured framework that supports efficient system operations and centralized data processing. The presentation layer is the user interface through which system users interact with the platform via web browsers and developed using React and Vite. This layer provides access to modules including the dashboard, job order management, service management, sales reporting, the released calendar, and user management.

The application layer serves as the system's core processing component. It is built using Python with the FastAPI framework. It processes user requests, manages system logic, handles service completion updates, and performs machine learning analysis for data analytics. This layer ensures that operational processes are executed efficiently before data is stored or retrieved from the database.

The data management layer manages the centralized database that stores service records, job orders, transaction data, and user account information. It uses a cloud-based PostgreSQL database to securely store all operational and transactional data of the system. This layer ensures secure storage, efficient retrieval, and consistent management of operational data required for reporting and analytics.

System Components and Modules

The proposed Web-Based Service Management System with Data Analytics for Shoelotskey comprises several functional modules designed to support the business's daily service management operations. Each module performs specific system functions and, when integrated, provides a centralized platform for managing service transactions, monitoring operational performance, controlling inventory, and generating analytical insights. The modular design of the system ensures that business processes are organized into manageable components, allowing the platform to maintain efficiency, scalability, and maintainability.

The Dashboard Module serves as the central monitoring interface of the system. This module provides users with a summarized view of the business's operational activities through visual indicators, charts, and statistical summaries. The dashboard displays information such as the total number of completed services, pending job orders, scheduled service releases, available inventory levels, and sales performance indicators. By presenting real-time operational data, the dashboard allows the owner and authorized staff to quickly assess the current status of business operations and identify areas requiring immediate attention.

The Job Order Management Module is responsible for managing customer service requests and recording service transactions. Staff members use this module to create new job orders by entering customer information, selected service type, service cost, item details, and expected release date. The module also allows users to update the status of job orders as they progress from initial intake to completion and release. By centralizing job order records in the database, this module improves organization and enables efficient tracking of service requests.

The Service Management Module allows the owner or administrator to manage the list of services offered by the business. Through this module, authorized users can add new service categories, update service descriptions, modify pricing, and maintain accurate service records. This ensures consistency in service transactions and designed to reduce errors during job order creation.

The Released Calendar Module provides a scheduling interface that allows users to monitor service completion timelines and customer release schedules. This module helps track when services are due for completion or ready for customer pickup. By organizing service schedules in calendar format, staff can monitor deadlines, prioritize workloads, and design to reduce delays in service delivery.

The Inventory Management Module is responsible for monitoring and controlling the stocks of cleaning materials, restoration supplies, and other consumables used in daily operations. This module allows authorized users to record stock-in and stock-out transactions, update available quantities, and track inventory usage based on completed services. It also provides low-stock alerts whenever supplies reach critical levels. By automating inventory control, the module helps prevent shortages, overstocking, and inaccurate manual stock records.

The Sales Reporting Module generates reports related to service transactions and business performance. This module compiles operational data and produces summaries of daily, weekly, and monthly sales activities. Reports may include revenue totals, service demand trends, top-performing services, inventory usage summaries, and transaction frequency. These reports help the owner evaluate business performance and support strategic planning.

The User Management Module allows the owner to manage system users and control access privileges. Through this module, user accounts can be created for staff members, login credentials can be managed, and roles can be assigned based on responsibilities. Role-based access control ensures that users can only access modules relevant to their assigned duties. This module enhances system security by preventing unauthorized access to sensitive operational data.

The system also includes a Data Analytics Module uses historical transaction data to generate business insights such as service trends, transaction patterns, and performance summaries. Through machine learning algorithm, this module can identify service completion time/date prediction and support decision-making based on patterns in operational data.

By integrating these modules, the proposed system provides a centralized and efficient platform for service management, inventory control, accurate record keeping, and operational monitoring. The modular architecture enables the business to streamline workflows, designed to reduce manual processes, improve service coordination, and support data-driven decision-making for Shoelotskey.

Process Modeling and System Workflow

Context Diagram

The diagram illustrates the interaction between the system called Shoelotskey, the owner, and the staff, showing how data flows within the system to support business operations.

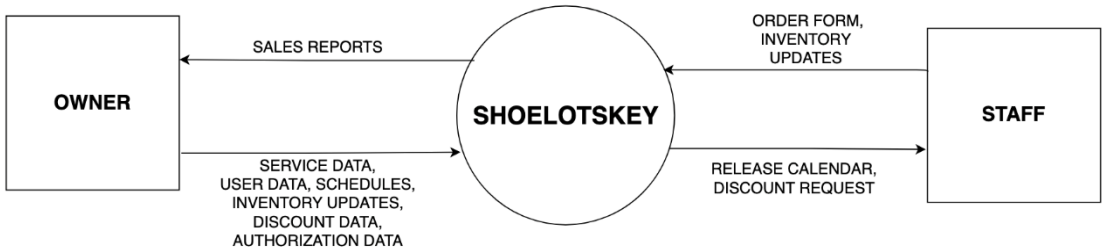


Figure 3: Context Diagram

The Owner interacts with the system by providing operational and administrative information, including service data, user account data, schedule data, inventory updates, discount configuration data, and authorization data. These inputs allow the Owner to manage system users, services, schedules, inventory, discounts, and restricted operations. In return, the system generates sales reports that enable the Owner to monitor sales transactions, revenue, and overall business performance.

The Staff interacts with the system by providing job order data and inventory transaction information generated during daily operations. Job order data includes the information necessary to record and process customer service transactions, while inventory transactions reflect the materials used in connection with service operations. In return, the system provides the release calendar, which allows Staff

to monitor expected service completion and release dates and organize ongoing customer orders. The system also provides applicable discount information or authorization results based on the discount rules and permissions configured by the Owner.

Overall, the Context Diagram presents the Shoelotskey Service Management System as the central point of interaction between the Owner and Staff. It shows only the major information entering and leaving the system, while the detailed processes, data stores, and internal data movements are presented in the succeeding Data Flow Diagrams and database models. The terminology used in the Context Diagram is maintained consistently with the system modules and other design artifacts to clearly represent the roles, information flows, and boundaries of the proposed system.

Data Flow Diagram (Level 0)

The diagram illustrates a more detailed data flow of the Shoelotskey system, highlighting how the owner and staff interact with the system to manage operations efficiently.

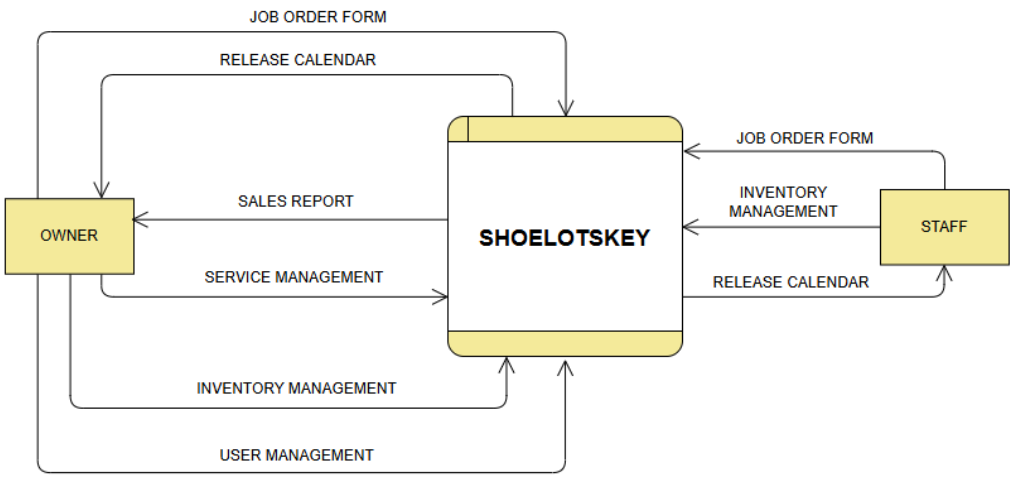


Figure 4: Data Flow Diagram Level 0

The Owner performs administrative and management-related activities within the system. The Owner provides service data for maintaining the services offered by the shop, inventory data for managing materials and supplies, and user account data for creating and maintaining Staff accounts and their corresponding permissions. The Owner may also create and manage job order data as part of customer service transactions. In return, the system provides the Owner with sales report data for monitoring transactions and business performance, as well as release calendar information for monitoring the expected completion and release of job orders.

The Staff primarily handles day-to-day operational transactions. Staff provides job order data containing the necessary customer, service, and transaction information for processing customer orders. Staff also records authorized inventory transactions associated with the materials used for job orders. In return, the system provides release calendar information, which allows Staff to monitor expected completion and release dates and assists in organizing customer orders according to their scheduled release.

Overall, DFD Level 0 presents the Shoelotskey Service Management System as a centralized platform through which the Owner and Staff exchange the information necessary to manage job orders, services, inventory, user accounts, release schedules, and sales reporting. The succeeding DFD levels further decompose these major data flows into their corresponding processes, data stores, and detailed system operations.

Data Flow Diagram (Level 1)

The Level 1 Data Flow Diagram (DFD) illustrates the internal structure and major functional components of the proposed Web-Based Service Management

System by breaking down the overall system into detailed processes, data flows, and data stores.

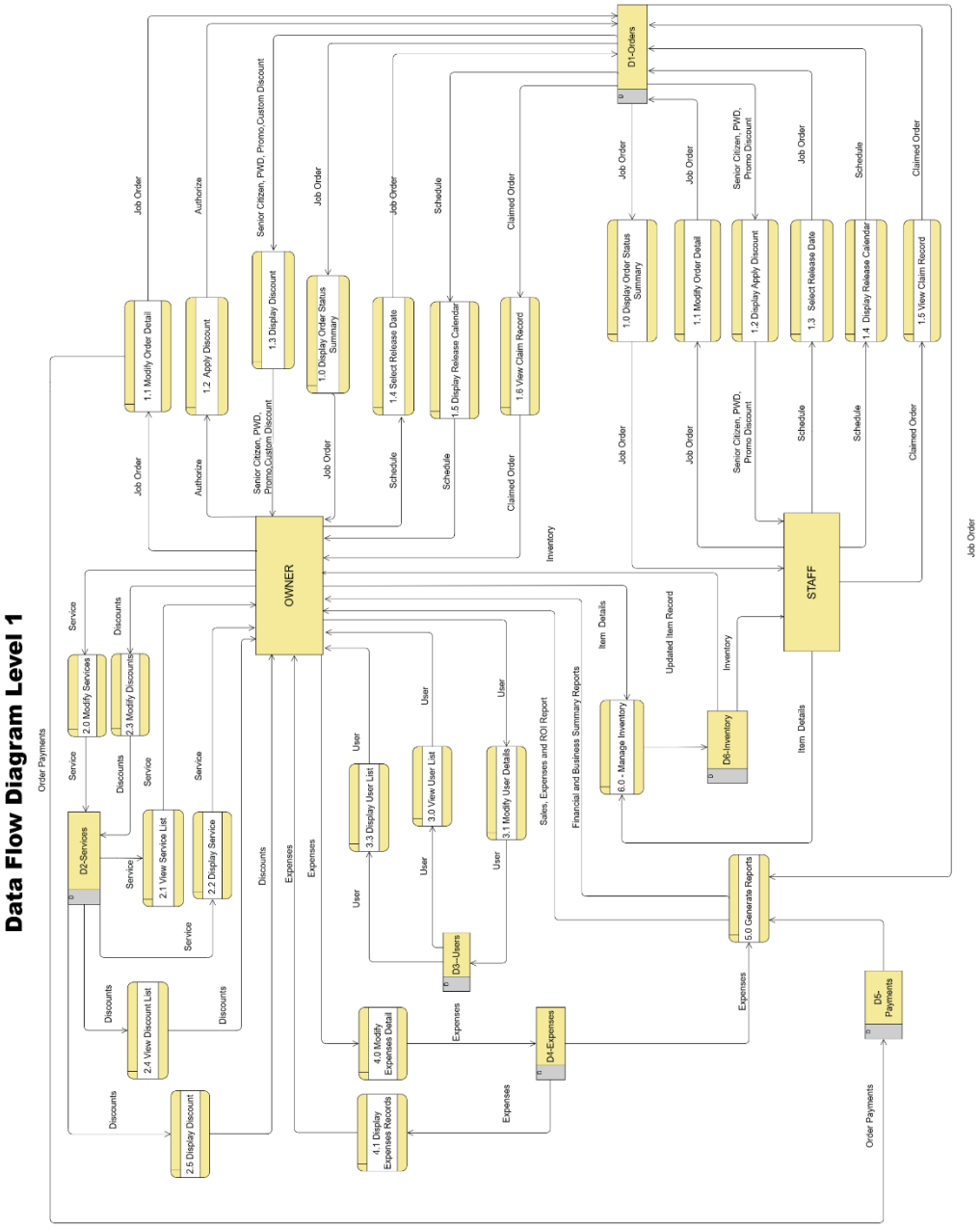


Figure 5: Data Flow Diagram Level 1

Figure 5 presents the Level 1 Data Flow Diagram (DFD) of the developed Web-Based Service Management System for Shoelotskey Villamor–Pasay. The diagram provides a detailed representation of how data flows between the system's two primary users, the Owner and Staff, the different system processes, and the corresponding data stores. It expands the major processes presented in the higher-level DFD by showing how operational information is entered, processed, retrieved, updated, and stored within the system.

The job order processes handle information related to customer service transactions. Both the Owner and Staff can access job order information according to their respective system permissions. These processes allow authorized users to view and modify job order details, apply applicable discounts, select release dates, view the release calendar, and access claim records. Job order information is stored in the Orders data store (D1), which provides the necessary records when users retrieve or update transaction information. Through these processes, the system supports the movement and monitoring of job orders throughout the service workflow.

The services and discounts processes allow the system to retrieve and maintain information concerning the services offered by the shop and the discounts that may be applied to transactions. Service information is maintained in the Services data store (D2). The Owner is provided with management functions for modifying service and discount information, while authorized users can retrieve the available information when processing job orders. This ensures that the service and discount information used in transactions corresponds to the configurations maintained within the system.

The user management processes enable the Owner to view and modify registered user information. User account information is retrieved from and stored

in the Users data store (D3). These processes support the administration of system users and help ensure that access to system functions is provided according to the assigned user role and permissions.

The system also handles expense management through processes that allow authorized expense information to be recorded, modified, and retrieved. Expense records are maintained in the Expenses data store (D4) and are subsequently used as part of the financial information required for report generation.

The report generation process consolidates relevant operational and financial data, including sales, payments, expenses, and other business information. The system processes these records to provide the Owner with financial and business summary reports, supporting the monitoring and evaluation of shop operations. Financial information intended for management is restricted to the Owner based on the system's role and permission controls.

Payment information generated from completed transactions is maintained in the Payments data store (D5). These records contribute to the system's sales and financial reporting functions and provide a digital record of payments associated with job orders.

The inventory management process handles information concerning the materials and supplies used in shop operations. Inventory information is stored in the Inventory data store (D6). The system retrieves item details and records authorized inventory updates to maintain current stock information. Inventory information can therefore be monitored alongside the shop's service operations.

Overall, the Level 1 DFD illustrates how the Owner and Staff interact with the system according to their authorized functions and how data moves between

system processes and data stores. It demonstrates the integration of job order processing, services and discounts, user management, expenses, payments, inventory, release scheduling, claims, and report generation within a centralized system. The terminology used in the diagram should remain consistent with the Context Diagram, other DFD levels, procedural flow, Entity Relationship Diagram, logical database design, module descriptions, and the final implemented system, thereby maintaining consistency among the system-analysis and design artifacts.

Procedural Workflow

The Procedural Workflow Diagram illustrates the step-by-step workflow of the Shoelotskey system, detailing how a customer service request is processed from start to finish.

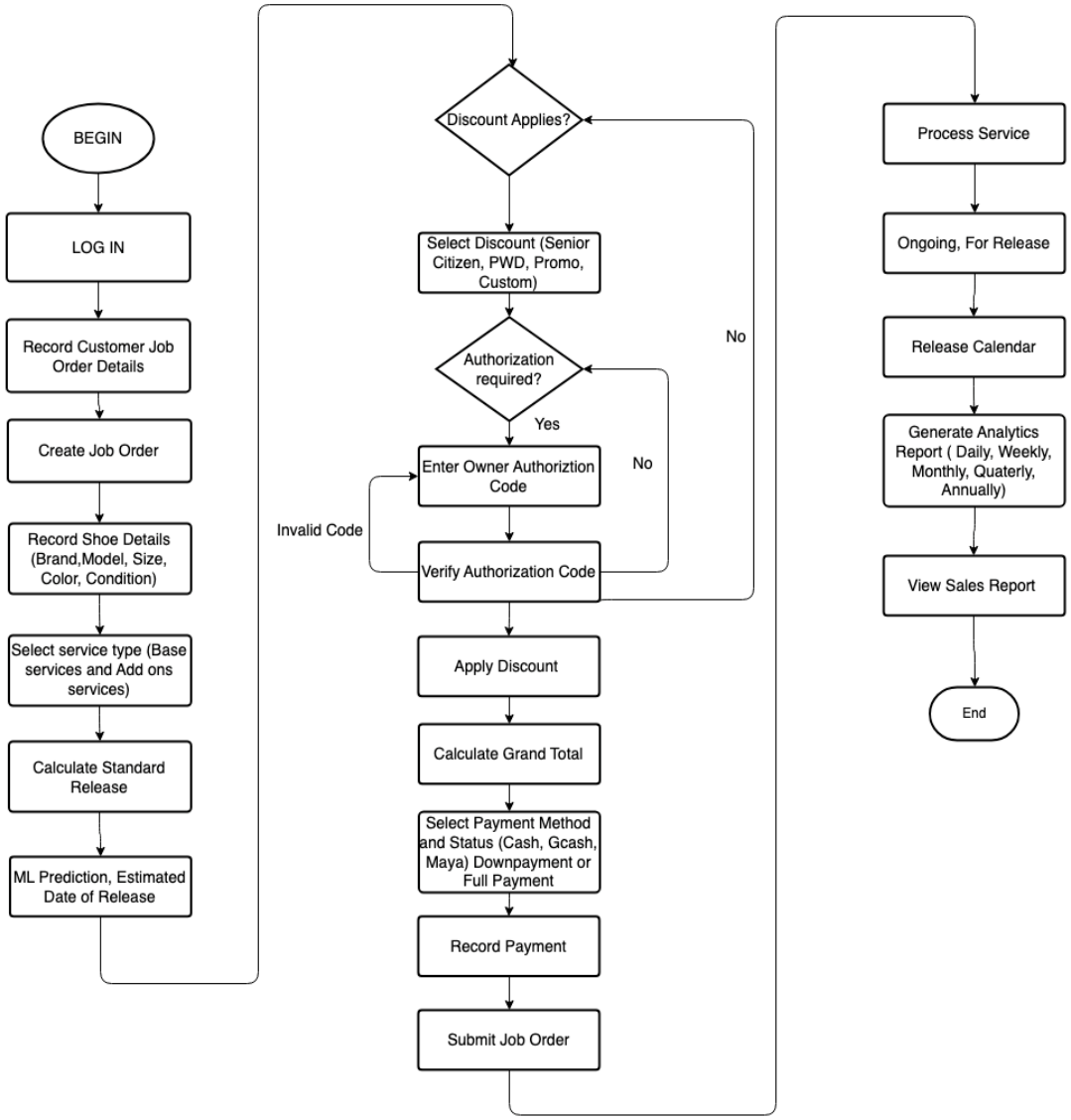


Figure 6: Procedural Workflow

The procedure begins when an authorized user logs in to the system. After successful authentication, the user records the customer's job order details and creates a new job order. The necessary shoe information, including the brand, model, size, color, and condition, is then recorded. The appropriate service type, consisting of base services and applicable add-on services, is selected based on the customer's requirements.

After the services are selected, the system calculates the standard release date based on the configured service duration. The system also generates an ML-predicted estimated release date, which serves as an advisory prediction of the expected service completion based on the information associated with the job order.

The workflow then determines whether a discount applies to the transaction. If no discount applies, the transaction proceeds to the calculation of the grand total. If a discount is applicable, the appropriate discount type, such as Senior Citizen, PWD, promotional, or custom discount, is selected. The system then determines whether the selected discount requires Owner authorization. When authorization is required, an Owner authorization code must be entered and verified. An invalid code prevents the discount from being applied and requires another authorization attempt. Once the authorization is successfully verified, the applicable discount is applied to the transaction.

After any applicable discount has been processed, the system calculates the grand total. The user then selects the customer's payment method and payment status, such as cash, GCash, or Maya, together with either downpayment or full-payment status. The payment information is recorded before the completed job order is submitted to the system.

Once submitted, the job order proceeds to service processing. Its status is updated as it moves through the operational workflow, including Ongoing and For Release. The corresponding release information is reflected in the Release Calendar, allowing authorized users to monitor orders that are expected or ready for release.

The information generated from completed system transactions is subsequently used for analytics and reporting. The system generates analytics reports according to selected periods, including daily, weekly, monthly, quarterly, and annual reports. The Owner can then view the Sales Report to monitor sales transactions and support business performance evaluation. The procedure ends after the relevant report has been accessed.

Overall, the procedural flow demonstrates how the system integrates job order creation, service selection, release-date calculation, machine-learning prediction, controlled discount application, payment recording, service processing, release scheduling, analytics, and sales reporting into a structured workflow.

Data Modeling and Database Design

Conceptual Data Model

The conceptual data model illustrates the key entities and relationships of the proposed Web-Based Service Management System with Data Analytics using Machine Learning Algorithms.

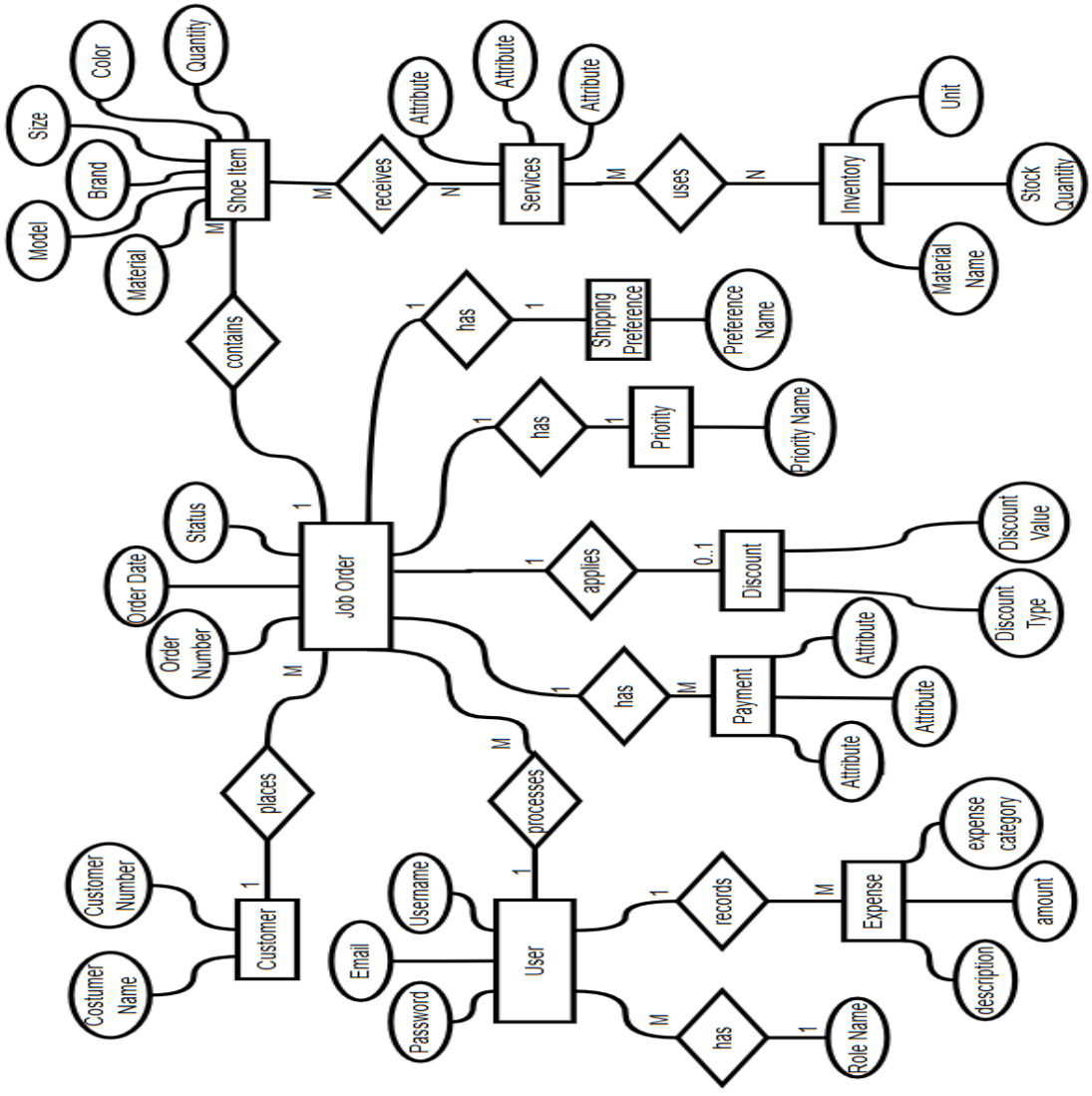


Figure 7: Conceptual Data Model

The Customer entity stores customer-related information, such as the customer's name and contact number. A customer places one or more Job Orders, while each job order is associated with a particular customer. The Job Order serves as the central transaction entity of the model and contains essential information such as the order number, order date, and status. It connects the customer transaction to other information required throughout the service process.

The User entity represents authorized users of the system, particularly the Owner and Staff. It contains account information such as username, password, and email and is associated with a role that determines the user's responsibilities and system access. Authorized users process Job Orders and may also record Expenses incurred during shop operations. The Expense entity contains information such as the expense description, amount, and expense category, allowing operational expenses to be documented for financial reporting purposes.

Each Job Order may contain one or more Shoe Items submitted by the customer. The Shoe Item entity describes the shoes associated with an order through attributes such as model, brand, size, color, material, and quantity. Each shoe item may receive one or more Services, representing the cleaning, restoration, reglue, color renewal, unyellowing, or other services requested for the customer's shoes.

The Services entity represents the service offerings available in the system. Services are also related to the Inventory entity because performing a particular service may require materials or supplies. The Inventory entity maintains information such as the material name, unit of measurement, and stock quantity. This relationship supports the monitoring of materials required in providing shoe services and helps maintain updated inventory information as materials are used during shop operations.

The Payment entity represents payment information associated with a Job Order. It allows the system to maintain a record of the financial transaction corresponding to a customer's service order. The Discount entity represents applicable discounts and contains information such as discount type and discount value. A Job Order may have an applicable discount, while an order without a discount can proceed without one, as represented by the optional relationship in the model.

The Priority entity identifies the processing priority assigned to a Job Order, while the Shipping Preference entity represents the customer's selected release or delivery-related preference when applicable. These entities provide additional information needed to manage how an order is processed and released.

Overall, the Conceptual Data Model shows that the Job Order functions as the central entity connecting the major operational components of the system. Through these relationships, the system can organize customer transactions, shoe items and requested services, payments and discounts, inventory requirements, user activities, and expenses in an integrated structure. The model serves as the conceptual foundation for the more detailed Entity Relationship Diagram and Logical Database Design, ensuring that the system's major data requirements and relationships are represented consistently.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) of the Shoelotskey system illustrates the structured relationships among the various entities that support the overall functionality of the shoe service management application.

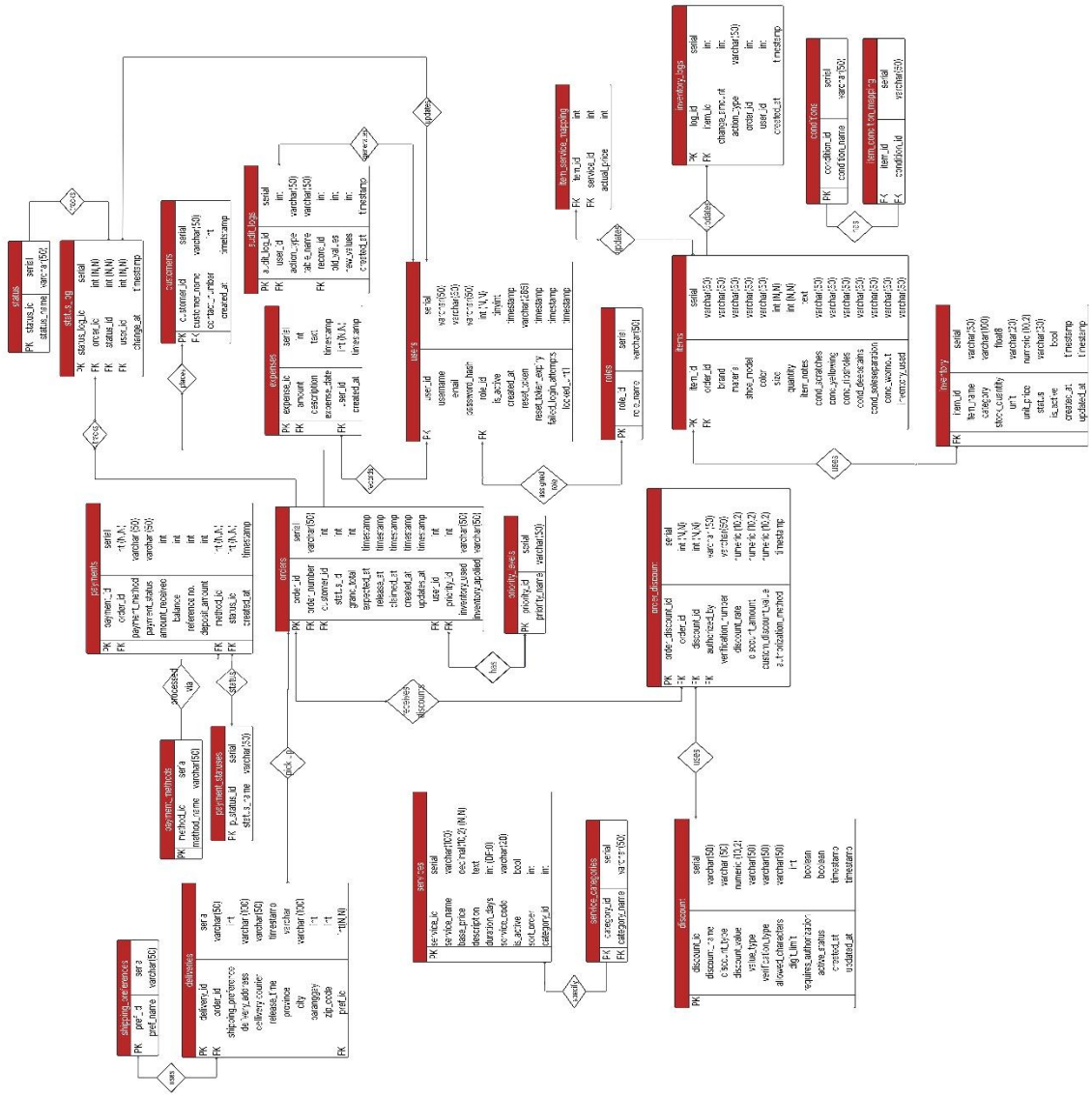


Figure 8: Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates how the data required by the Shoelotskey Service Management System is organized and related within the database. Each entity represents a major type of information maintained by the system, while primary keys (PK) uniquely identify individual records and foreign keys (FK) establish relationships between related entities. The design supports the system's major functions, including job order processing, service management, payment recording, discount application, inventory monitoring, user management, order-status tracking, delivery information, and audit logging.

At the center of the database is the Orders entity, which represents the primary transaction processed by the system. It stores essential job order information such as the order number, customer reference, current status, grand total, expected and actual release information, claimed date, assigned user, priority, and inventory-related information. Each order is associated with a record in Customers, allowing customer information to be maintained separately from individual transactions. The Status and Status Log entities support the monitoring of order progress and preserve the history of status changes as a job order moves through the service workflow.

Service-related information is maintained through the Services and Service Categories entities. The Services entity contains information such as the service name, base price, description, duration, service code, and active status, while Service Categories organizes the available services into their corresponding classifications. These records provide the service information needed when processing customer job orders.

The database also supports controlled discount management through the Discount and Order Discount entities. The Discount entity stores predefined discount configurations, including the discount name, type, value, verification

requirements, and whether authorization is required. The Order Discount entity associates an applied discount with a particular order and records information such as the authorizing user, verification details, discount rate or amount, custom discount value, and authorization method. This structure allows discounts applied to transactions to be documented and traced rather than directly altering the original discount configuration.

Payment transactions are represented by the Payments, Payment Methods, and Payment Statuses entities. Payments are associated with job orders and contain information such as the selected payment method, payment status, amount received, remaining balance, reference number, deposit amount, and creation date. Payment Methods and Payment Statuses provide standardized values for recording how a customer paid and the current state of the payment.

The Users and Roles entities support account and access management. User records contain account information and authentication-related data, while Roles identifies the role assigned to each account. This structure supports role-based access for the Owner and Staff and allows system functions to be restricted according to authorized responsibilities.

Inventory-related information is maintained through the Inventory, Inventory Logs, and Item Service Mapping entities. Inventory stores the materials and supplies available to the shop, including quantities, units, unit prices, status, and activity information. Inventory Logs maintains records of inventory movements or changes, including the amount changed, action performed, associated order, responsible user, and date of the transaction. The Item Service Mapping entity associates inventory items with particular services and records the expected or actual quantity required. Together, these entities support inventory monitoring and provide traceability for materials used during service operations.

The Items, Conditions, and Item Condition Mapping entities maintain information about customer shoes or items included in job orders. Item records contain identifying and descriptive characteristics such as brand, material, model, color, size, quantity, notes, and recorded shoe conditions. The Conditions entity standardizes the possible conditions that can be associated with an item, while Item Condition Mapping establishes the relationship between individual items and their identified conditions.

Additional operational information is maintained through the Expenses, Deliveries, and Shipping Preferences entities. Expenses records business expenditures and identifies the user responsible for recording them. Delivery information stores shipping and release-related details for orders requiring delivery, while Shipping Preferences provides standardized delivery preferences used by the corresponding delivery records.

Finally, the Audit Logs entity provides accountability and traceability by recording important activities performed within the system. Audit records identify the responsible user, type of action, affected table or record, previous and updated values, and the time the activity occurred. This supports the system's security and monitoring requirements by providing a historical record of significant changes.

Overall, the ERD demonstrates how the Shoelotskey Service Management System maintains integrated but logically separated data structures for customers, job orders, services, discounts, payments, users, inventory, expenses, deliveries, status history, and audit records. The relationships established through primary and foreign keys help maintain data integrity, traceability, and consistency across the system while supporting its operational, administrative, and reporting functions.

Logical Database Design

The logical database design of the Shoelotskey system presents a detailed and structured representation of how data is organized, related, and managed within the application to support its full range of operations.

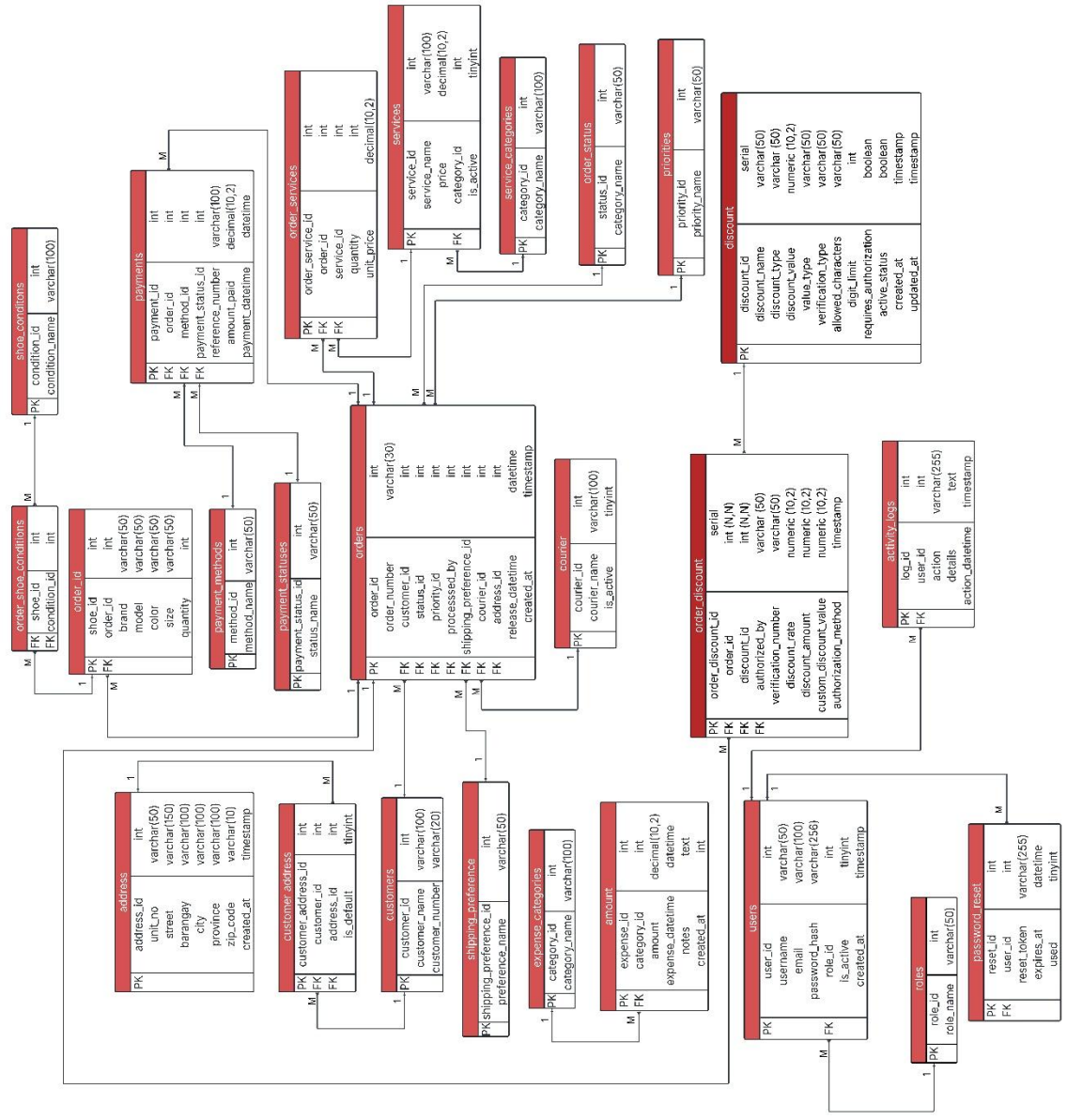


Figure 9: Logical Database Design

At the center of the design is the orders table, which serves as the primary transaction table for job orders. It contains the order number and references other tables through foreign keys, including the customer, order status, priority, user who processed the order, shipping preference, courier, and address. It also records the release date and time and the date when the order was created. Through these relationships, a job order can be associated with the information necessary for processing and monitoring the customer's service transaction.

The customer-related tables consist of customers, customer_address, and address. The customers table stores basic customer information, such as the customer's name and contact number. The customer_address table functions as an associative table connecting customers with their corresponding addresses, while the address table stores detailed location information such as unit number, street, barangay, city, province, and ZIP code. This structure separates customer information from address information and reduces unnecessary duplication of data.

Information regarding the shoes included in each job order is maintained through the order_id table shown in the diagram, which contains the shoe identifier, associated order, brand, model, color, size, and quantity. The order_shoe_conditions associative table connects individual shoe items with the shoe_conditions table, allowing one shoe item to have one or more recorded conditions. This enables the system to document the condition of the customer's shoes as part of the service transaction.

The system's service information is represented by the services, service_categories, and order_services tables. The services table stores the available services, their prices, categories, and active status. The service_categories table classifies the available services, while order_services

connect particular services to individual job orders. It also stores the quantity and unit price applicable to the transaction. This allows a single job order to contain multiple services while preserving the service price associated with the order.

The payment structure consists of the payments, payment_methods, and payment_statuses tables. The payments table associates a payment with a specific order and records the payment method, payment status, reference number, amount paid, and payment date and time. The separate payment-method and payment-status tables standardize the possible values used for payment transactions and support consistent financial records.

The discount structure is represented by the discount and order_discount tables. The discount table maintains the configured discount information, including the discount name, type, value, value type, verification requirements, authorization requirements, and active status. The order_discount table records the actual discount applied to a specific job order, including the discount rate, discount amount, custom discount value, verification information, and authorization information. Separating the configured discount from the applied order discount enables the system to maintain a record of which discount was used for each transaction and supports authorization and audit requirements.

Several reference tables support the job-order workflow. The order_status table stores the available order statuses used to monitor the progression of job orders. The priorities table identifies the applicable order priorities, while the shipping_preference and courier tables maintain information related to the customer's selected release or shipping arrangement. These reference tables help standardize values rather than repeatedly storing textual descriptions directly in every order record.

The user-management structure consists primarily of the users and roles tables. The users table stores system account information, including username, email, password hash, role, active status, and creation date. The roles table defines the roles assigned to users, such as Owner and Staff. The use of a hashed password rather than a plain-text password also reflects the system's implementation of account-security measures.

The password_reset table supports account recovery by maintaining reset tokens, their expiration information, and whether a token has already been used. Meanwhile, the activity_logs table records user actions and corresponding details and timestamps. This provides an audit trail that assists in monitoring important activities performed within the system.

Expense information is represented through the amount and expense_categories tables shown in the diagram. The expense record stores the expense amount, date and time, notes, and corresponding category, while expense_categories standardize the classifications assigned to expenses. These records provide the financial information required for expense monitoring and reporting.

Overall, the Logical Database Design demonstrates how the major information requirements of the system are converted into a relational database structure. Primary and foreign keys establish relationships among customers, job orders, shoe items, services, payments, discounts, users, expenses, and supporting reference data. The separation of information into related tables helps minimize unnecessary duplication, maintain referential integrity, and support reliable retrieval and updating of records. The logical database design therefore provides the structural basis for implementing the system's operational, financial, security, and audit-related functions.

System Security and Data Privacy Design

The System Security and Data Privacy Design of the proposed Web-Based Service Management System with Analytics and Machine Learning for Shoelotskey Villamor–Pasay aims to support the confidentiality, integrity, and appropriate handling of operational and personal data stored and processed within the system. Since the system manages business and customer information, including job orders, service transactions, customer records, and user credentials, security controls were incorporated to reduce risks associated with unauthorized access, modification, disclosure, and misuse of system data. These controls are treated as implemented safeguards rather than as evidence of complete security or regulatory compliance.

The system's security framework integrates multiple protection mechanisms, including JWT-based authentication, Bcrypt password hashing, role- and permission-based access control, input validation, account lockout and inactivity controls, audit logging, and backup procedures. These mechanisms are intended to restrict system functions and information according to authorized user access and to provide traceability of relevant system activities. The effectiveness of selected implemented controls is verified through documented security test cases that specify the test condition, expected result, actual result, and pass-or-fail outcome. Therefore, security claims in this study are limited to controls that were implemented and, where applicable, verified through testing.

The system does not claim complete information-security or data-privacy compliance. Advanced encryption at rest and a comprehensive regulatory compliance program are outside the implemented scope of the study. Instead, the system design considers applicable responsibilities under Republic Act No. 10173, or the Philippine Data Privacy Act of 2012, and its implementing rules, particularly

in relation to the lawful and legitimate processing of personal information, transparency through an appropriate privacy notice and consent where required, data minimization, appropriate retention and secure deletion or disposal, reasonable security measures, personal data breach management, and the exercise of applicable data-subject rights. Organizational implementation of these responsibilities remains the responsibility of Shoelotskey as the entity controlling the personal information processed through the system.

Only information necessary for legitimate service-management and related business purposes should be collected and processed through the system. Access to personal and operational information is restricted according to authorized roles and assigned permissions, while audit records provide accountability for relevant user activities. Personal information should not be retained longer than necessary for its declared purpose and should be securely deleted or disposed of according to an established retention policy. Appropriate procedures should also be established by Shoelotskey for responding to personal data breaches and requests from data subjects concerning their personal information.

Through these measures, the system provides technical safeguards and design considerations that support responsible security and privacy practices without representing that the system, by itself, guarantees complete protection or full legal compliance. The combination of implemented technical controls, documented security testing, and organizational privacy responsibilities provides a more appropriate basis for protecting the business and customer information processed through the platform.

Security functional testing was conducted separately from the system's use case testing to verify selected security controls implemented in the developed web-based system. The testing focused on authentication, authorization, password

protection, session management, server-side input validation, audit logging, error handling, transport protection, and backup and recovery. Selected negative test scenarios were also performed to determine whether the system appropriately rejected invalid or unauthorized requests. For each security test, the expected behavior was defined before execution and compared with the actual observed result to determine a Pass or Fail outcome. Testing was limited to the implemented controls and did not constitute a comprehensive penetration test, vulnerability assessment, or certification of compliance with the Philippine Data Privacy Act.

Table 15. System Security Functional Testing

ID	Security Control	Security Test Procedure	Expected Result	Actual Result	Status
ST -01	Authentication – Invalid Credentials	Attempt to log in using a registered account with an incorrect password.	Authentication is rejected and no authenticated session/token is issued.	Server returned **HTTP 401 Unauthorized** with an invalid-credentials response. No JWT token or authenticated session was issued.	**Pass**
ST -02	Authentication – Protected API	Send a request to a protected API endpoint without a JWT access token.	Server rejects the request and does not return protected data.	Request to the protected /api/orders endpoint without authentication returned **HTTP 401 Unauthorized**, and protected data were not returned.	**Pass**
ST -03	Authentication – Invalid/Expired JWT	Access a protected endpoint using an invalid or expired JWT token.	Server rejects the token and requires valid authentication.	An invalid-signature JWT was rejected with **HTTP 401 Unauthorized**, and access to the protected resource was denied.	**Pass**
ST -04	Account Lockout	Enter an incorrect password three consecutive times for the same account.	Account is temporarily locked for 15 minutes and subsequent authentication attempts are restricted according to the	After three consecutive failed attempts, the account was assigned a **15-minute lockout**; the subsequent attempt returned **HTTP 403 Forbidden**.	**Pass**

ID	Security Control	Security Test Procedure	Expected Result	Actual Result	Status
			configured policy.		
ST -05	Session Inactivity	Authenticate successfully, remain inactive for 30 minutes, and then attempt to access a protected function.	Inactive session is no longer accepted and reauthentication is required.	Server-side session tracking recorded the user's last_active timestamp and enforced the configured session timeout; expired authentication was rejected with **HTTP 401**.	**Pass**
ST -06	Password Hashing	Create/change a test account password and inspect the corresponding password field in the database using authorized administrative access.	Password is stored as a Bcrypt hash and the original plaintext password is not stored.	Database inspection confirmed password values stored using the **Bcrypt $2b$12$ hash format**; plaintext passwords were not stored in the password field.	**Pass**
ST -07	Role-Based Authorization	Authenticate as Staff and directly request an Owner-only endpoint/page.	Access is denied even when the restricted URL/API is requested directly.	A Staff request to the restricted /api/activities endpoint returned **HTTP 403 Forbidden** and did not expose the restricted resource.	**Pass**
ST -08	Permission-Based Authorization	Remove a specific permission from a test Staff account and attempt the corresponding protected action through the API.	Server prevents the action because the account lacks the required permission.	A Staff account with the applicable permission revoked received **HTTP 403 Access Denied** when the protected action was attempted.	**Pass**
ST -09	Server-Side Input Validation	Submit malformed, missing, or invalid values directly to an API endpoint,	Server rejects invalid input and does not save an invalid record.	Pydantic server-side validation rejected invalid payloads with **HTTP 422** field-level validation errors; invalid	**Pass**

ID	Security Control	Security Test Procedure	Expected Result	Actual Result	Status
		bypassing front-end validation.		records were not persisted.	
ST -10	SQL Injection Resistance	Submit non-destructive SQL-injection test strings into applicable inputs and authentication parameters.	Input does not alter query logic, bypass authentication, or expose unintended records.	SQL-injection test strings were handled as literal values through SQLAlchemy parameterized queries; no authentication bypass or unintended record exposure was observed.	**Pass**
ST -11	Cross-Site Scripting (XSS) Handling	Submit harmless script test strings into text fields that are subsequently displayed.	Stored/displayed input does not execute as JavaScript.	Script payloads were neutralized by application sanitization and React's contextual output escaping; the submitted scripts did not execute in the rendered interface.	**Pass**
ST -12	Unauthorized Record Modification	Authenticate as a user without modification permission and directly attempt to modify a protected record.	Server rejects the request and the record remains unchanged.	The unauthorized modification request returned **HTTP 403 Forbidden**, and database verification confirmed that the target record remained unchanged.	**Pass**
ST -13	Audit Logging	Perform defined auditable actions using a test account and inspect the audit log.	Audit trail records the event with identifying details such as user, action, and timestamp.	An audit record was created containing the actor, action type, affected record ID, timestamp, IP address, and recorded change details.	**Pass**
ST -14	Failed/Restricted Action Logging	Trigger a security-relevant event included within the implemented audit policy and inspect the audit trail.	Event is recorded when it falls within the configured audit-logging scope.	Failed-login, account-lockout, and forbidden-access events were recorded in the audit trail, including AUTH_LOGIN_FAILED, ACCOUNT_LOCKED, and 403_FORBIDDEN.	**Pass**

ID	Security Control	Security Test Procedure	Expected Result	Actual Result	Status
ST -15	Logout/Token Invalidation Behavior	Log out and attempt to reuse the previous client session/token to access a protected resource.	An invalidated session cannot continue accessing protected resources.	Logout marked the associated server-side session as **revoked (is_revoked=True)**; subsequent use of the revoked session token returned **HTTP 401**.	**Pass**
ST -16	Sensitive Error Handling	Submit malformed requests and request nonexistent or restricted resources.	Controlled errors are returned without exposing sensitive internal information.	Controlled JSON error responses were returned without exposing traceback information, server file paths, database credentials, or other sensitive implementation details.	**Pass**
ST -17	Backup and Restoration	Create a controlled database backup and restore it in an isolated test environment.	Backup can be restored and selected records can be retrieved accurately.	The backup archive passed SHA-256 integrity verification and was restored to an **isolated SQLite test instance**, with all tested relational tables and row counts recovered.	**Pass**
ST -18	HTTPS Transport Protection	Inspect the deployed production site's HTTPS/TLS transport configuration.	Production traffic is served through HTTPS/TLS and authenticated traffic is not transmitted through plaintext HTTP.	The report identified HTTPS/TLS enforcement in the Heroku production configuration and sslmode=require for the database connection.	**Pass**

As shown in the Table 15, The system security functional testing evaluated 18 selected security controls covering authentication, session management, password protection, authorization, input validation, injection handling, audit logging, backup and restoration, error handling, and transport protection. Based on the documented test results, all 18 security controls satisfied their defined expected outcomes under the specified test conditions. The tests demonstrated rejection of invalid and unauthenticated requests, enforcement of account lockout

and access restrictions, server-side validation of invalid inputs, protection against the tested non-destructive SQL injection and XSS inputs, recording of security-relevant events, revocation of server-side sessions following logout, and controlled handling of application errors. The backup-restoration test was conducted in an isolated SQLite test environment rather than through a destructive restoration of the production PostgreSQL database; therefore, its result is limited to the tested restoration environment.

Overall, the results provide technical evidence that the selected security controls operated as expected under the defined test conditions. However, the findings should not be interpreted as proof that the system is completely secure, free from vulnerabilities, formally OWASP-compliant, or formally certified under ISO/IEC 25010. Rather, the security functional testing supplements the perception-based assessment of the IT experts with observable evidence concerning specific implemented security controls.

Access Control and Authentication Mechanisms

The system implements secure authentication and role-based access control mechanisms to regulate user access to system functionalities. Authentication ensures that users must verify their identity before accessing the platform, thereby preventing unauthorized access to system resources.

Each user is required to log in to the system using a unique username and password assigned during account creation. The authentication process verifies the provided credentials against the stored user records within the system database. Only users with valid credentials are granted access to the system interface.

The system supports two primary user roles:

Owner– The owner functions as the primary system administrator and decision-maker. This role has full access to system modules, including dashboard monitoring, service management, job order review, report generation, and user account administration. The owner can also analyze business performance using data analytics generated by the system.

Staff– Staff users are responsible for managing day-to-day service operations. Their responsibilities include recording customer service requests, creating job orders, updating service status, and monitoring service completion schedules.

Role-based access control ensures that users can only access functions that correspond to their assigned responsibilities. This approach enhances system security while maintaining operational efficiency by preventing unauthorized users from modifying sensitive system information.

Data Encryption and Protection Measures

The proposed system implements security and data protection measures to safeguard sensitive operational information stored in the database and transmitted across network connections. For data transmission, encrypted communication protocols are used to protect information transmitted between users and the application server. These security mechanisms are designed to maintain the confidentiality, integrity, and availability of business data throughout the operation of the system. For stored data, protection is primarily provided through authentication, role-based access control, controlled database access, and regular

backup procedures. Advanced encryption mechanisms for data at rest are not included in the current implementation.

Sensitive operational data, including job orders, service records, inventory information, transaction history, and user credentials, are stored in a centralized relational database with controlled access permissions. Only authorized users and validated system processes are granted access to view, modify, or manage records. This helps prevent unauthorized access, accidental modification, and misuse of critical business information. In addition, the system implements regular backup and recovery procedures to protect data from accidental deletion, hardware failure, or security incidents. Periodic database backups ensure that records can be restored quickly in case of unexpected system disruptions.

To further strengthen security, the system follows OWASP Top 10 hardening practices that address common web application vulnerabilities. Broken access control is mitigated through JWT token validation and Role-Based Access Control (RBAC), ensuring that only authorized users can access specific system functions based on their roles. Cryptographic failures are prevented by securing passwords using Bcrypt hashing with a 12-round salt configuration. Injection attacks, such as SQL injection, are mitigated through the use of parameterized queries implemented via SQLAlchemy ORM.

The system also addresses insecure design by applying Third Normal Form (3NF) relational database constraints to preserve data consistency and integrity. Security misconfiguration risks are reduced through the use of hardened HTTP security headers, restricted CORS policies, and environment-based credential management. To minimize risks from vulnerable or outdated components, continuous dependency auditing is performed on major frameworks and libraries such as FastAPI, Pydantic, and React. Identification and authentication failures

are mitigated through multi-layer token validation and automated session expiration controls.

Moreover, the system protects software and data integrity through a real-time monitoring component developed in Node.js that detects unauthorized file modifications or suspicious changes within the environment. Logging and monitoring failures are addressed through centralized real-time logs of API requests, user actions, and data modifications for auditing and incident response purposes. Lastly, server-side request forgery (SSRF) attacks are mitigated by restricting and validating outgoing server-side requests to prevent unauthorized communication with external resources.

Through these integrated security measures, the proposed system ensures reliable protection of operational data while providing a secure and stable environment for managing the daily business processes of Shoelotskey.

Data Privacy and Ethical Considerations

The proposed system is designed to follow responsible data privacy and ethical data handling practices to protect the privacy of individuals whose information may be recorded during service transactions. Customer information and service records are treated as confidential data and are stored securely within the system database.

Access to operational data is restricted only to authorized users based on their assigned roles within the system. This restriction ensures that sensitive information cannot be accessed, shared, or modified by unauthorized individuals.

All data collected by the system is used exclusively for service management, operational monitoring, and business reporting purposes. The system does not

allow unauthorized disclosure or misuse of stored information. Ethical considerations in data management ensure that information is handled responsibly while supporting transparency and accountability in business operations. These privacy and ethical safeguards strengthen user trust in the system and ensure responsible management of business data within the Shoelotskey service environment.

Audit Logging and Activity Monitoring

To maintain transparency and accountability within system operations, the proposed platform incorporates audit logging and activity monitoring mechanisms. Audit logs record significant system activities performed by users and system processes.

The system logs events such as user authentication attempts, job order creation, service status updates, report generation, and user account modifications. Each log entry includes relevant details, such as the user responsible for the action, the activity performed, and the event timestamp.

Audit logs provide a reliable record of system usage and allow administrators to track operational activities within the platform. These logs can also be used to investigate potential security incidents or unusual system behavior.

Activity monitoring tools enable administrators to detect unauthorized access attempts, abnormal system activity, or potential misuse of system resources. By maintaining detailed activity records, the system improves operational transparency while strengthening system security.

Implementation Environment and Technology Stack

The implementation environment and technology stack describe the technological infrastructure used to develop, deploy, and operate the proposed web-based system. The system is built using modern web technologies that support efficient application development, secure data management, and reliable system performance.

The technology stack integrates server-side programming frameworks, relational database systems, and client-side interface technologies to enable the system to function as a fully web-based platform accessible through internet browsers. This architecture allows users to interact with the system through a graphical interface while the application server processes system logic and communicates with the database server to retrieve or store operational data.

Software Requirements

The software requirements specify the system components needed to develop and operate the web-based service management platform.

The system requires a web server platform capable of hosting the application and processing user requests. Server-side software handles system logic, data processing, and machine-learning analytics to analyze historical service records.

A relational database management system is required to store and manage system data, including job orders, service records, transaction details, and user account information. Users access the system through modern web browsers, which provide an interface for interacting with system modules such as dashboards, job order forms, service management pages, and sales reports.

These software components collectively provide the necessary environment for operating the web-based service management system efficiently and securely.

Hardware Requirements

The hardware requirements for the proposed system include computing devices capable of accessing web-based applications and supporting internet connectivity.

The owner and staff members will use desktop or laptop computers equipped with modern web browsers to interact with the system interface. These devices allow users to record service transactions, update job orders, and monitor business operations.

The system will be hosted on a server environment capable of handling application processing, database management, and network communication. The server infrastructure does not require specialized hardware, making the system suitable for small- and medium-scale business environments.

Stable internet connectivity is required to enable communication between users and the web server, ensuring smooth system access and data synchronization.

Development Tools and Programming Technologies

The Web-Based Service Management System for Shoelotskey is developed using a modern and reliable technology stack suited for high-performance web applications.

The frontend of the system is built using React with TypeScript to ensure structured and maintainable code. Vite is used to support fast development and efficient system performance, while Tailwind CSS provides a clean, responsive, and user-friendly interface design. Lucide React is integrated to maintain a consistent and professional set of icons across the system.

The backend is developed using Python with the FastAPI framework. This setup allows the system to handle multiple user requests efficiently while enforcing business rules and processing system logic. FastAPI also supports smooth integration with mathematical and analytical components.

For data storage, the system uses a PostgreSQL relational database. SQLAlchemy ORM is applied to manage database operations, ensuring data accuracy, consistency, and integrity throughout the system.

The system's predictive analytics feature is implemented using the Scikit-Learn library. A Random Forest regression model is used to estimate service completion timelines based on a simulated operational dataset representing verified Shoelotskey service configurations and operational conditions, helping improve operational planning and decision-making.

To ensure security, user passwords are protected using the Bcrypt hashing algorithm, preventing unauthorized access and enhancing overall system protection.

Overall, this technology stack ensures the system is secure, scalable, efficient, and suitable for real-world business operations, making it appropriate for a capstone-level implementation.

Deployment Environment

The proposed system is deployed within a secure, cloud-based web environment to ensure wide accessibility, reliability, and efficient system performance. As a web-based application, it enables authorized users such as the business owner and staff to access the platform through standard internet browsers using either the organization's local network or an internet connection. The deployment process involves installing the application on a secure web server and configuring a centralized database server to manage operational data effectively. This architecture supports seamless communication between system components while maintaining data integrity and accessibility.

For production, the system is hosted on the Heroku cloud platform, where both frontend and backend components are deployed using dedicated dynos to ensure stable and efficient execution. The application's database is managed Heroku Postgres, which provides high availability, automated backups, and secure database connections. Additionally, the system integrates the Mailgun API to handle notifications and administrative communications, including transactional emails such as password reset requests and system alerts.

To ensure data security and confidentiality, the entire system is protected by SSL/TLS encryption, safeguarding all data transmitted between users and the server from unauthorized access and potential tampering. The deployment environment also incorporates regular system monitoring, maintenance procedures, and software updates to maintain optimal performance and address potential vulnerabilities. Overall, this deployment strategy delivers a secure, reliable, and scalable platform that effectively supports service operations and informed decision-making for Shoelotskey, making it suitable for real-world implementation in a capstone-level project.

CHAPTER 4

Research Methodology, Results and Discussions

Research Design

This study used a developmental-descriptive research design. It focused on the design, development, and evaluation of a Web-Based Service Management System with Data Analytics using Machine Learning Algorithms. The developmental aspect was used to create the proposed system, while the descriptive aspect was applied to assess its functionality, usability, and relevance to the business operations. The study followed the Waterfall Software Development Life Cycle (SDLC), which includes requirements gathering, system design, development, testing, deployment, and maintenance.

Data Gathering Technique

The researchers gathered data through interviews, focus group discussions, direct observation, document review, system testing, and evaluation survey questionnaires. All data gathering activities were conducted with the permission of the business owner and with the voluntary participation of respondents. Participants were informed of the purpose of the study, and all collected information was used strictly for academic and system development purposes only. Confidential business records and personal information were treated with privacy and were not disclosed to unauthorized individuals.

Interviews were conducted with the business owner and co-owner, and four (4) staff members of Shoelotskey from March 5 to March 10, 2026 to identify current problems, operational challenges, and system requirements. Each

interview lasted approximately 20 to 30 minutes using a semi-structured interview guide. The responses were documented and analyzed to determine the features needed in the proposed system.

A focus group discussion involving the same six (6) participants was conducted on March 12, 2026 to allow respondents to collectively share their experiences, concerns, and suggestions regarding the existing manual process and the features needed in the proposed system. The discussion was facilitated by the researchers and recorded through written notes for analysis.

Direct observation was conducted from March 13 to March 15, 2026 during regular business hours to examine the actual workflow, service transactions, customer handling procedures, and record-keeping practices of the business. Observation helped validate the information gathered during the interviews and discussions.

Comparative performance testing was conducted to compare the existing manual paper-based process with the developed centralized digital system. The researchers measured the time required to retrieve selected customer and service records using the manual process and the developed system. The same or equivalent retrieval tasks were performed using both methods to provide comparable measurements. Data accuracy and completeness were also examined based on the selected records, while operational efficiency was assessed based on the time and number of steps required to complete the identified tasks.

Dashboard reporting performance was evaluated by comparing the time required to prepare selected reports using the existing manual process with the time required to generate equivalent reports using the developed analytics dashboard. The evaluation also considered the accessibility and usefulness of the

information provided by the dashboard for monitoring sales, service performance, and operational activities.

After the proposed system was developed, system testing was conducted from June 1 to June 10, 2026 with the participation of the owner and staff to verify functionality, usability, and accuracy of the developed modules. This included testing of login functions, transaction encoding, inventory updates, report generation, and prediction outputs.

Document review was applied to examine Shoelotskey's historical records. A total of 714 paper receipts from August 2025 to January 2026 were digitized into monthly CSV files for the machine learning component. Records with unreadable completion dates, incomplete fields, or inconsistent data were excluded through filtering and human validation. This resulted in 119 validated orders with verified Date Received and Claimed Date values for machine learning development and evaluation.

Prior to the administration of the Technology Acceptance Model (TAM) questionnaire, the developed system underwent a two-week pilot testing period from September 2 to September 16, 2026. During this period, the shop owner and staff used the system for actual and simulated operational activities, including creating job orders, monitoring service statuses, testing inventory functions using simulated data, and accessing system information based on their assigned roles. This provided the respondents with hands-on experience and familiarity with the system before its evaluation. After the pilot testing period, the TAM questionnaire was administered to assess Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Behavioral Intention (BI), and Actual System Use (ASU).

Finally, separate evaluation questionnaires were distributed to the intended evaluator groups from September 17 to September 19, 2026. The Shoelotskey owner, co-owner, and staff, consisting of six (6) respondents, evaluated the user acceptance of the developed system using the Technology Acceptance Model (TAM), while selected (5) IT experts evaluated the software quality of the system using the ISO/IEC 25010 criteria.

To address ethical considerations, the identities of respondents and confidential business information were kept private and were not disclosed in any part of the study. Participation was voluntary, and respondents were free to decline answering any question not related to the study. No fabricated or manipulated empirical respondent or historical business data were used. Historical records and evaluation responses were obtained from authorized Shoelotskey sources and actual respondents. A separate simulated operational dataset was generated specifically for machine-learning model development and evaluation and is explicitly identified as simulated throughout the study.

Research Instruments

The researchers used different research instruments to gather the information necessary for the development and evaluation of the proposed Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor–Pasay. These instruments consisted of an interview guide, observation guide, ISO/IEC 25010-based system evaluation questionnaire, and Technology Acceptance Model (TAM)-based user acceptance questionnaire.

The interview and observation instruments were used during the requirements gathering stage of the study. The interview guide was used to gather

information from the management and staff of Shoelotskey Villamor–Pasay regarding the existing service process, difficulties encountered in managing job orders and records, inventory practices, and other operational requirements. The observation guide was used to document the actual workflow of the establishment and to verify how service transactions, job orders, service monitoring, inventory activities, and release of completed orders were performed. The information gathered from these instruments served as a basis for identifying the requirements and designing the functions of the proposed system.

For the evaluation of the developed system, the researchers used a structured questionnaire based on the applicable software quality characteristics of ISO/IEC 25010. The instrument evaluated the system in terms of Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety. These criteria were evaluated by IT experts to determine the overall software quality of the developed system.

The researchers also incorporated the Technology Acceptance Model (TAM) to evaluate the acceptance of the system among its intended users. The TAM portion measured Perceived Usefulness, Perceived Ease of Use, Behavioral Intention to Use, and Actual System Use. The validation instrument specifically assessed whether these constructs were appropriately represented in the questionnaire.

Validation of the Research Instrument

Prior to their use and administration, the research instruments underwent expert content validation to establish their relevance, clarity, appropriateness, comprehensiveness, and alignment with the objectives of the study. The instruments subjected to validation consisted of the interview guide, observation

guide, ISO/IEC 25010-based system evaluation questionnaire, and Technology Acceptance Model (TAM)-based user acceptance questionnaire. Five validators with academic and professional backgrounds related to information systems and information technology reviewed the instruments.

The validators were Mr. Erickson A. Antonio, MIT, Director for Admission and Scholarship of the National Aviation Academy of the Philippines; Assoc. Prof. Bely B. Apostol, MEAM, IS Permanent Faculty of the Institute of Computer Studies; Ms. Ruth Jenneth L. Cacho, IT Faculty of the Institute of Computer Studies; Mr. Surewin Legaspi, IS Program Coordinator of the Institute of Computer Studies; and Ms. Hazel Villarica, IS Faculty Member of the Institute of Computer Studies. Their positions and academic or professional roles provided relevant expertise for examining the content and suitability of the research instruments.

The validators evaluated the research instruments using a four-point validation rating scale, where 4 represented Highly Acceptable, 3 represented Acceptable, 2 represented Less Acceptable, and 1 represented Not Acceptable. Fourteen validation criteria were used to examine the relevance of the research instruments to the study objectives, alignment with the functions of the developed system, adequacy and comprehensiveness of the items, coverage of ISO/IEC 25010 and TAM, appropriateness of terminology and language, organization and layout, clarity of wording, appropriateness of the response scale, and overall suitability of the instrument.

The Content Validity Index (CVI) was used to summarize the results of the expert validation. The Item-Level Content Validity Index (I-CVI) was determined based on the proportion of validators who rated each validation criterion as either 3 or 4, while the Scale-Level Content Validity Index/Average (S-CVI/Ave) represented the overall level of expert agreement across the validation criteria.

The results showed that all 14 validation criteria obtained an I-CVI of 1.00, indicating that all five validators rated every criterion either Acceptable or Highly

ITEM NO.	ITEM STATEMENT	Criteria (Rate 1-4)					I-CVI
		Validator 1	Validator 2	Validator 3	Validator 4	Validator 5	
1	The survey items are relevant to achieving the objectives of the study and the system's goal of improving service management operations for Shoelotskey Villamor-Pasay.	4	4	4	4	3	1
2	The survey items are aligned with the proposed system's core functions, including job order management, service monitoring, inventory management, data analytics, and machine learning-based service completion prediction.	4	4	4	4	3	1
3	The survey contains an appropriate number of items for evaluating the major features and functionalities of the Web-Based Service Management System.	4	4	4	4	3	1
4	The survey items are comprehensive and adequately cover the intended aspects of the developed system without being too narrow or too broad.	4	4	3	4	3	1
5	The survey covers the ISO/IEC 25010 software quality characteristics relevant to evaluating the developed system.	4	4	3	4	4	1
6	The survey includes appropriate items for measuring user acceptance based on the Technology Acceptance Model (TAM), including perceived usefulness, perceived ease of use, behavioral intention to use, and actual system use.	4	4	3	4	3	1
7	The survey items and response choices are designed to produce consistent, reliable, and measurable responses relevant to evaluating the Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay.	3	4	4	4	3	1
8	The terminology, language, and concepts used in the survey are appropriate, understandable, and relevant for Shoelotskey management, staff, and IT evaluators.	4	4	4	4	3	1
9	The layout and structure of the survey instrument are logically organized, visually clear, and technically sound.	4	4	4	4	3	1
10	The wording of the survey items is clear, precise, and understandable to the intended respondents.	4	4	4	4	3	1
11	The total number of survey items is appropriate and can be completed within the allotted time without causing respondent fatigue or disengagement.	4	4	3	4	3	1
12	The response scale provides an appropriate measure of the respondents' evaluation of the developed Web-Based Service Management System.	4	4	3	4	3	1
13	The survey instrument provides an accurate basis for evaluating the Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms in relation to the objectives of the study.	4	4	4	4	3	1
14	Overall, the survey instrument is suitable for evaluating the effectiveness, quality, and user acceptance of the Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay.	4	4	4	4	3	1
				S-CVI / AVERAGE			1

Your instrument demonstrates very strong content validity based on the expert ratings.

Items 1–14: I-CVI = 1.00, meaning all five validators rated these items 3 or 4. Overall S-CVI / Ave = 1.00, indicating a very high level of agreement regarding the relevance of the instrument's items.

Acceptable. The overall S-CVI/Ave was also 1.00, demonstrating a very high level of agreement among the validators regarding the relevance and suitability of the research instruments.

Figure 11: Content Validation Results of the Research Instrument

Revisions Based on Expert Validation

The researchers reviewed the questionnaire based on the ratings and recommendations provided during expert validation. Most of the validation forms contained no additional written recommendations and indicated that the questionnaire was acceptable for use. One validator, Ms. Hazel Villarica, provided specific recommendations concerning the technical wording and content of the instrument. These recommendations emphasized the use of appropriate IT terminology, ensuring that technical-expert items were understandable to IT professionals, clearly explaining the function of machine learning within the system, and ensuring that the items could generate technically relevant feedback that would support system decision-making and improvement.

Based on the expert validation results and recommendations, the researchers incorporated the applicable revisions before administering the final questionnaire. The completed validation established that the instrument was considered appropriate, relevant, aligned with the objectives of the study, clear, organized, and suitable for collecting the necessary evaluation data.

Reliability of the Research Instrument

The researchers did not conduct a separate statistical reliability test using Cronbach's alpha for the present study. The system evaluation was conducted within the specific operational setting of Shoelotskey Villamor–Pasay using purposively selected respondents based on their direct involvement with the developed system or their relevant information technology expertise. The ISO/IEC 25010-based questionnaire was administered to five IT experts, while the TAM-based questionnaire was administered to six intended users of the system.

Cronbach's alpha is commonly used to assess the internal consistency of multiple questionnaire items measuring the same construct. However, given the limited and specialized respondent groups in the present study, a reliability coefficient derived from five IT experts and six intended users would require substantial caution in interpretation. Instead, prior to administration, the research instruments underwent expert content validation by five qualified validators who assessed their relevance, clarity, appropriateness, comprehensiveness, terminology, organization, alignment with the study objectives, and coverage of the applicable ISO/IEC 25010 and TAM constructs. The validation resulted in an I-CVI of 1.00 for all 14 validation criteria and an overall S-CVI/Ave of 1.00.

Accordingly, the study establishes content validity through expert validation but does not claim that internal consistency reliability was statistically established through Cronbach's alpha. The absence of a separate reliability analysis is acknowledged as a limitation of the study. Future studies involving larger respondent samples may conduct Cronbach's alpha or another appropriate reliability analysis to further assess the internal consistency of the multi-item constructs.

Statistical Treatment of Data

The data gathered in the study were analyzed using descriptive statistics and appropriate evaluation measures based on the objectives of the study. For the comparative analysis of the manual paper-based process and the developed digital system, the researchers used measured task completion time, record retrieval time, data accuracy, and operational efficiency as comparison indicators. The mean or average time required to complete the identified tasks was computed

for each process. Percentage improvement or reduction in time was also determined to describe the difference between the two processes.

For the machine learning component, the Random Forest Regression model was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2) to determine its predictive performance. For the real-time analytics dashboard, reporting efficiency was assessed by comparing the time required to prepare selected reports using the manual process and the time required to generate equivalent reports using the developed dashboard. The accessibility and usefulness of dashboard information were also summarized based on the evaluation responses. For the ISO/IEC 25010 and TAM evaluation questionnaires, weighted mean was used to summarize the responses of the IT experts and Shoelotskey personnel, respectively. The resulting means were interpreted using the established four-point Likert scale interpretation adopted in the study.

Formula:

Weighted Mean

$$WM = \frac{\Sigma fx}{N}$$

Where:

WM = Weighted Mean
f = Frequency of responses
x = Weight assigned to each response
N = Total number of respondents

Percentage Reduction

$$\%\boldsymbol{Reduction} = \frac{\boldsymbol{Manual\ Time - Digital\ Time}}{\boldsymbol{Manual\ Time}} \times \mathbf{100}$$

Where:

Manual Time = average time using the existing manual process
Digital Time = average time using the developed system

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Where:

MAE = Mean Absolute Error
n = Number of observations
y_i = Actual value
$\hat{y}_i$= Predicted value

Interpretation: Lower MAE indicates that the predictions are, on average, closer to the actual completion times.

Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where:

RMSE = Root Mean Square Error
n = Number of observations
y_i = Actual value
$\hat{y}_i$= Predicted value

Interpretation: Lower RMSE indicates smaller prediction errors, with larger errors receiving greater weight.

Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Where:

R^2 = Coefficient of Determination
$\boldsymbol{y_i}$ = Actual value
$\widehat{y_i}$= Predicted value
$\overline{y}$ = Mean of the actual values
n = Number of observations

Interpretation: A higher R^2 indicates that the model explains a greater proportion of the variation in service completion time.

Frequency and Percentage Distribution

$$P = \frac{f}{N} \times 100$$

Where:

- **P** = Percentage
- **f** = Frequency of a particular comment or suggestion
- **N** = Total number of respondents in the corresponding evaluator group
- **100** = Percentage constant

Objective Technical Test Analysis

Objective technical test results were analyzed separately from the Likert-scale expert evaluation. For tests producing dichotomous outcomes, each trial was classified as either Pass or Fail according to a predefined acceptance criterion. The pass rate was calculated using:

Pass Rate (%) = (Number of Passed Trials / Total Number of Trials) × 100

For quantitative performance measures such as response time, the mean was used to represent the average observed performance, while the standard deviation was used to describe variation across repeated trials. Where appropriate, the minimum and maximum observed values were also reported.

No Likert interpretation such as Acceptable or Highly Acceptable was applied to objective test measurements. Instead, technical test results were evaluated directly against their predefined acceptance criteria.

Population and Sampling Technique

The respondents of this study consisted of individuals directly involved in the operation and evaluation of the proposed system for Shoelotskey. The target population for the internal evaluation consisted of six (6) Shoelotskey personnel, specifically the business owner, co-owner, and four (4) staff who are directly involved in customer transactions, service processing, shoe cleaning and restoration services, and inventory-related activities. These respondents are considered essential because they are the intended end-users of the developed web-based service management system and can provide firsthand feedback regarding its usability, functionality, and overall acceptance.

In addition to the internal respondents, the study included selected Information Technology (IT) experts or Information Technology (IT) practitioners to evaluate the technical quality of the system based on ISO/IEC 25010 standards. These experts may include IT instructors, software developers, system analysts, programmers, or professionals with experience in software evaluation and system development. Their participation is necessary to assess the system in terms of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety.

The study used total population sampling for the six (6) Shoelotskey personnel because the population is small and manageable. This allows all employees and management personnel to participate in evaluating the system.

Purposive sampling was used to select qualified IT professionals with relevant experience in software development, system analysis, programming, or software quality evaluation. By combining operational users and technical evaluators, the study ensures a comprehensive assessment of the proposed system from both user and expert perspectives. The internal respondents will assess user acceptance using the Technology Acceptance Model (TAM), while the IT experts, will evaluate the software quality using ISO/IEC 25010 criteria.

Respondents of the Study

The respondents of the study were divided into two groups: the intended users of the developed system and selected Information Technology (IT) experts. The intended users consisted of the owner, co-owner, and staff of Shoelotskey Villamor–Pasay, while the IT experts were selected to evaluate the software quality of the developed system.

Shoelotskey Villamor–Pasay is a small-scale establishment with a limited number of personnel. The establishment consists of two (2) owners and four (4) staff members, for a total of six (6) personnel who are directly involved in the operations of the shop. Since the population of intended users was relatively small, the researchers employed total enumeration, wherein all qualified members of the establishment were included as respondents for the Technology Acceptance Model (TAM) evaluation.

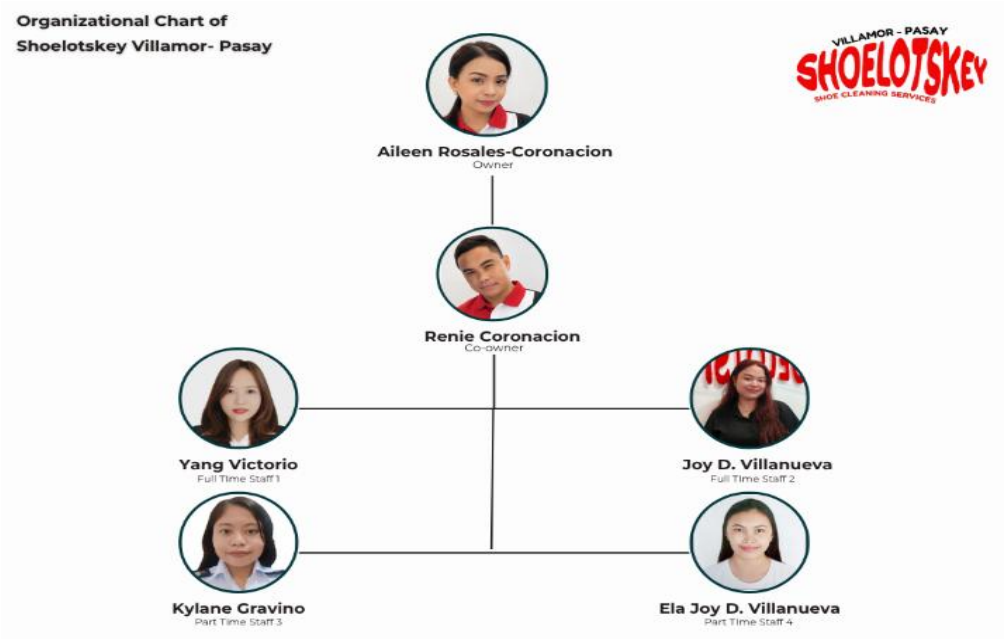


Figure 12: Organizational Chart of Shoelotskey Villamor–Pasay

Figure 12 presents the organizational structure of Shoelotskey Villamor–Pasay. The structure illustrates the relatively small workforce of the establishment and identifies the personnel considered as the intended users of the developed system. All two (2) owners and four (4) staff members were included in the evaluation because they represent the personnel directly involved in the shop's operations and are the intended users of the system. Therefore, the six (6) TAM respondents represent the actual population of qualified users within the establishment rather than a sample drawn from a larger employee population.

In addition to the intended users, five (5) IT experts were purposively selected to evaluate the developed system based on the ISO/IEC 25010 software quality characteristics. The IT experts were selected separately from the shop personnel based on the qualifications established by the researchers, including relevant professional experience and knowledge in information technology and

software systems. Therefore, the study involved a total of eleven (11) respondents: five (5) IT experts and six (6) intended system users.

4.1 Demographic Profile of the Respondents:

Table 16

FREQUENCY AND PERCENTAGE DISTRIBUTION OF RESPONDENTS ACCORDING TO CATEGORY

Category	Frequency	Percentage
IT Expert	5	45.45%
Shop Owner	2	18.18%
Shop Staff	4	36.36%
TOTAL	11	100%

Table 16 shows the frequency and percentage distribution of respondents according to category.

There are 5 or 45.45% respondents are IT Experts. There are 4 or 36.36% are Shop Staff while there are only 2 or 18.18% are Shop Owner.

Findings revealed that the most of the respondents are IT Expert with a frequency of 5 or 45.45%.

4.2 Comparative Analysis of Manual and Developed System

The comparative analysis evaluated the existing manual paper-based process against the developed centralized digital system. The comparison focused on service record accessibility, retrieval time, and data accuracy. These indicators were selected because they directly correspond to the operational challenges identified in Chapter 1 and the first specific objective of the study.

Record Retrieval Testing Procedure

The researchers conducted a comparative record retrieval test to determine the difference in retrieval time between the existing manual process and the developed Web-Based Service Management System. A total of 10 paired retrieval trials were conducted by the researchers, Charmaine Rose G. Angeles and Kylane C. Gravino. Both researchers jointly conducted all 10 trials using the same established testing procedure.

For each trial, a customer/service record was selected for retrieval, and the same selected record was used for both the manual and digital methods to ensure direct comparability. The manual retrieval was performed first, followed by the digital retrieval of the same selected customer/service record. For the manual method, the researchers searched for the selected record from the existing paper-based records. For the digital method, the same record was searched and retrieved using the developed system.

For the digital retrieval trials, the developed system was accessed using an ASUS Vivobook 15 equipped with an AMD Ryzen 5 7520U processor and 16 GB of RAM. The laptop was connected to a PLDT Wi-Fi connection with a subscribed speed of 50 Mbps. Retrieval time for both the manual and digital methods was measured using a smartphone stopwatch.

Timing began when the researchers started searching for the selected customer/service record. For the manual method, timing ended when the complete requested record was located from the paper-based records. For the digital method, timing ended when the corresponding record was successfully located and displayed in the developed system. The same timing procedure was consistently applied throughout all 10 paired trials. These testing conditions

provided a consistent basis for comparing the record retrieval time of the existing manual process and the developed system.

Table 17

Comparative Analysis of Record Retrieval Time (Manual vs. Digital)

Trial	Record Used	Manual Time (min)	Digital Time (min)	Paired Difference (min)
1	ORD-2026-09-16-001	3	0.5	2.50
2	ORD-2026-09-15-001	4	0.4	3.60
3	ORD-2026-09-15-002	5	0.3	4.70
4	ORD-2026-09-18-002	5	0.3	4.70
5	ORD-2026-09-08-001	4	0.5	3.50
6	ORD-2026-08-17-001	5	0.4	4.60
7	ORD-2026-09-22-001	10	0.5	9.50
8	ORD-2026-09-09-001	3	0.4	2.60
9	ORD-2026-09-22-002	6	0.3	5.70
10	ORD-2026-09-22-005	7	0.5	6.50
Mean	-	**5.20**	**0.41**	**4.79**
Median	-	**5.00**	**0.40**	**4.65**
SD	-	**2.10**	**0.09**	**2.08**
Range	-	**3.00-10.00**	**0.30-0.50**	**2.50-9.50**

Percentage Reduction:

$$\%Reduction = \frac{5.20 - 0.41}{5.20} \times 100 = 92.12\%$$

As shown in Table 17, the manual process recorded a mean retrieval time of 5.20 minutes (SD = 2.10) and a median of 5.00 minutes, with retrieval times ranging from 3.00 to 10.00 minutes. In comparison, the developed system recorded a mean retrieval time of 0.41 minutes (SD = 0.09) and a median of 0.40 minutes, with retrieval times ranging from 0.30 to 0.50 minutes.

The mean paired difference between the two methods was 4.79 minutes, indicating that the developed system required less time to retrieve the same selected customer/service records during the conducted trials. The overall reduction in mean retrieval time was 92.12%. All 10 paired trials showed shorter retrieval times using the developed system, with observed time savings ranging from 2.50 to 9.50 minutes.

The largest absolute difference was observed in Trial 7, where retrieval time decreased from 10.00 minutes using the manual process to 0.50 minutes using the developed system, representing a difference of 9.50 minutes. The smallest difference was observed in Trial 1, where retrieval time decreased from 3.00 minutes to 0.50 minutes, representing a difference of 2.50 minutes.

These findings indicate that, under the conditions of the conducted retrieval test, the developed system provided faster retrieval of the selected customer/service records compared with the existing manual process. Since the comparison was based on 10 paired retrieval trials, the findings are interpreted specifically within the conditions and records included in the test.

Table 18

COMPARATIVE ANALYSIS OF DATA ACCURACY (MANUAL VS. DIGITAL)

The data accuracy assessment examined duplicate entries, incomplete records, records requiring correction, and data-validation mechanisms in the

existing manual process and the developed system. For the manual baseline, the available historical paper records consisted of 119 audited receipts from January 2024 to April 2025. For the developed system, the formal two-week pilot period covered September 2 to September 16, 2026, during which 31 digital job orders were examined. Because the manual and digital records originated from different observation periods and did not provide equivalent denominators for every indicator, the results were analyzed descriptively rather than as a matched statistical comparison.

Data Accuracy Indicator	Manual Process	Developed System – Formal Pilot (N = 31)
Records Examined	**119 historical paper receipts**	**31 digital job orders**
Duplicate Entries	**2–3 per day***	**0 of 31 (0.00%)**
Incomplete Records	**50 of 119 (42.0%)** missing claim/completion dates	**0 of 31 (0.00%)**
Records Requiring Correction	**2–3 per day***	**3 of 31 (9.68%); 0.20/day**
Data Validation	Manual checking	Automated field validation

**The available manual evidence for duplicate entries and record corrections was documented as daily observational frequency rather than transaction-level counts; therefore, exact manual percentages for these indicators could not be computed.*

As shown in Table 18, the historical manual-record audit examined 119 paper receipts, of which 50 (42.0%) lacked claim/completion dates, providing an exact measure of missing information in the available manual records. The available manual observations also documented approximately 2–3 duplicate entries and 2–3 record corrections per day; however, exact transaction-level denominators were not available for these two indicators.

For the developed system, 31 digital job orders processed during the formal two-week pilot period were examined. The audit identified 0 duplicate records (0.00%) and 0 incomplete records (0.00%). Three records required post-entry

correction, corresponding to 9.68% of the 31 transactions and an average of 0.20 corrections per day.

The audit applied predefined criteria to distinguish actual corrections from normal system operations. Routine status transitions such as New Order → On-Going → For Release → Claimed were not classified as corrections. Only post-entry modifications made to correct erroneous transaction information were counted.

These findings indicate that the developed system maintained complete and non-duplicate digital records during the formal pilot period and incorporated automated validation controls. However, because the manual and digital evidence came from different observation periods and equivalent transaction-level denominators were unavailable for all manual indicators, the findings are interpreted as a descriptive comparison of the available evidence rather than a statistically matched comparison of error rates.

Table 19

SUMMARY OF COMPARATIVE ANALYSIS (MANUAL VS. DIGITAL)

Evaluation Area	Manual System	Developed System	Result
Record Accessibility	Physical paper-based records requiring manual searching	Centralized digital records accessible through the system	Improved accessibility
Mean Record Retrieval Time	**5.20 min**	**0.41 min**	**92.12% reduction**
Median Record Retrieval Time	**5.00 min**	**0.40 min**	Lower observed retrieval time

Evaluation Area	Manual System	Developed System	Result
Retrieval Time Range	**3.00–10.00 min**	**0.30–0.50 min**	Narrower observed retrieval range
Duplicate Records	**2–3 per day***	**0 of 31 (0.00%)**	No duplicates detected during digital pilot
Incomplete Records	**50 of 119 (42.0%)** missing claim/completion dates	**0 of 31 (0.00%)**	No incomplete records detected during digital pilot
Records Requiring Correction	**2–3 per day***	**3 of 31 (9.68%); 0.20/day**	Lower observed daily correction frequency
Data Validation	Manual checking	Automated field validation	Automated validation implemented
Service Status Monitoring	Manual checking of paper records	Real-time digital status monitoring	Centralized digital monitoring

**Manual duplicate and correction figures were recorded as daily observational frequencies; exact transaction-level denominators were unavailable for these indicators.*

As shown in Table 19, the developed system demonstrated improvements in service-record accessibility and retrieval performance compared with the existing manual process. Based on the 10 paired record-retrieval trials, the mean retrieval time decreased from 5.20 minutes to 0.41 minutes, representing a 92.12% reduction. The median retrieval time decreased from 5.00 minutes to 0.40 minutes, while the observed range decreased from 3.00–10.00 minutes to 0.30–0.50 minutes.

For data accuracy, the historical manual-record audit identified 50 of 119 paper receipts (42.0%) with missing claim/completion dates. The available manual observations additionally documented duplicate entries and records requiring correction as daily frequency ranges. In comparison, analysis of the 31 digital job

orders from the formal September 2–16, 2026 pilot period identified 0 duplicate records (0.00%), 0 incomplete records (0.00%), and 3 records requiring correction (9.68%), equivalent to 0.20 corrections per day.

Overall, the findings indicate that the developed system provided faster record retrieval, centralized access to service records, automated field validation, and complete and non-duplicate digital transaction records during the formal pilot period. Nevertheless, because the available manual and digital data were collected from different periods and equivalent denominators were unavailable for some manual accuracy indicators, the data-accuracy results are interpreted descriptively and do not represent a matched statistical comparison.

.4.3 Machine Learning Model Evaluation

The machine learning component was evaluated in two distinct stages corresponding to the two datasets described in the Machine Learning Development section. Dataset A consisted of 119 validated historical service orders and was used for the empirical assessment of Order-to-Claim Turnaround Days, while Dataset B consisted of 10,000 simulated operational service orders and was used for the development and evaluation of the finalized Random Forest model for predicting Received-to-Ready-for-Release Duration. The two evaluation stages are presented separately because the datasets used different target definitions and served different research purposes.

Historical Evaluation – Dataset A

Dataset A was evaluated using 93 development records and a chronologically separate 26-record temporal holdout set. As described in the Machine Learning Development section, five-fold expanding-window cross-validation was conducted using only the 93 development records. The fold MAEs

were 7.97, 5.52, 8.34, 11.41, and 7.54 days, resulting in a mean cross-validation MAE of 8.1541 days, an aggregate RMSE of 10.3225 days, and an R^2 of 0.2679. The 26-record temporal holdout was not included in model development or cross-validation and was used only for final historical evaluation.

The historical Random Forest used the pre-fixed configuration of 200 trees, unrestricted maximum depth, max_features='sqrt', and random_state=42. The Dataset B hyperparameter search was not applied to this historical experiment.

Table 20

BASELINE AND MODEL PERFORMANCE ON HISTORICAL DATASET A

Model / Benchmark	MAE (Days)	RMSE (Days)	R^2	±1 Day	±2 Days	±3 Days	±5 Days
Mean-Duration Baseline	9.34	10.13	−0.0006	3.9%	3.9%	7.7%	11.5%
Median-Duration Baseline	8.96	10.36	−0.0469	7.7%	7.7%	15.4%	34.6%
Shoelotskey Business Rules	**6.62**	**8.45**	**0.3033**	**23.1%**	**30.8%**	**38.5%**	**42.3%**
Linear Regression	18.50	20.51	−3.1065	0.0%	0.0%	0.0%	7.7%
Decision Tree Regressor	10.23	13.29	−0.7237	19.2%	23.1%	26.9%	42.3%
Random Forest Regressor	8.35	10.19	−0.0125	11.5%	23.1%	30.8%	42.3%

As shown in Table 20, the historical Random Forest obtained an MAE of 8.35 days, RMSE of 10.19 days, and R^2 of −0.0125. Although its MAE was lower than those of the mean-duration and median-duration baselines, it did not outperform Shoelotskey's existing business-rule estimate, which obtained an MAE

of 6.62 days and RMSE of 8.45 days. The negative R^2 is not interpreted as "negative accuracy"; rather, it indicates that the Random Forest did not explain variation in the unseen historical Order-to-Claim target better than the relevant reference prediction.

Tolerance analysis showed that 11.5% of the historical Random Forest predictions were within ±1 day, 23.1% within ±2 days, 30.8% within ±3 days, and 42.3% within ±5 days. These results indicate that the historical Random Forest was not sufficiently reliable to replace the existing business-rule estimate for operational release-date determination.

Limitation of the Historical Target

The historical evaluation must also be interpreted in relation to the target available in Dataset A. The paper-based records contained Date Received and Claimed Date but did not contain a reliable actual Ready-for-Release timestamp. Consequently:

Order-to-Claim Turnaround Days = Claimed Date − Date Received

The resulting duration may contain both actual service-processing time and the period between service completion and customer collection. Therefore, the historical Random Forest was partly attempting to predict customer pickup behavior, which is outside Shoelotskey's operational control.

Dataset A was retained as empirical evidence of this limitation rather than discarded. The historical findings consequently informed the distinction between Order-to-Claim Turnaround and the operational Received-to-Ready-for-Release Duration used in Dataset B. Your manuscript already establishes this target limitation in the Machine Learning Development section.

Finalized Operational Model – Dataset B

As detailed in the Machine Learning Development section, Dataset B contained 10,000 simulated operational service orders divided into 7,000 training, 1,500 validation, and 1,500 final test records. The final test set remained separate from model training, hyperparameter development, and model selection and was used only after the selected model had been finalized.

Linear Regression, Decision Tree Regressor, and Random Forest Regressor were compared using the validation partition. Linear Regression obtained an MAE of 2.4764 days, Decision Tree obtained 0.0616 days, and Random Forest obtained 0.0641 days. Although Decision Tree produced the slightly lower MAE, Random Forest obtained the lower RMSE (0.2994 days), higher R^2 (0.9987), and higher ±1-day accuracy (97.73%), supporting its selection as the finalized operational model.

The hyperparameter search and complete search space are presented in the Machine Learning Development section. The selected Random Forest configuration consisted of 300 trees, maximum depth of 12, minimum samples split of 10, minimum samples leaf of 4, and max_features=1.0, with bootstrap=True, criterion='squared_error', and random_state=20261002. Your existing methodology documents both the search space and the distinction from Dataset A's fixed configuration.

Table 21

FINAL RANDOM FOREST PERFORMANCE ON DATASET B

Evaluation Metric	Result
Final Test Records	**1,500**
MAE	**0.0815 days**

Evaluation Metric	Result
RMSE	**0.4042 days**
R^2	**0.9976**
Within ±1 Day	**1,458 (97.20%)**
Within ±2 Days	**1,484 (98.93%)**
Within ±3 Days	**1,493 (99.53%)**
Within ±5 Days	**1,500 (100.00%)**

As shown in Table 21, the finalized Random Forest obtained an MAE of 0.0815 days, RMSE of 0.4042 days, and R^2 of 0.9976 on the independent 1,500-record final test partition. Tolerance analysis showed that 97.20% of predictions were within ±1 day, 98.93% within ±2 days, 99.53% within ±3 days, and 100.00% within ±5 days.

These results demonstrate strong predictive performance within the simulated operational conditions represented by Dataset B. However, because Dataset B consists of simulated operational service orders, these findings should not be interpreted as equivalent to external validation using a large independent collection of actual Ready-for-Release transactions. Actual Ready-for-Release observations accumulated through continued system use may subsequently be used for real-world validation and future model refinement.

Residual and Error Analysis by Service Type

To further examine whether predictive errors varied across the services represented in Dataset B, residual and error analysis was conducted using the final 1,500-record test set. An order was included in a service category when the corresponding service quantity was greater than zero. Since a job order may contain more than one service, the service categories were treated as non-mutually-exclusive. Consequently, the combined service-specific sample sizes

exceeded the 1,500 unique final-test records, with a cumulative subgroup count of 2,664 service-category observations.

Residual error was calculated as:

Residual = Actual Ready-for-Release Duration – Predicted Ready-for-Release Duration

A positive residual indicates underprediction, wherein the actual duration was longer than the predicted duration, while a negative residual indicates overprediction, wherein the predicted duration was longer than the actual duration. A mean residual close to zero indicates limited systematic directional bias.

Table 22

RESIDUAL AND ERROR ANALYSIS BY SERVICE TYPE – DATASET B FINAL TEST SET

Service Type	N	MAE (Days)	RMSE (Days)	Mean Residual (Days)	±1 Day	±2 Days	±3 Days	±5 Days
Basic Cleaning	1,483	0.0806	0.4058	+0.0025	97.17%	98.92%	99.53%	100.00%
Full Reglue	569	0.1123	0.4620	−0.0205	95.61%	98.77%	99.47%	100.00%
Minor Reglue	118	0.0835	0.4178	−0.0173	97.46%	99.15%	99.15%	100.00%
Unyellowing	185	0.1158	0.5419	+0.0100	95.68%	98.38%	98.92%	100.00%
Minor Restoration	274	0.1193	0.5363	+0.0033	97.08%	97.45%	99.27%	100.00%
Color Renewal	35	0.0006	0.0012	+0.0001	100.00%	100.00%	100.00%	100.00%

Note. Service categories are non-mutually-exclusive because a single order may contain multiple services. Therefore, the service-specific sample sizes sum to 2,664 and should not be interpreted as 2,664 unique orders. Residual was calculated as Actual Ready-for-Release Duration minus Predicted Ready-for-Release Duration.

As shown in Table 22, the service-specific analysis demonstrated generally low prediction errors across the six service categories represented in the Dataset B final test set. Color Renewal recorded the lowest MAE at **0.0006 days** and the lowest RMSE at **0.0012 days**. However, this result should be interpreted cautiously

because only 35 final-test orders contained Color Renewal. Among the other service categories, Basic Cleaning recorded an MAE of 0.0806 days, Minor Reglue 0.0835 days, Full Reglue 0.1123 days, Unyellowing 0.1158 days, and Minor Restoration 0.1193 days. Minor Restoration therefore recorded the highest MAE, while Unyellowing recorded the highest RMSE at **0.5419 days**.

The mean residual values were close to zero across all service categories, indicating limited systematic directional bias. Full Reglue and Minor Reglue recorded mean residuals of **−0.0205 days** and **−0.0173 days**, respectively, indicating slight overprediction. In contrast, Basic Cleaning (**+0.0025 days**), Unyellowing (**+0.0100 days**), Minor Restoration (**+0.0033 days**), and Color Renewal (**+0.0001 days**) showed slight underprediction. These residual values were small in magnitude, indicating that no service category demonstrated a substantial overall directional prediction bias within the Dataset B final test set.

Tolerance analysis further showed that the percentage of predictions falling within ±1 day ranged from **95.61% for Full Reglue** to **100.00% for Color Renewal**, while Basic Cleaning achieved 97.17%, Minor Reglue 97.46%, Unyellowing 95.68%, and Minor Restoration 97.08%. At the ±2-day threshold, all service categories achieved at least **97.45%**, while ±3-day accuracy ranged from **98.92% to 100.00%**. All six service categories achieved **100.00% of predictions within ±5 days**.

Because the service categories were non-mutually-exclusive, the subgroup results should not be averaged to reproduce the overall Dataset B performance metrics. Orders containing multiple services contributed to each applicable service category; therefore, the subgroup analysis was intended to identify whether particular categories exhibited noticeably different error patterns rather than to

provide mutually exclusive comparisons among services. The analysis did not identify a severe service-specific prediction-error pattern within the simulated operational final test set.

Overall, the residual and error analysis supplement the aggregate evaluation of the finalized Random Forest model by demonstrating its prediction-error distribution across the service categories represented in Dataset B. The findings should nevertheless be interpreted within the context of the simulated operational dataset and should not be considered equivalent to external validation using actual future Ready-for-Release transactions. Continued collection of actual Ready-for-Release records through the developed system may provide the basis for subsequent real-world validation and further refinement of the model.

4.4 Analytics Dashboard Evaluation

The analytics dashboard was evaluated in two ways: (1) by comparing the time required to prepare selected reports using the existing manual process with the report-generation time of the developed analytics dashboard, and (2) by evaluating the perceptions of the shop owners and staff regarding the dashboard's contribution to business information access, performance monitoring, and decision-making.

Analytics Dashboard Reporting Efficiency

To address the reproducibility of the dashboard performance measurements, a technical benchmark was conducted for seven report types: Daily Sales, Weekly Sales, Monthly Sales, Quarterly Sales, Annual Sales, Expenses, and Return on Investment (ROI). Fixed reporting periods containing actual records from the system database were used for the evaluation. The tested

local database contained 103 orders, 105 payment records, and 11 expense records.

For the developed system, dashboard report-generation time was operationally defined as the elapsed application-level response time beginning when the test client initiated a request to the FastAPI reporting endpoint and ending when the completed response was received and parsed by the test client. The measurement therefore included authentication and role-based access control processing, database queries through SQLAlchemy, report calculations, and response serialization. It did not include browser rendering, user navigation, login credential entry, or the time required by a user to interpret the displayed information.

A high-resolution monotonic timer, time.perf_counter(), was used for the measurements. For each report type, three warm-up executions were performed followed by 10 measured executions. All 10 measured trials were retained in the analysis. analytics_dashboard_reproducibi… The mean dashboard generation time for each report was subsequently calculated from these repeated measurements.

To establish equivalence between the report information generated by the dashboard and the underlying data, the dashboard output was independently compared with direct database calculations using the same stored records and corresponding fixed reporting periods. The verification covered material financial and operational fields for the evaluated reports. All seven report categories passed the content-equivalence verification, with no differences reported between the tested dashboard values and their corresponding independently calculated values.

Table 23

ANALYTICS DASHBOARD REPORTING EFFICIENCY (MANUAL VS. DASHBOARD)

Report Type	Manual Time (min)	Dashboard Time (sec)	SD (sec)	Content Equivalence	Observed Time Reduction
Daily Sales	10	0.0657	0.0066	PASS	99.9891%
Weekly Sales	18	0.0808	0.0416	PASS	99.9925%
Monthly Sales	30	0.0694	0.0059	PASS	99.9961%
Quarterly Sales	90	0.0972	0.0376	PASS	99.9982%
Annual Sales	120	0.0835	0.0055	PASS	99.9988%
Expenses	15	0.0996	0.0350	PASS	99.9889%
ROI	23	0.0907	0.0086	PASS	99.9934%
Overall Mean	**43.71**	**0.0838**	**—**	**7/7 PASS**	**99.9968%**

Note. Manual preparation times represent researcher-observed human timings from the comparative manual reporting test. Dashboard times represent application/API report-generation response times and exclude browser rendering, user navigation, login credential entry, and human interpretation time. Dashboard statistics were obtained from 10 measured executions per report following three warm-up executions. The observed time reduction was calculated as: [(Manual Preparation Time − Dashboard Mean Time in Minutes) / Manual Preparation Time] × 100.

As shown in Table 23, The repeated technical benchmark showed an overall mean dashboard report-generation time of approximately 0.0838 second across the seven evaluated reports. The individual mean generation times ranged from 0.0657 second for Daily Sales to 0.0996 second for Expenses. The Annual Sales report, despite covering the complete 2026 reporting period, had a mean generation time of 0.0835 second, while the ROI report had a mean of 0.0907

second. The benchmark retained all measured trials, including slower observations, rather than removing them as outliers. The reported standard deviations further describe the variability observed across the repeated executions.

The report-content verification further demonstrated that the dashboard was producing the intended information from the underlying records rather than merely retrieving information more quickly. For example, the Monthly Sales test for September 2026 produced identical dashboard and independently calculated values for gross sales, net sales, order counts, payments received, balance due, refunds, and retained payments. Similarly, the Annual Sales verification produced matching values between the dashboard and independent calculation for the evaluated financial and operational fields. The Expenses and ROI calculations likewise produced matching values in the independent verification.

When compared with the researcher-observed manual preparation times, the repeated dashboard benchmark resulted in an overall calculated time reduction of approximately 99.9968% for the report-generation comparison. However, this percentage should be interpreted specifically within the scope of the evaluation. The manual measurements represent human report-preparation activities, whereas the dashboard measurements represent automated application/API processing after transaction records have already been digitized and stored in the system. Consequently, the percentage should not be interpreted as indicating that the entire human reporting workflow is universally 99.9968% faster. Rather, it quantifies the substantial reduction observed in the report-generation stage under the specified testing conditions.

The findings therefore indicate that the developed analytics dashboard can rapidly generate the evaluated business reports from digitized transaction records while

maintaining consistency with independently calculated report values. Within the tested dataset, reporting periods, system version, and local testing environment, the dashboard demonstrated substantial improvement in report-generation efficiency compared with the existing manual preparation process.

Table 24

SHOP OWNERS' AND STAFF'S EVALUATION OF THE ANALYTICS DASHBOARD IMPACT

Analytics Dashboard Impact	N	Mean	SD	Interpretation
1. The analytics dashboard makes important business information easier to access.	6	4.00	0.00	Highly Acceptable
2. The analytics dashboard provides useful information for monitoring sales, completed services, pending job orders, inventory levels, and service trends across different reporting periods.	6	4.00	0.00	Highly Acceptable
3. The analytics dashboard supports informed decision-making based on sales, service, inventory, and operational performance information.	6	3.50	0.55	Acceptable
General Weighted Mean	**6**	**3.83**	**0.38**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

As shown in Table 24, the Analytics Dashboard Impact obtained a general weighted mean of 3.83 with a standard deviation of 0.38, interpreted as Highly Acceptable. This indicates that the six shop owners and staff respondents generally perceived the analytics dashboard as useful for improving access to

business information, monitoring operational performance, and supporting informed decision-making.

The first indicator, which assessed whether the analytics dashboard makes important business information easier to access, obtained a mean of 4.00 and a standard deviation of 0.00, interpreted as Highly Acceptable. Similarly, the second indicator, which assessed the usefulness of the dashboard for monitoring sales, completed services, pending job orders, inventory levels, and service trends across different reporting periods, obtained a mean of 4.00 and a standard deviation of 0.00, also interpreted as Highly Acceptable. These results indicate unanimous positive ratings among the six respondents for these two indicators.

The third indicator, which assessed whether the analytics dashboard supports informed decision-making based on sales, service, inventory, and operational performance information, obtained a mean of 3.50 and a standard deviation of 0.55, interpreted as Acceptable. Although this indicator received a comparatively lower rating than the first two indicators, the result still indicates a positive assessment of the dashboard's contribution to decision-making.

Overall, the results demonstrate that the intended users evaluated the analytics dashboard positively, particularly in terms of information accessibility and operational monitoring. The findings complement the technical reporting-efficiency results presented in Table 23 by providing a user-based evaluation of the dashboard's perceived impact, rather than serving as evidence of report-generation speed.

4.5 ISO/IEC 25010-Based System Quality Evaluation

The quality evaluation of the developed Web-Based Service Management System was conducted using two complementary approaches: expert perception-based evaluation and objective technical testing. This distinction was established to ensure that the subjective ratings provided by the IT experts were not interpreted as direct empirical proof of technical compliance with ISO/IEC 25010.

The perception-based evaluation involved five (5) qualified IT experts who assessed the system across nine software quality characteristics using a four-point Likert scale. Each quality characteristic consisted of five indicators. The individual ratings were summarized using the mean and standard deviation. The mean was used to describe the overall level of acceptability perceived by the experts, while the standard deviation was used to describe the variability of their ratings.

The following scale was used in interpreting the mean scores: 1.00–1.50 = Not Acceptable; 1.51–2.50 = Less Acceptable; 2.51–3.50 = Acceptable; and 3.51–4.00 = Highly Acceptable.

Because only five IT experts participated in the evaluation, the results are presented primarily using descriptive statistics and are interpreted specifically as the experts' assessment of the developed system. The mean and standard deviation values therefore describe the evaluators' perceptions and should not be interpreted independently as objective proof of ISO/IEC 25010 conformity or certification.

To supplement the expert assessment, the researchers conducted objective technical tests on measurable system properties associated with selected ISO/IEC 25010 quality characteristics. These tests included response time, concurrent

access, database integrity, browser compatibility, access control, failed-login handling, backup restoration, error recovery, input validation, and system reliability. For transparency and reproducibility, each technical test documented the test condition, number of trials, expected output, actual output, and corresponding pass-or-fail result.

The expert ratings and objective technical tests were therefore treated as complementary but separate sources of evidence. Expert ratings measured perceived software quality and acceptability, whereas the technical tests provided observable evidence regarding selected operational and technical behaviors of the developed system.

Table 25

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF FUNCTIONAL SUITABILITY

Functional Suitability	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system provides the necessary functions for managing job orders, customer information, inventory, reports, and release schedules.	4	4	3	4	4	3.80	0.45	Highly Acceptable
2. Each system module produces the expected result based on the user's operation.	3	3	3	4	4	3.40	0.55	Acceptable

Functional Suitability	E1	E2	E3	E4	E5	Mean	SD	Interpretation
3. The Machine Learning component uses a Random Forest Regressor to analyze job order data, identify service patterns, and generate a predicted release date for operational decision-making.	3	4	3	3	4	3.40	0.55	Acceptable
4. The system's data analytics functions provide information for monitoring sales, services, and business performance.	4	4	3	4	4	3.80	0.45	Highly Acceptable
5. The system supports the complete job order workflow, including New Order, On-Going, For Release, and Claimed statuses.	4	4	4	3	4	3.80	0.45	Highly Acceptable
General Weighted Mean						**3.64**	**0.49**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Dr. Eduardo Yu:** "The prediction feature power of machine learning provides transformation to the customer and motivates the service provider to achieve the target."

- **Edison Yu:** "Job order workflow, CRUD and other functions are working as intended."
- **Ken Carlo Javier:** "The Machine Learning integration for predicting release dates calculates predictions synchronously (~1.2s) during order creation without blocking the UI, adding immediate, practical value to the workflow."

As shown in Table 26, the Functional Suitability of the Web-Based Service Management System achieved a general weighted mean of 3.64 with a standard deviation of 0.49, which indicated that IT experts rated the system as Highly Acceptable in terms of providing the necessary functions to perform daily tasks. The three indicators were rated Highly Acceptable while the two indicator was rated Acceptable, reflecting a high level of satisfaction with the system's functional capabilities. The higher rating was observed in the system providing the necessary functions for managing job orders, customer information, inventory, reports, and release schedules; each system module producing the expected result; the Machine Learning component using a Random Forest Regressor; the system's data analytics functions; and the system supporting the complete job order workflow. The IT experts' remarks confirm that the machine learning prediction feature adds practical value to the workflow, the job order and CRUD functions work as intended, and the synchronous ML prediction (~1.2s) enhances operational efficiency. This suggests that the system is was positively evaluated in terms of functional suitability and meets the operational requirements of Shoelotskey.

Table 26

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF PERFORMANCE EFFICIENCY

Performance Efficiency	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system retrieves database records and loads pages within an acceptable response time.	3	3	4	4	4	3.60	0.55	Highly Acceptable
2. Job order processing and saving transactions are completed without noticeable delays.	3	4	4	4	4	3.80	0.45	Highly Acceptable
3. Sales and expense reports can be generated for daily, weekly, monthly, quarterly, and annual periods within an acceptable response time.	4	4	4	3	4	3.80	0.45	Highly Acceptable
4. The Machine Learning component generates a predicted release date from valid input data without noticeably delaying the transaction workflow.	3	3	4	3	4	3.40	0.55	Acceptable
5. Navigation between system modules occurs without noticeable delay, supporting efficient single-page application (SPA) routing.	4	4	3	4	4	3.80	0.45	Highly Acceptable
General Weighted Mean						**3.68**	**0.49**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Ken Carlo Javier:** "The Machine Learning integration calculates predictions synchronously (~1.2s) during order creation without blocking the UI."
- **Dr. Eduardo Yu:** "The overall performance and usability of the developed system is aligned to the ISO 25010 software product quality."

As shown in Table 26, the Performance Efficiency of the Web-Based Service Management System obtained a general weighted mean of **3.68** with a standard deviation of **0.49**, interpreted as **Highly Acceptable** based on the assessment of the five IT experts. Four of the five indicators were rated Highly Acceptable, while one indicator was rated Acceptable. The experts gave favorable assessments to the system's retrieval of database records and page loading, processing and saving of job order transactions, generation of sales and expense reports, generation of the Machine Learning-predicted release date, and navigation between system modules. The experts also observed that the Machine Learning prediction was generated synchronously without blocking the user interface. These findings indicate that the IT experts perceived the system's performance efficiency as highly acceptable during the evaluation. However, the mean and standard deviation represent expert perceptions and should not be interpreted as objective measurements of system response time, concurrent-processing capability, or formal compliance with ISO/IEC 25010. Therefore, the perception-based findings were supplemented by separate objective technical tests of response time and concurrent access, in which predefined test conditions, numbers of trials, expected outputs, actual outputs, and pass-or-fail results were documented.

Table 27

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF COMPATIBILITY

Compatibility	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system operates properly and maintains its functional layout across commonly used web browsers.	4	4	3	4	4	3.80	0.45	Highly Acceptable
2. Modifications made to existing records, such as job order status changes, are immediately reflected in all related system modules.	4	4	4	4	4	4.00	0.00	Highly Acceptable
3. Information entered in one module is correctly synchronized across other system modules.	4	4	3	4	4	3.80	0.45	Highly Acceptable
4. The Machine Learning output is correctly integrated into the Job Order module to display the predicted release date.	3	4	3	4	4	3.60	0.55	Highly Acceptable
5. The system's data analytics functions retrieve and consolidate transaction data to display current operational metrics.	3	4	3	4	4	3.60	0.55	Highly Acceptable
General Weighted Mean						**3.76**	**0.40**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Iheo Corona:** "No major functional issues encountered. Minor observations: the predicted release date did not generate ('ML: unavailable'), and currency values display inconsistently (P912.5 instead of P912.50)."
- **Edison Yu:** "Data analysis/validation. The non-persistent/duplicate. Some errors did not provide clear message in UI."

As shown in Table 26, the Compatibility of the Web-Based Service Management System achieved a general weighted mean of 3.76 with a standard deviation of 0.40, which indicated that IT experts rated the system as Highly Acceptable in terms of working properly with other system components and supported platforms. All five indicators were rated Highly Acceptable, reflecting a high level of satisfaction with the system's compatibility. The highest rating was observed in the system operating properly across commonly used web browsers, modifications made to existing records being immediately reflected in all related system modules, information entered in one module being correctly synchronized across other system modules, the Machine Learning output being correctly integrated into the Job Order module, and the system's data analytics functions retrieving and consolidating transaction data. The IT experts' remarks noted minor observations such as the ML prediction occasionally showing "ML: unavailable" and currency display inconsistencies. These observations indicate areas for improvement in future iterations. This suggests that the system is highly compatible and demonstrated successful integration among the evaluated modules with its various modules and platforms.

Table 28

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF INTERACTION CAPABILITY

Interaction Capability	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system's menus, icons, and buttons are clearly labeled and easy to recognize.	4	4	3	4	4	3.80	0.45	Highly Acceptable
2. The system's job order, inventory, service, and user management features are easy to learn.	4	4	3	4	4	3.80	0.45	Highly Acceptable
3. The system provides validation messages and confirmation prompts to help prevent input errors and unintended actions.	4	3	3	4	4	3.60	0.55	Highly Acceptable
4. The layout and navigation make it easy to move between different modules.	4	3	4	4	4	3.80	0.45	Highly Acceptable
5. Frequently used tasks can be completed with minimal steps.	4	3	3	4	4	3.60	0.55	Highly Acceptable
General Weighted Mean						**3.72**	**0.49**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Jheo Corona:** "The job order form performs best. The customer's name field automatically suggests existing customer records as you type, so returning customers can be selected in one click instead of re-entering their details. This speeds up encoding and reduces typing errors."
- **Ken Carlo Javier:** "Enhance the interface to comply with WCAG standards. Improve color contrast (the current gray text is difficult to read) and ensure proper font sizing for better readability."

As shown in Table 28, the Interaction Capability of the Web-Based Service Management System achieved a general weighted mean of 3.72 with a standard deviation of 0.49, which indicated that IT experts rated the system as Highly Acceptable in terms of being easy to learn, understand, and use. All five indicators were rated Highly Acceptable, reflecting a high level of satisfaction with the system's interaction capability. The highest rating was observed in the system's menus, icons, and buttons being clearly labeled and easy to recognize; the system's job order, inventory, service, and user management features being easy to learn; the system providing validation messages and confirmation prompts; the layout and navigation making it easy to move between different modules; and frequently used tasks being completed with minimal steps. The IT experts' remarks highlighted the job order form as the best-performing feature due to its auto-suggest customer records, which speeds up encoding and reduces typing errors. However, recommendations for WCAG compliance, improved color contrast, and proper font sizing were noted for future enhancement. This suggests that the

system is highly user-friendly and was evaluated as highly acceptable in interaction capability.

Table 29

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF RELIABILITY

Reliability	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system performs consistently without unexpected interruptions during operation.	4	4	4	4	4	4.00	0.00	Highly Acceptable
2. The system handles invalid or incomplete input without unexpected failure.	3	3	3	4	4	3.40	0.55	Acceptable
3. Transactions are processed without data loss or unintended changes to existing records.	4	3	3	4	4	3.60	0.55	Highly Acceptable
4. The system supports manual PostgreSQL database backups through the Heroku platform to facilitate data recovery.	4	4	3	4	4	3.80	0.45	Highly Acceptable
5. The Machine Learning component consistently generates a predicted release date when provided with a complete and valid job order input.	4	4	4	4	4	4.00	0.00	Highly Acceptable
General Weighted Mean						**3.72**	**0.49**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Edison Yu:** "Data analysis/validation. The non-persistent/duplicate. Some errors did not provide clear message in UI."
- **Ken Carlo Javier:** "Backend API validation is overly permissive. While the UI correctly blocks empty fields, direct API testing shows it accepts empty or malformed payloads and assigns default values (e.g., 0.00 for totals) instead of rejecting them, lacking defense-in-depth."

As shown in Table 29, the Reliability of the Web-Based Service Management System obtained a general weighted mean of 3.72 with a standard deviation of 0.49, interpreted as Highly Acceptable based on the assessment of the five IT experts. Four of the five indicators were rated Highly Acceptable, while one indicator was rated Acceptable. The experts provided favorable assessments of the system's consistency during operation, handling of invalid or incomplete inputs without unexpected failure, processing of transactions without data loss or unintended changes, support for manual PostgreSQL database backups through the Heroku platform, and consistent generation of predicted release dates by the Machine Learning component. However, the IT experts also identified concerns regarding overly permissive backend API validation and the lack of clear user-interface messages for some errors, indicating areas for improvement in input validation, error handling, and defense-in-depth. Overall, the findings indicate that the IT experts perceived the system's reliability as highly acceptable. These results represent expert assessment and should not be interpreted independently as direct measurements of system failure rate, recovery capability, data integrity, or operational reliability. Accordingly, the perception-based findings were

supplemented by separate objective technical tests involving database integrity, backup restoration, error recovery, input validation, and repeated system operation, with predefined test conditions, numbers of trials, expected outputs, actual outputs, and pass-or-fail results documented for each test.

Table 30

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF SECURITY

Security	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system uses JSON Web Token (JWT) authentication with bcrypt password hashing, enforces a three-attempt login limit with a 15-minute lockout, and automatically terminates inactive sessions after 30 minutes, addressing OWASP A07: Identification and Authentication Failures.	4	4	4	3	4	3.80	0.45	Highly Acceptable
2. The system enforces Role-Based Access Control (RBAC) for Staff and Owner roles and prevents direct access to unauthorized modules through URL requests, addressing OWASP A01: Broken Access Control.	4	4	4	4	4	4.00	0.00	Highly Acceptable
3. The system uses parameterized queries	4	4	3	4	2	3.40	0.89	Acceptable

Security	E1	E2	E3	E4	E5	Mean	SD	Interpretation
and input validation to prevent SQL injection and invalid data submission, addressing OWASP A03: Injection.								
4. The system records critical user activities and system events in audit logs, including actions, user information, timestamps, and relevant changes, addressing OWASP A09: Security Logging and Monitoring Failures.	4	3	3	4	4	3.60	0.55	Highly Acceptable
5. The system uses HTTPS with SSL/TLS encryption to protect data transmitted between the client and application server, addressing OWASP A02: Cryptographic Failures.	4	3	4	3	4	3.60	0.55	Highly Acceptable
General Weighted Mean						**3.68**	**0.49**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Ken Carlo Javier:** "Enforce strict backend validation to reject incomplete payloads at the API level. Implement Redis caching for the Sales/Analytics modules to reduce database load as data scales, and add rate-limiting to the login endpoint to prevent brute-force attacks."

- **Iheo Corona:** "Standardize currency display to two decimals and fix the negative pending payments value."

As shown in Table 30, the Security of the Web-Based Service Management System obtained a general weighted mean of 3.68 with a standard deviation of 0.49, interpreted as Highly Acceptable based on the assessment of the five IT experts. Four of the five indicators were rated Highly Acceptable, while one indicator was rated Acceptable. The experts provided favorable assessments of the system's implemented security mechanisms, including JSON Web Token (JWT) authentication with bcrypt password hashing, the three-attempt login limit with a 15-minute lockout, automatic termination of inactive sessions after 30 minutes, Role-Based Access Control (RBAC) for Staff and Owner roles, parameterized queries and input validation, audit logging of critical user activities and system events, and HTTPS with SSL/TLS encryption. The IT experts also recommended stricter backend validation, login rate-limiting, Redis caching to reduce database load as data scales, and standardization of currency displays. These recommendations identify areas for further improvement and should not be interpreted as being negated by the overall favorable expert rating. Overall, the findings indicate that the IT experts perceived the implemented security mechanisms as highly acceptable. However, these ratings represent expert assessment of the security features presented during system evaluation and do not constitute formal verification of OWASP compliance, penetration testing, or comprehensive security certification. Accordingly, the perception-based findings were supplemented by separate objective technical tests involving access control, authentication, failed-login handling, input validation, and other applicable security controls, with predefined test conditions, numbers of trials, expected outputs, actual outputs, and pass-or-fail results documented as supplementary technical evidence.

Table 31

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF MAINTAINABILITY

Maintainability	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system provides clear error messages identifying the specific field and cause of invalid input to assist troubleshooting.	4	4	3	4	4	3.80	0.45	Highly Acceptable
2. The system allows incorrect data entries to be corrected without affecting completed transaction records.	4	4	3	4	4	3.80	0.45	Highly Acceptable
3. The system remains functional when existing records, services, or configurations are modified.	3	4	3	4	4	3.60	0.55	Highly Acceptable
4. The system logs administrative actions and critical errors to facilitate system monitoring and maintenance.	4	3	3	4	4	3.60	0.55	Highly Acceptable
5. The system's modules can be updated or modified without disrupting the job order workflow.	3	4	4	4	4	3.80	0.45	Highly Acceptable
General Weighted Mean						**3.72**	**0.46**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Edison Yu:** "Transactional update. More rigid DB model constraints."
- **IT Expert 4:** "There should be logs audit trail for inventory."
- **Ken Carlo Javier:** "Automated pipeline to retrain the ML model weekly with new historical data."

As shown in Table 31, the Maintainability of the Web-Based Service Management System achieved a general weighted mean of 3.72 with a standard deviation of 0.46, which indicated that IT experts rated the system as Highly Acceptable in terms of being easy to maintain, update, and improve. All five indicators were rated Highly Acceptable, reflecting a high level of satisfaction with the system's maintainability. The highest rating was observed in the system providing clear error messages, allowing incorrect data entries to be corrected without affecting completed transaction records, remaining functional when existing records, services, or configurations are modified, logging administrative actions and critical errors, and the system's modules being updated or modified without disrupting the job order workflow.

The IT experts' remarks recommended transactional updates, more rigid database model constraints, inventory audit trails, and an automated ML retraining pipeline. These recommendations provide valuable guidance for future enhancements. This suggests that the system is highly maintainable and can be easily updated or modified to accommodate future changes.

Technical Artifact Inspection for Maintainability

To strengthen the maintainability evaluation, qualified IT experts were provided access to relevant technical artifacts of the developed system. The inspection focused on the organization and understandability of artifacts that directly affect software maintenance, rather than relying solely on observation of the user interface.

Table 32

TECHNICAL ARTIFACT INSPECTION FOR SYSTEM MAINTAINABILITY

Technical Artifact	Aspect Examined	Evidence Inspected	Expert Finding
Source-Code Structure	Modularity and organization	React/TypeScript frontend and FastAPI/Python backend	The system maintains clear separation between the frontend and backend and uses dedicated modules for reporting, database operations, and Machine Learning. However, maintainability is affected by large and highly concentrated files, particularly the main backend controller and selected frontend components, indicating a need for further modularization.
Source-Code Naming	Readability and analyzability	Components, functions, classes, variables, routes, models, and schemas	Naming conventions were generally consistent and descriptive across the Python and TypeScript codebase. Minor inconsistencies were identified in primary-key naming, including the use of item_id for different entities, mixed id and entity-specific ID conventions, and similarly named page and component files.
Database Schema	Structural organization	SQLAlchemy ORM models, local SQLite database, database	The database demonstrates a structured relational design with defined models, relationships,

	and modifiability	configuration, and schema documentation	foreign keys, and cascading operations. However, schema maintainability could be improved through a formal migration framework and more consistent naming. The production PostgreSQL database was not directly inspected during this artifact review.
Dependency Organization	Dependency maintainability	Frontend and backend dependency and configuration files	Frontend and backend dependencies were clearly separated; however, the inspection identified unused and redundant dependencies, including unused frontend packages and database drivers, indicating that dependency cleanup would improve maintainability.
Technical Documentation	Analyzability	README, database and architecture documentation, disaster-recovery documentation, API documentation, and source-code documentation	The project contains substantial database, architecture, disaster-recovery, and API documentation. Minor documentation deficiencies were identified, including an outdated script reference, incomplete environment-variable documentation, and redundant or unorganized files within the documentation directory.
Test Documentation	Testability	Backend test scripts, objective technical testing reports, ML testing evidence, and use-case testing documentation	The project contains substantial backend testing evidence and reproducible technical testing reports. However, automated frontend unit/component tests, a unified backend test framework, continuous integration, and automated code-coverage tracking were not present in the inspected repository.
Error and Audit Logging	Analyzability	Audit-log models, centralized audit function, activity-history interface, inventory logs, and status logs	Audit logging was implemented with traceability from system actions to persistent database records and administrative retrieval. Critical activities include actor, timestamp,

			affected record, and change information. A minor limitation is that runtime application errors primarily rely on console logging rather than persistent rotating application logs.

Table 32 presents the results of the technical artifact inspection conducted to supplement the perception-based evaluation of system maintainability. The inspection examined the source-code structure, naming conventions, database schema, dependency organization, technical documentation, test documentation, and error and audit logging of the developed system. The findings showed strengths in the separation of frontend and backend components, relational database organization, technical documentation, backend testing evidence, and audit-trail implementation. However, the inspection also identified maintainability concerns, particularly the presence of large and highly concentrated source-code files, unused or redundant dependencies, the absence of a formal database migration framework, incomplete environment-variable documentation, and the absence of automated frontend unit testing and continuous integration.

Overall, the technical artifact inspection indicates that the system possesses several maintainability-supporting characteristics while also containing identifiable areas for improvement in modularity, dependency management, documentation organization, database migration management, and automated testing. These findings supplement the IT experts' perception-based assessment by providing artifact-level evidence concerning the system's analyzability, modifiability, modularity, and testability. The results are limited to the artifacts and repository state examined during the inspection and should not be interpreted as formal certification of conformity with ISO/IEC 25010.

Table 33

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF FLEXIBILITY

Flexibility	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system supports different user roles, such as Staff and Owner, with appropriate access permissions and functions.	4	4	4	4	4	4.00	0.00	Highly Acceptable
2. The system supports different types of shoe services within the same workflow.	4	3	4	4	4	3.80	0.45	Highly Acceptable
3. The system handles different customer transactions, such as single-service and multi-service orders within the same workflow.	4	4	3	4	4	3.80	0.45	Highly Acceptable
4. The system accommodates different payment methods and payment statuses within the same transaction workflow.	4	3	3	4	4	3.60	0.55	Highly Acceptable
5. The system allows new services to be added without affecting existing transactions.	4	4	4	4	4	4.00	0.00	Highly Acceptable
General Weighted Mean						**3.84**	**0.37**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Dr. Eduardo Yu:** "More flexibility on the drop-down option by providing a facility to add/update the data. More detailed information on the predicted date for explainable and transparency."
- **Edison Yu:** "Some of the options are hardwired, it is wise to have more flexibility to add new brands etc., courier etc., model etc. Avoid direct modification on your products & services."
- **Ken Carlo Javier:** "Add SMS or email notifications when an order is ready for pickup, and low-stock alerts in Inventory Management. A customer history view showing past orders per client would also help."

As shown in Table 33, the Flexibility of the Web-Based Service Management System achieved a general weighted mean of 3.84 with a standard deviation of 0.37, which indicated that IT experts rated the system as Highly Acceptable in terms of being able to adapt to future changes and business needs. All five indicators were rated Highly Acceptable, reflecting a high level of satisfaction with the system's flexibility.

The highest rating was observed in the system supporting different user roles, supporting different types of shoe services, handling different customer transactions, accommodating different payment methods and payment statuses, and allowing new services to be added without affecting existing transactions. The

IT experts' remarks recommended more flexibility in dropdown options, more detailed information on predicted dates for explainable, SMS/email notifications, low-stock alerts, and a customer history view. These recommendations provide valuable guidance for future enhancements. This suggests that the system is highly flexible and can easily adapt to the evolving operational needs of Shoelotskey.

Table 34

IT EXPERTS' ASSESSMENT OF SYSTEM QUALITY USING ISO/IEC 25010 SOFTWARE QUALITY STANDARDS IN TERMS OF SAFETY

Safety	E1	E2	E3	E4	E5	Mean	SD	Interpretation
1. The system helps prevent accidental loss or deletion of important records.	4	3	4	4	4	3.80	0.45	Highly Acceptable
2. The system requires explicit user confirmation before executing critical data deletions or modifications.	3	4	4	4	4	3.80	0.45	Highly Acceptable
3. The system enforces mandatory field requirements before allowing a transaction to proceed.	3	4	4	4	2	3.40	0.89	Acceptable
4. The system aborts transactions that violate defined business logic or validation rules.	4	4	4	4	2	3.60	0.89	Highly Acceptable
5. The system prevents duplicate submissions	4	3	3	4	4	3.60	0.55	Highly Acceptable

Safety	E1	E2	E3	E4	E5	Mean	SD	Interpretation
during transaction processing.								
General Weighted Mean						**3.64**	**0.64**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

IT Experts' Remarks:

- **Ken Carlo Javier:** "Automated SMS/Email notifications for customers when their order status changes. Auto-deduction of raw materials from inventory when a job order status changes to 'On-Going'."
- **Edison Yu:** "Real time notifies. Back-up / restore. Acceptable reports."

As shown in Table 34, the Safety of the Web-Based Service Management System achieved a general weighted mean of **3.64** with a standard deviation of **0.64**, which indicated that the IT experts rated the system as **Highly Acceptable** in terms of helping prevent errors and protecting important data. Four of the five indicators were rated Highly Acceptable, while one indicator was rated Acceptable, indicating an overall positive assessment of the system's safety features.

The highest mean ratings were observed in the system helping prevent accidental loss or deletion of important records and requiring explicit user confirmation before executing critical data deletions or modifications, both with a mean of **3.80** and a standard deviation of **0.45**. These were followed by the system aborting transactions that violate defined business logic or validation rules, with a mean of **3.60** and a standard deviation of **0.89**, and preventing duplicate submissions during transaction processing, with a mean of **3.60** and a standard deviation of **0.55**. The lowest-rated indicator was the enforcement of mandatory

field requirements before allowing a transaction to proceed, which obtained a mean of **3.40** and a standard deviation of **0.89**, interpreted as **Acceptable**.

The IT experts' remarks recommended automated SMS or email notifications, auto-deduction of raw materials from inventory, real-time notifications, and backup and restore features. These recommendations identify areas that may be considered for future enhancement. Overall, the findings suggest that the system provides safety measures that support the protection of important business data and the prevention of unintended transactions, while the enforcement of mandatory field requirements remains an area for further improvement.

4.5.10 Objective Technical Testing of Selected ISO/IEC 25010 Quality Characteristics

To supplement the perception-based evaluation conducted by the five IT experts, the researchers performed objective technical tests on selected measurable characteristics of the developed system. Unlike the questionnaire results, which represent expert judgments, these tests examined observable system behavior under predefined test conditions.

Each test was conducted using a defined procedure and acceptance criterion. The number of successful trials divided by the total number of trials was used to determine the pass rate:

Pass Rate (%) = (Number of Passed Trials / Total Number of Trials) × 100

For response-time testing, the researchers should additionally report the mean response time and standard deviation across repeated trials rather than relying solely on a single measurement.

Table 35

OBJECTIVE TECHNICAL TESTING OF SELECTED SYSTEM QUALITY CHARACTERISTICS

Technical Test	Related Quality Characteristic	Test Condition	No. of Trials	Expected Output / Acceptance Criterion	Actual Output	Result
Response Time	Performance Efficiency	Execute six predefined API operations repeatedly under the same controlled test environment	180 trials (30 per operation)	Mean response time <500 ms per operation; P95 <1,000 ms; 100% successful HTTP responses	Mean ± SD ranged from 17.82 ± 17.90 ms (ML Prediction) to 517.10 ± 58.39 ms (Login); 180/180 requests completed successfully. Login exceeded the <500 ms mean criterion.	**Partially Passed**
Concurrent Access	Performance Efficiency / Reliability	Multiple requests execute predefined operations simultaneously at 1, 5, 10, and 20 concurrent workers	36 requests	Requests complete without application failure, HTTP 500 errors, unintended overwriting, record collision, or data loss	36/36 requests succeeded (100%); mean latency ranged from 90.36 ms at 1 worker to 911.69 ms at 20 workers; no record collisions reported.	**Passed**
Database Integrity	Reliability	Create/update-controlled records and verify relational constraints and transaction integrity in the local test database	10 checks	Stored values and relationships remain consistent after transactions and defined database constraints operate correctly	10/10 checks passed (100%); PK, FK, NOT NULL, rollback, relational consistency, and persistence behaviors	**Passed**

Technical Test	Related Quality Characteristic	Test Condition	No. of Trials	Expected Output / Acceptance Criterion	Actual Output	Result
					were verified.	
Browser Compatibility	Compatibility	Execute the same nine-step predefined workflow using each available tested browser	36 planned workflow trials across 4 browsers	Core functions and workflows operate correctly in all tested browsers; unavailable browsers are reported as not tested	Chrome: 9/9 passed; Edge: 9/9 passed; Firefox: Not Tested; Safari: Not Tested.	**Partially Passed**
Access Control	Security	Attempt protected functions through direct API access using unauthenticated, Staff, and Owner access states	21 trials	Unauthorized requests are denied and protected data/functions are not exposed	20/21 trials matched the predefined expected response (95.2%); all unauthorized requests were blocked. One authorized Owner request returned HTTP 422 instead of the expected HTTP 400; no privilege leakage was reported.	**Passed with Qualification**
Failed-Login Handling	Security	Submit consecutive invalid credentials and verify account-lockout behavior	5 controlled attempts	Failed attempts follow the implemented policy; third failed attempt triggers 15-minute lockout and subsequent access remains blocked	5/5 trials passed (100%); lockout was triggered on the third failed attempt and subsequent access was blocked.	**Passed**
Backup Restoration	Reliability	Restore a verified system backup in an isolated controlled	4 verification phases	Backup is verified and restored while required tables and	4/4 phases passed (100%); 29 tables were restored with	**Passed**

Technical Test	Related Quality Characteristic	Test Condition	No. of Trials	Expected Output / Acceptance Criterion	Actual Output	Result
		database environment and compare restored records		records remain retrievable and consistent	reported record-count parity, including 103 orders and 1,048 customers.	
Error Recovery	Reliability	Introduce predefined recoverable error conditions and execute subsequent valid operations	6 fault injections	System handles errors without server crash or unintended data loss and resumes normal operation	6/6 tests passed (100%); no server crashes occurred; appropriate error responses were returned; subsequent valid requests succeeded.	**Passed**
Input Validation	Functional Suitability / Reliability / Safety	Submit malformed, invalid, attack, and valid-boundary inputs through the API	15 test vectors	Invalid inputs are rejected, valid inputs are processed correctly, and no database corruption occurs	15/15 vectors passed (100%); 13 invalid/attack payloads were rejected and 2 valid-boundary payloads were accepted; no database corruption was reported.	**Passed**
System Reliability	Reliability	Execute the predefined critical service lifecycle workflow repeatedly under the same test environment	30 consecutive cycles	At least 95% of critical workflows complete without unexpected failure or data corruption	30/30 cycles passed (100%); mean workflow latency was 315.51 ms; no database corruption	**Passed**

Technical Test	Related Quality Characteristic	Test Condition	No. of Trials	Expected Output / Acceptance Criterion	Actual Output	Result
					was reported.	

Note. The results represent objective technical testing conducted under the specified controlled test environment and should not be interpreted as formal ISO/IEC 25010 certification. Browser compatibility was classified as Partially Passed because Firefox and Safari were not tested. Response-time testing was classified as Partially Passed because the mean authentication response time of 517.10 ms exceeded the predefined mean threshold of less than 500 ms, although all 180 requests completed successfully and the reported P95 values remained below 1,000 ms. Access-control testing was classified as Passed with Qualification because all unauthorized access attempts were blocked, although one authorized Owner request produced HTTP 422 rather than the predefined HTTP 400 response.

4.5.11 Summary of ISO/IEC 25010-Based Evaluation Evidence

To provide an integrated presentation of the ISO/IEC 25010-based system evaluation, Table 36 summarizes the different sources of evidence used in assessing the developed system. The evaluation distinguished between perception-based evidence, obtained from the assessment of the five IT experts, and objective or technical evidence, obtained through technical testing and technical artifact inspection. This distinction was maintained because the mean and standard deviation derived from the expert questionnaire describe the evaluators' perceptions of system quality, whereas the technical tests and artifact inspection provide observable evidence of selected system behaviors and technical characteristics. The two forms of evidence were therefore treated as complementary rather than interchangeable measures of system quality.

Table 36

SUMMARY OF ISO/IEC 25010-BASED EVALUATION EVIDENCE

Quality Characteristic	Perception-Based Evidence	Objective/Technical Evidence	Basis of Interpretation
Functional Suitability	Five IT experts' questionnaire ratings	Input-validation testing using 15 API test vectors	Expert-perceived functional suitability supplemented by observable input-validation results
Performance Efficiency	Five IT experts' questionnaire ratings	Response-time testing across six API operations and concurrent-access testing at 1, 5, 10, and 20 workers	Expert-perceived performance efficiency supplemented by measured response-time and concurrent-access results
Compatibility	Five IT experts' questionnaire ratings	Browser-compatibility testing using Google Chrome and Microsoft Edge; Firefox and Safari were not tested	Expert-perceived compatibility supplemented by testing of the available browser environments
Interaction Capability	Five IT experts' questionnaire ratings	No separate objective technical test conducted	Interpretation limited to the perception-based assessment of the IT experts
Reliability	Five IT experts' questionnaire ratings	Concurrent-access, database-integrity, backup-restoration, error-recovery, input-validation, and repeated system-reliability testing	Expert-perceived reliability supplemented by observable technical results under controlled test conditions
Security	Five IT experts' questionnaire ratings	Access-control, failed-login handling, and input-validation testing	Expert-perceived security supplemented by technical testing of selected authentication,

Quality Characteristic	Perception-Based Evidence	Objective/Technical Evidence	Basis of Interpretation
			authorization, and validation controls
Maintainability	Five IT experts' questionnaire ratings	Technical artifact inspection of source-code structure and naming, database schema, dependency organization, technical and test documentation, and error and audit logging	Expert-perceived maintainability supplemented by artifact-level inspection of the system's modularity, analyzability, modifiability, and testability
Flexibility	Five IT experts' questionnaire ratings	No separate objective technical test conducted	Interpretation limited to the perception-based assessment of the IT experts
Safety	Five IT experts' questionnaire ratings	Input-validation and database-integrity testing applicable to the prevention of invalid or inconsistent data	Expert-perceived safety supplemented by observable validation and data-integrity results

Table 36 summarizes the complementary evidence used in evaluating the developed system based on the selected ISO/IEC 25010 quality characteristics. The questionnaire administered to the five IT experts served as the perception-based component of the evaluation, while objective technical testing and technical artifact inspection provided separate evidence concerning selected observable system behaviors and technical characteristics. The expert questionnaire results were therefore interpreted as levels of perceived acceptability and were not treated independently as proof of technical conformity or formal ISO/IEC 25010 certification.

For Functional Suitability, Performance Efficiency, Compatibility, Reliability, Security, and Safety, the perception-based findings were supplemented by

applicable objective technical tests. These tests examined selected measurable system behaviors, including input validation, response time, concurrent access, database integrity, browser compatibility, access control, failed-login handling, backup restoration, error recovery, and repeated system operation. For Maintainability, the perception-based assessment was supplemented by a technical artifact inspection covering source-code structure and naming, database schema, dependency organization, technical documentation, test documentation, and error and audit logging. The artifact inspection identified both maintainability-supporting characteristics and areas requiring improvement, particularly in source-code modularity, dependency organization, database migration management, documentation organization, and automated testing.

For Interaction Capability and Flexibility, no separate objective technical tests were conducted; therefore, conclusions for these characteristics were limited to the perception-based assessments of the five IT experts. Overall, the combined evaluation provides a clearer distinction between subjective expert assessment and objective or artifact-based technical evidence. The technical findings supplement, rather than replace, the expert questionnaire results and are limited to the specific test conditions and technical artifacts examined. Accordingly, the findings should not be interpreted as formal certification of conformity with ISO/IEC 25010. The maintainability inspection itself explicitly makes this limitation.

4.6 Technology Acceptance Model Evaluation

The TAM evaluation was conducted with six (6) shop owners and staff who assessed the system across five constructs: Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Behavioral Intention to Use (BI), Actual System Use (ASU), and Analytics Dashboard Impact.

Table 37

SHOP OWNERS' AND STAFF'S EVALUATION OF THE SYSTEM IN TERMS OF PERCEIVED USEFULNESS (PU)

Perceived Usefulness	N	Mean	Standard Deviation	Interpretation
1. The system makes it easier to manage customer records and job orders.	6	4.0000	.00000	Highly Acceptable
2. The system helps track services and inventory more efficiently.	6	4.0000	.00000	Highly Acceptable
3. The system reduces the need for paperwork and manual record-keeping.	6	4.0000	.00000	Highly Acceptable
General Weighted Mean	**6**	**4.0000**	**.00000**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

As shown in Table 37, the Perceived Usefulness of the Web-Based Service Management System achieved a general weighted mean of 4.00 with a standard deviation of 0.00, which indicated that respondents rated the system as Highly Acceptable in terms of its usefulness. All three indicators were rated Highly Acceptable, reflecting a high level of satisfaction with the system's usefulness. The highest rating was observed in the system's ability to make it easier to manage customer records and job orders, help track services and inventory more efficiently, and reduce the need for paperwork and manual record-keeping. This suggests that the system is highly effective in improving operational efficiency and reducing administrative workload for Shoelotskey personnel.

Table 38

SHOP OWNERS' AND STAFF'S EVALUATION OF THE SYSTEM IN TERMS OF PERCEIVED EASE OF USE (PEOU)

Perceived Ease of Use	N	Mean	Standard Deviation	Interpretation
1. The menus, buttons, and labels are easy to understand.	6	3.83	. 0.41	Highly Acceptable
2. The system is easier to use than the manual paper-based system.	6	3.83	0.41	Highly Acceptable
3. Adding and updating the records in the system is easy.	6	4.0000	.00000	Highly Acceptable
General Weighted Mean	**6**	**3.89**	**0.32**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

As shown in Table 38, the Perceived Ease of Use of the Web-Based Service Management System achieved a general weighted mean of 3.89 with a standard deviation of 0.32, which indicated that respondents rated the system as Highly Acceptable in terms of ease of use. All three indicators were rated Highly Acceptable, reflecting a high level of satisfaction with the system's usability. The highest rating was observed in the system's menus, buttons, and labels being easy to understand, the system being easier to use than the manual paper-based system, and adding and updating records being easy. This suggests that the system is user-friendly and was evaluated as highly acceptable in interaction capability for Shoelotskey personnel to operate effectively.

Table 39

SHOP OWNERS' AND STAFF'S EVALUATION OF THE SYSTEM IN TERMS OF BEHAVIORAL INTENTION TO USE (BI)

Behavioral Intention to Use	N	Mean	Standard Deviation	Interpretation
1. The user intends to continue using the system in performing work at Shoelotskey.	6	3.67	0.52	Highly Acceptable
2. The system is better than the previous way of managing shop work.	6	3.67	0.52	Highly Acceptable
3. The user intends to use the system regularly for managing daily tasks at Shoelotskey.	6	3.50	0.55	Acceptable
General Weighted Mean	**6**	**3.61**	**0.50**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

As shown in Table 39, Behavioral Intention to Use obtained a general weighted mean of 3.61 with a standard deviation of 0.50, interpreted as Highly Acceptable. Although the result remained within the Highly Acceptable range, Behavioral Intention obtained the lowest general weighted mean among the TAM constructs. This indicates that the respondents generally expressed an intention to continue using the developed system, but their responses were comparatively less favorable and showed greater variation than their ratings for Perceived Usefulness, Perceived Ease of Use, Actual System Use, and Analytics Dashboard Impact. The result suggests that favorable perceptions of the system's usefulness and ease of use did not translate into an equally strong level of intention for continued future use. Given the limited two-week pilot-testing period and the small group of six respondents, the finding should be interpreted cautiously. The comparatively lower

Behavioral Intention score may indicate some reservations regarding continued system use; however, the specific reasons for these reservations were not directly measured by the questionnaire. Therefore, the result provides positive but comparatively more cautious evidence of intended continued use and should not be interpreted as confirmation of sustained future adoption.

Table 40

SHOP OWNERS' AND STAFF'S EVALUATION OF THE SYSTEM IN TERMS OF ACTUAL SYSTEM USE (ASU)

Actual System Use	N	Mean	Standard Deviation	Interpretation
1. The user consistently uses the system for receiving shoes.	6	4.00	0.00	Highly Acceptable
2. The user uses the system to record and manage customer job orders.	6	4.00	0.00	Highly Acceptable
3. The user uses the system to check inventory, service status, and reports.	6	4.00	0.00	Highly Acceptable
General Weighted Mean	**6**	**4.00**	**0.00**	**Highly Acceptable**

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

As shown in Table 40, Actual System Use obtained a general weighted mean of 4.00 with a standard deviation of 0.00, interpreted as Highly Acceptable. All six respondents gave the highest rating to the three indicators concerning the use of the system for receiving shoes, recording and managing customer job orders, and accessing inventory, service-status, and report information. The zero standard deviation indicates complete agreement among the six respondents on these questionnaire items during the evaluation period. However, the Actual System Use findings were obtained following a two-week pilot-testing period that

involved both actual and simulated operational activities. Therefore, the result represents the respondents' reported system use and experience within the limited pilot period and should not be interpreted as definitive evidence of sustained long-term system use, complete organizational adoption, or continued use beyond the study period. Longer-term observation of actual operational usage would be necessary to establish sustained system adoption.

Table 41

SUMMARY OF THE TAM AND ISO/IEC 25010 EVALUATION RESULTS

Evaluation	Respondent Group	N	General Weighted Mean	Standard Deviation	Interpretation
TAM Evaluation	Shop Owners and Staff				
Perceived Usefulness (PU)		6	4.0000	0.00	Highly Acceptable
Perceived Ease of Use (PEOU)		6	3.89	0.32	Highly Acceptable
Behavioral Intention to Use (BI)		6	3.61	0.50	Highly Acceptable
Actual System Use (ASU)		6	4.00	0.00	Highly Acceptable
Analytics Dashboard Impact		6	3.83	0.38	Highly Acceptable
ISO/IEC 25010 Evaluation	IT Experts				
Functional Suitability		5	3.64	0.49	Highly Acceptable
Performance Efficiency		5	3.68	0.49	Highly Acceptable

Evaluation	Respondent Group	N	General Weighted Mean	Standard Deviation	Interpretation
Compatibility		5	3.76	0.40	Highly Acceptable
Interaction Capability		5	3.72	0.49	Highly Acceptable
Reliability		5	3.72	0.49	Highly Acceptable
Security		5	3.68	0.49	Highly Acceptable
Maintainability		5	3.72	0.46	Highly Acceptable
Flexibility		5	3.84	0.37	Highly Acceptable
Safety		5	3.64	0.64	Highly Acceptable

Legend: 1.00–1.50 Not Acceptable | 1.51–2.50 Less Acceptable | 2.51–3.50 Acceptable | 3.51–4.00 Highly Acceptable

As shown in Table 41, all TAM constructs and ISO/IEC 25010 quality characteristics were interpreted as Highly Acceptable according to their respective rating scales. Under the TAM evaluation, Perceived Usefulness and Actual System Use obtained the highest general weighted means of 4.00, followed by Perceived Ease of Use at 3.89, Analytics Dashboard Impact at 3.83, and Behavioral Intention to Use at 3.61. Although Behavioral Intention remained Highly Acceptable, its comparatively lower mean indicates that respondents expressed somewhat more reservation regarding continued future use than regarding the system's immediate usefulness, ease of use, and use during the pilot period. Furthermore, the perfect mean and zero standard deviation obtained for Perceived Usefulness and Actual System Use should be interpreted cautiously because the TAM evaluation involved only six respondents and Actual System Use was assessed following a limited two-week pilot period involving actual and simulated activities.

The TAM results represent user perceptions and reported experience during the evaluation period and should not be interpreted as objective evidence of technical system performance or sustained long-term adoption. Similarly, the ISO/IEC 25010 questionnaire results represent the IT experts' perceptions of software quality and are supplemented separately by applicable objective technical tests. Machine-learning predictive performance is evaluated independently using MAE, RMSE, R^2, and prediction-tolerance measures. Accordingly, user acceptance, perceived software quality, measured technical performance, and machine-learning predictive performance are treated as distinct but complementary forms of evidence and are not combined into a single claim of overall system effectiveness.

Table 42

IT EXPERTS' SUMMARY AND RANKING OF SUGGESTIONS / COMMENTS TO THE SYSTEM

Suggestion / Comment	Frequency	Percentage (%)	Rank	Remarks / Implementation
More flexibility in dropdown options, including adding or updating brands, couriers, models, and other entries	3	60.0%	1	Indicates the need for customizable data entry options to avoid hardwired selections and improve system adaptability to changing business needs.
Detailed audit logs for inventory	2	40.0%	2	Highlights the importance of tracking inventory movements for accountability, stock control, and auditability.
Automated SMS/email	2	40.0%	2	Suggests improving customer

Suggestion / Comment	Frequency	Percentage (%)	Rank	Remarks / Implementation
notifications for order status changes				communication through notifications when order statuses change or orders are ready for pickup.
Automated ML retraining pipeline using new historical data	1	20.0%	4	Reflects the recommendation for continuous model improvement as additional historical data become available.
Strict backend validation to reject incomplete payloads at the API level	1	20.0%	4	Indicates the need for backend-level validation to strengthen data integrity beyond user-interface validation.
Redis caching for Sales/Analytics modules and rate limiting on login	1	20.0%	4	Emphasizes recommendations for performance optimization and additional protection against repeated login attempts as the system scales.
WCAG compliance, including color contrast and font sizing	1	20.0%	4	Highlights the importance of improving system accessibility and readability.
Standardize currency display to two decimal places and correct negative pending payments	1	20.0%	4	Indicates the need for consistent monetary formatting and correction of payment computation issues.

Suggestion / Comment	Frequency	Percentage (%)	Rank	Remarks / Implementation
Customer history view showing past orders per client	1	20.0%	4	Reflects the recommendation to provide easier access to customers' previous service records.
Automatic deduction of raw materials from inventory when a job order status changes to “On-Going”	1	20.0%	4	Suggests automating inventory consumption tracking to improve stock monitoring.

Legend: Frequency = number of IT experts who mentioned the suggestion/comment; Percentage = (Frequency ÷ 5) × 100; n = 5 IT experts.

As shown in Table 42, the users' summary and ranking of suggestions and comments revealed that the **most frequently cited suggestion among IT experts** was **more flexibility in dropdown options** (frequency = 3, 60.0%, Rank 1), indicating the need for customizable data entry options to avoid hardwired selections and improve system adaptability. This was followed by **detailed audit logs for inventory** and **automated SMS/email notifications for order status changes** (frequency = 2, 40.0%, Rank 2), highlighting the importance of inventory accountability and improved customer communication. Other recommendations, including automated ML retraining, strict backend validation, Redis caching and login rate limiting, WCAG compliance, currency standardization, customer history view, and automatic deduction of raw materials, each received a frequency of 1 (20.0%, Rank 4).

The highest-ranked suggestions, particularly **dropdown flexibility, inventory audit logs, and SMS/email notifications**, reflect opportunities to further improve **system flexibility, inventory accountability, and customer**

communication. The other recommendations also identify potential areas for future enhancement involving **machine learning maintenance, data validation, system performance, accessibility, data presentation, customer record management, and inventory automation**. These findings suggest that although the system received a **Highly Acceptable** evaluation, the IT experts identified areas that may be considered for further development and improvement of the system's functionality and user experience.

6. Recommend possible system enhancements by identifying improvements and additional features that may be implemented by future researchers to further enhance system functionality, data analytics capabilities, and user experience.

Based on the suggestions and comments presented in Table 24, several possible system enhancements were identified for consideration by future researchers and developers. The most frequently recommended enhancement was **greater flexibility in dropdown options**, mentioned by three IT experts (60.0%), particularly the ability to add or update information such as brands, couriers, models, and other system entries. **Detailed inventory audit logs** and **automated SMS/email notifications for order status changes** were each recommended by two IT experts (40.0%). Other recommendations, each mentioned by one IT expert (20.0%), included automated ML retraining using new historical data, strict backend validation, Redis caching and login rate limiting, WCAG compliance, standardized currency display and correction of negative pending payments, a customer history view, and automatic deduction of raw materials from inventory.

These recommendations identify possible areas for future enhancement in terms of **system functionality, data analytics capabilities, security, accessibility, inventory management, customer communication, and user experience**. The findings may serve as a basis for future researchers and developers in determining which additional features and improvements can be implemented to further enhance the Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay.

CHAPTER 5

Summary, Conclusions, and Recommendations

This chapter presents the summary of findings, conclusions, and recommendations of the study entitled “A Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay.” The study aimed to design, develop, and evaluate a centralized web-based service management system intended to improve the establishment’s operational efficiency, data accuracy, service monitoring, reporting, and managerial decision-making.

The developed system integrated centralized job order management, service-status monitoring, inventory management, sales and operational reporting, descriptive analytics, user and permission management, and predictive analytics using a Random Forest Regressor. The system distinguished the Official Expected Release Date, generated through Shoelotskey’s predefined business rules, from the ML Predicted Release Date, generated by the machine learning component. The Random Forest component predicts Ready-for-Release Duration and uses the resulting predicted duration to derive the corresponding ML Predicted Release Date.

The study evaluated the developed system through manual-versus-digital comparative testing, machine learning performance evaluation, analytics dashboard assessment, ISO/IEC 25010-based software quality evaluation, and Technology Acceptance Model (TAM)-based user acceptance evaluation.

Summary of Findings

Based on the results presented and discussed in Chapter 4, the following findings were obtained according to the research questions and specific objectives of the study.

1. Comparative Analysis of the Existing Manual Process and the Developed System

The comparative analysis showed that the developed centralized digital system improved record accessibility and substantially reduced record retrieval time compared with the existing manual paper-based process.

Based on ten paired retrieval trials, the existing manual process obtained a mean retrieval time of 5.20 minutes, whereas the developed system obtained a mean retrieval time of 0.41 minute, representing a 92.12% reduction in mean retrieval time. The median retrieval time decreased from 5.00 minutes to 0.40 minute, while the observed retrieval-time range decreased from 3.00–10.00 minutes to 0.30–0.50 minute.

The data-accuracy assessment also demonstrated favorable results during the formal digital pilot period. The historical manual-record audit examined 119 paper receipts, of which 50 records or 42.0% lacked claim/completion dates. In comparison, the 31 digital job orders examined during the September 2–16, 2026 pilot period contained zero duplicate records and zero incomplete records. Three digital records required post-entry correction, corresponding to 9.68% of the 31 records or 0.20 correction per day. The developed system also incorporated automated field validation and centralized digital service-status monitoring.

However, the data-accuracy findings were interpreted descriptively because the historical manual records and digital pilot records came from different observation periods and equivalent transaction-level denominators were unavailable for some manual indicators. Thus, the findings indicate improved record accessibility, completeness, validation, and retrieval performance within the evaluated conditions rather than a statistically matched comparison of identical manual and digital transactions.

2. Predictive Performance of the Random Forest Regressor

The machine learning component was developed and evaluated through two distinct but related stages involving Dataset A and Dataset B.

Dataset A consisted of 119 validated historical service orders derived from 714 digitized paper receipts. The historical records contained Date Received and Claimed Date but did not contain a reliable actual Ready-for-Release timestamp. Consequently, the historical target was defined as Order-to-Claim Turnaround Days, which included both the service-processing period and possible customer pickup delay. The study appropriately recognized that this target did not directly represent the actual service-completion duration the operational model was intended to predict.

On the historical 26-record temporal holdout set, the Random Forest Regressor obtained an MAE of 8.35 days, RMSE of 10.19 days, and R^2 of −0.0125. Its prediction-tolerance results were 11.5% within ±1 day, 23.1% within ±2 days, 30.8% within ±3 days, and 42.3% within ±5 days. The historical Random Forest did not outperform Shoelotskey's existing business-rule estimate. These findings demonstrated the limitation of using customer Claim Date as a proxy for actual service completion.

Based on the limitation identified in Dataset A, the prediction target for the finalized operational model was refined to Received-to-Ready-for-Release Duration. Dataset B was subsequently developed as a 10,000-record simulated operational dataset representing the verified service configurations and operational conditions required for the finalized prediction task. It was divided into 7,000 training records, 1,500 validation records, and 1,500 final test records.

Linear Regression, Decision Tree Regressor, and Random Forest Regressor were compared during model selection. Although the Decision Tree obtained a slightly lower validation MAE of 0.0616 day compared with 0.0641 day for Random Forest, the Random Forest obtained a lower RMSE of 0.2994 day, a higher R^2 of 0.9987, and a higher ±1-day prediction tolerance of 97.73%. Based on the combined evaluation measures and its ensemble structure, Random Forest was selected as the finalized operational model.

The finalized Random Forest was subsequently evaluated on the independent 1,500-record final test partition and obtained an MAE of 0.0815 day, RMSE of 0.4042 day, and R^2 of 0.9976. Prediction-tolerance analysis showed that 97.20% of predictions were within ±1 day, 98.93% within ±2 days, 99.53% within ±3 days, and 100.00% within ±5 days.

Therefore, the study achieved the machine learning objective of quantitatively evaluating the predictive performance the Random Forest predictive component using the specified performance measures. The finalized model demonstrated strong predictive performance under the simulated operational conditions represented by Dataset B and was integrated into the developed system to generate an advisory ML Predicted Release Date.

However, Dataset B consists of simulated operational service orders. Consequently, the obtained predictive performance should not be interpreted as equivalent to external validation using a large independent dataset of actual Ready-for-Release transactions. The manuscript appropriately recognizes this limitation and provides for future validation and retraining using actual Ready-for-Release observations accumulated through continued system use.

3. Impact of the Real-Time Analytics Dashboard

The real-time analytics dashboard demonstrated substantial improvement in report-generation efficiency under the evaluated conditions.

Seven report categories were tested: Daily Sales, Weekly Sales, Monthly Sales, Quarterly Sales, Annual Sales, Expenses, and Return on Investment. The existing manual reporting procedures obtained an overall mean preparation time of 43.71 minutes, whereas the developed dashboard obtained an overall mean application-level report-generation response time of approximately 0.0838 second. All seven evaluated reports passed content-equivalence verification against independently calculated values.

The comparison resulted in an observed 99.9968% reduction in report-generation time. However, this percentage applies specifically to the evaluated report-generation stage because the manual measurements represented human preparation activities while the digital measurements represented automated application/API processing after the required transaction data had already been digitized and stored. Therefore, the percentage does not indicate that the entire human reporting workflow became 99.9968% faster.

The intended users also evaluated the Analytics Dashboard Impact as Highly Acceptable, with a general weighted mean of 3.83 and a standard deviation

of 0.38. The dashboard received particularly favorable ratings for making important business information easier to access and providing useful information for monitoring sales, completed services, pending job orders, inventory levels, and service trends.

The findings therefore indicate that the analytics dashboard improved the efficiency of generating the evaluated reports and provided accessible operational information to support monitoring and decision-making within the conditions of the study.

4. Software Quality Based on ISO/IEC 25010

The developed system was evaluated by five IT experts using nine applicable ISO/IEC 25010 software quality characteristics.

The results were Functional Suitability, 3.64; Performance Efficiency, 3.68; Compatibility, 3.76; Interaction Capability, 3.72; Reliability, 3.72; Security, 3.68; Maintainability, 3.72; Flexibility, 3.84; and Safety, 3.64. All nine characteristics were interpreted as Highly Acceptable.

Flexibility obtained the highest general weighted mean of 3.84, followed by Compatibility at 3.76. Interaction Capability, Reliability, and Maintainability each obtained 3.72; Performance Efficiency and Security each obtained 3.68; and Functional Suitability and Safety each obtained 3.64.

The expert questionnaire was supplemented by applicable objective technical testing and technical-artifact inspection. These procedures examined selected measurable system behaviors, including input validation, response time, concurrent access, database integrity, browser compatibility, access control, failed-login handling, backup restoration, error recovery, and repeated system operation.

Maintainability was additionally examined through inspection of source-code structure, database schema, dependencies, technical documentation, testing artifacts, and error and audit logging.

The results therefore indicate a Highly Acceptable perceived level of software quality across all nine evaluated ISO/IEC 25010 characteristics, supported by separate technical evidence for applicable measurable characteristics. These findings represent a capstone-level system evaluation and do not constitute formal ISO/IEC 25010 certification.

5. User Acceptance Based on the Technology Acceptance Model

The Technology Acceptance Model evaluation was conducted among six intended users consisting of Shoelotskey management and staff.

Perceived Usefulness obtained a general weighted mean of 4.00, Perceived Ease of Use obtained 3.89, Behavioral Intention to Use obtained 3.61, and Actual System Use obtained 4.00. All four TAM constructs were interpreted as Highly Acceptable.

Perceived Usefulness and Actual System Use received the highest general weighted means, indicating favorable perceptions regarding the system's immediate operational usefulness and respondents' reported experience using its functions. Perceived Ease of Use also obtained a highly favorable evaluation.

Behavioral Intention to Use obtained the lowest general weighted mean among the TAM constructs at 3.61, although it remained Highly Acceptable. This indicates positive but comparatively more cautious intention regarding continued future use.

The Actual System Use results were obtained following the two-week pilot-testing period involving actual and simulated operational activities. Therefore, the findings represent respondents' reported use and experience during the evaluation period and should not be interpreted as definitive evidence of sustained long-term organizational adoption.

6. Recommended Improvements and Future Enhancements

The findings identified several areas for continued system improvement. The IT experts recommended increased flexibility in configurable dropdown options, more detailed inventory audit logs, and improvements to notification capabilities. The technical maintainability assessment also identified opportunities for improving source-code modularity, dependency organization, database migration management, technical documentation, and automated testing.

The machine learning component requires continued collection of actual Ready-for-Release observations. These records may eventually provide a sufficiently large real operational dataset for external validation, model retraining, and comparison of the current Random Forest model against alternative predictive approaches.

Longer-term observation is also recommended to determine whether the positive user-acceptance findings observed during the two-week pilot period continue under sustained operational use.

Conclusions

Based on the findings of the study, the following conclusions were drawn.

1. The developed Web-Based Service Management System improved service-record accessibility and retrieval efficiency compared with the existing manual paper-based process under the evaluated conditions. The reduction in mean retrieval time from 5.20 minutes to 0.41 minute demonstrates the practical benefit of centralized digital record management. The absence of duplicate and incomplete records among the 31 digital job orders examined during the formal pilot further indicates improved record completeness during the evaluated period. However, the data-accuracy results remain descriptive because the manual and digital evidence originated from different observation periods.

2. The machine learning objective was achieved in terms of quantitatively evaluating the predictive performance of the Random Forest Regressor using the specified MAE, RMSE, R^2, and prediction-tolerance measures. The historical Dataset A assessment revealed that Order-to-Claim Turnaround was not an appropriate operational target for predicting actual service completion because the target incorporated customer pickup behavior. This finding provided the methodological basis for refining the operational prediction target to Received-to-Ready-for-Release Duration.

The finalized Random Forest model developed and evaluated using Dataset B obtained an MAE of 0.0815 day, RMSE of 0.4042 day, R^2 of 0.9976, and 97.20% of final-test predictions within ±1 day. These findings demonstrate strong predictive performance under the simulated operational conditions represented by Dataset B. Nevertheless, because Dataset B was simulated, the results do not establish equivalent predictive performance on a large independent dataset of

actual Shoelotskey Ready-for-Release transactions. Therefore, the ML Predicted Release Date remains an advisory predictive output, while the Official Expected Release Date remains separately determined through Shoelotskey's predefined business rules.

3. The developed analytics dashboard substantially improved the efficiency of generating the evaluated business reports after the required operational data had been digitized and stored. All seven evaluated report types passed content-equivalence verification, and the intended users rated the overall impact of the dashboard as Highly Acceptable. The dashboard therefore serves as an effective tool for providing timely access to operational information and supporting business monitoring under the evaluated conditions.

4. The developed system obtained Highly Acceptable ratings across all nine evaluated ISO/IEC 25010 software quality characteristics. The results indicate positive IT-expert perceptions of the system's Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety. Applicable technical tests and artifact inspections provided complementary evidence for selected measurable system characteristics. These findings support the suitability of the developed system for its intended operational context but do not constitute formal ISO/IEC 25010 certification.

5. The developed system achieved a Highly Acceptable level of user acceptance based on the Technology Acceptance Model. The intended users positively evaluated its Perceived Usefulness, Perceived Ease of Use, Behavioral Intention to Use, and Actual System Use. The findings indicate that the system was considered useful and generally easy to use during the pilot period. However, the limited number of respondents and two-week pilot duration mean that sustained long-term adoption has not yet been established.

6. Overall, the study achieved its primary objective of designing, developing, implementing, and evaluating a centralized Web-Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor-Pasay. The system addressed the identified operational requirements through centralized record management, faster retrieval, automated reporting, real-time analytics, inventory and service-status monitoring, security and access controls, and advisory machine learning prediction. At the same time, limitations concerning historical data quality, the absence of historical Ready-for-Release timestamps, simulated operational data used for the finalized ML model, the limited number of evaluators, and the short pilot duration establish appropriate directions for future improvement and validation.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are proposed.

1. For Shoelotskey Villamor-Pasay. The establishment may continue using the developed system for centralized job-order management, service-status monitoring, inventory management, sales and operational reporting, analytics, and other supported internal functions. Consistent digital recording of transactions should be maintained to reduce dependence on fragmented paper records and establish a more complete operational database. Regular database backups, audit-log reviews, user-permission management, and system maintenance should also be maintained to support the integrity, availability, and security of business information.

2. For the machine learning component. The establishment should continuously record the actual timestamp at which every job order reaches the For

Release/Ready-for-Release stage. These observations should be accumulated to create a real operational dataset that directly measures Received-to-Ready-for-Release Duration. Once a sufficiently representative dataset has been collected, the current Random Forest model should be externally validated and subsequently retrained using actual operational observations. Future evaluations should continue reporting MAE, RMSE, R^2, prediction tolerance, residual behavior, and relevant baseline comparisons. Until sufficient real-world validation has been completed, the ML Predicted Release Date should remain advisory and should not replace the Official Expected Release Date.

3. For the analytics dashboard. Future versions may provide additional configurable visualizations, comparative period analyses, trend indicators, and other decision-support information that can assist management in interpreting sales, service demand, inventory usage, and operational performance. Particular attention may be given to strengthening decision-support functions while maintaining the reporting efficiency demonstrated in the present study.

4. For software quality and maintainability. Future development should address the improvement areas identified during technical assessment, particularly source-code modularity, dependency organization, database migration management, technical documentation, and automated testing. Regression, integration, functional, security, and performance testing should be repeated following significant system modifications.

5. For security and auditability. Role- and permission-based access control, authentication safeguards, account lockout, inactivity controls, audit logging, and database backup procedures should continue to be maintained and tested. More detailed inventory audit logs may be implemented to improve accountability for stock additions, usage, adjustments, and other inventory movements.

6. For system flexibility and notifications. Configurable dropdown options may be expanded so that authorized users can add or update business-specific entries such as brands, models, couriers, and other operational selections without requiring source-code modification. Notification functionality may likewise be enhanced based on the operational requirements of the establishment.

7. For user acceptance and implementation. A longer implementation and observation period is recommended to determine whether the Highly Acceptable TAM findings are sustained beyond the two-week pilot period. Future evaluations may examine actual frequency of use, task completion, user errors, continued usage, and satisfaction over several months.

8. For future researchers. Future studies may evaluate the system using larger respondent groups, longer implementation periods, additional branches or comparable service establishments, and larger collections of actual Ready-for-Release transactions. Researchers should maintain a clear distinction among historical, simulated, and actual operational datasets and should not interpret performance on simulated data as equivalent to external real-world validation.

9. For future machine learning research. Once sufficient actual Ready-for-Release records have been collected, future researchers may compare Random Forest with other regression and ensemble algorithms using the same real operational dataset, target definition, training-validation-test procedure, and evaluation metrics. This would provide stronger evidence regarding which predictive approach is most appropriate for Shoelotskey's actual operating environment.

10. For future system expansion. Features outside the present scope, including customer-facing service tracking, online booking, mobile application

support, external payment-gateway integration, accounting-system integration, third-party delivery integration, multi-branch management, and advanced enterprise analytics, may be considered based on future operational requirements and available resources.

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